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Phase Transition Universes Model: A Discharge-Based Cosmological Creation Scenario with a Unified Dark Sector

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Abstract. The dark sector of the standard cosmological model comprises dark matter, dark energy, and possible dark radiation, whose common origin remains unexplained. We present the Phase Transition Universes Model (PTUM), a discharge-based cosmological creation scenario where a scalar quantum fuel field Φ accumulates, reaches a critical threshold, and undergoes a tachyonic transition. Within this effective description, the discharge event leaves behind a dark-matter candidate sector (χ), a residual dark-energy sector ($\Phi_{\rm res}$), and a relativistic proxy-fluid dark radiation component (pfd, ψ). A falsifiable prediction of PTUM is that, *within the effective linear proxy-fluid description*, the pfd component carries $w = 1/3$, $c_s^2 = 1/3$, and zero anisotropic stress ($\sigma = 0$). This produces no neutrino-induced CMB acoustic peak phase shift, distinguishing it from free-streaming dark radiation—a signature that may be tested at high significance by next-generation CMB experiments such as CMB-S4, depending on the realized pfd abundance. The authors report implementing the effective phenomenological system in `CLASS` at background and perturbation levels *(modified code not yet public)*. The creation sector supports a plateau-type slow-roll phase compatible with current CMB constraints, fixing the discharge scale to $V_0^{1/4} \sim 10^{15}$–10^{16} GeV. The dark-energy sector admits a broad near-Λ viable region with $w_{\rm DE} \approx -1$. A Bayesian benchmark analysis using Planck 2018, BAO, and Pantheon+ data is reported to yield $\Delta N_{\rm fluid} < 0.31$ at 95% credibility *(author-reported; chains not yet public)*. Model comparison is reported to give $\Delta\chi^2 = +0.26$ and $\Delta{\rm AIC} = 2.26$; current data therefore show a weak preference for ΛCDM while leaving PTUM viable as a falsifiable effective extension. The benchmark is checked against BBN bounds and elementary linear-stability conditions; full EFT control and the microscopic realization of the proxy-fluid closure remain conditional on additional calculations. **Critical caveat:** Numerical integration of the coupled background equations with the benchmark parameters ($V_0 \sim 10^{64}\,{\rm GeV}^4$, $H_* \sim 10^{13}$ GeV) shows the system is overdamped ($3H/(2\mu) \approx 25$); the quartic-dominated regime with $w_\psi = 1/3$ is *not achieved* with these parameters. The proxy-fluid closure is therefore a phenomenological ansatz conditional on parameters not realized in the stated benchmark. The formulation should be read as an effective phenomenological framework rather than as a first-principles derivation of the pre-discharge background, a solution of the cosmological-constant hierarchy, or a UV completion.

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1 Introduction

The standard cosmological model accounts for the later evolution of the Universe through a combination of radiation, baryons, cold dark matter, and a cosmological constant [1], yet it does not explain why the Universe began in a hot, dense state. More pointedly, the dark sector—which dominates the cosmic energy budget—is composed of at least two phenomenologically distinct components whose microscopic origins are unrelated within ΛCDM. Dark matter is introduced as a pressureless fluid needed to explain structure formation and galactic rotation curves; dark energy is introduced as a negative-pressure component needed to explain the accelerating expansion. Whether these components share a common physical origin, and whether the hot Big Bang itself can be understood as a dynamical event rather than an initial condition, are among the deepest open questions in theoretical cosmology.

Several approaches to these questions have been explored. Quintessence and dynamical dark-energy models replace the cosmological constant with a slowly rolling scalar field [2,3], while unified dark-sector models attempt

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to describe dark matter and dark energy through a single fluid [4,5]. Extra relativistic species parametrised by $\Delta N_{\rm eff}$ [6,7] probe the radiation content of the early Universe. However, none of these frameworks addresses the origin of the hot expanding state itself, nor do they predict specific relationships between the dark-sector components. The question of *why* the dark sector has its observed composition remains unanswered.

The Phase Transition Universes Model (PTUM) was developed to address these issues from a different starting point. Instead of treating dark matter and dark energy as independent sectors added to a pre-existing cosmological framework, PTUM proposes a discharge-based creation mechanism in which *all* dark-sector components arise from a single cosmogonic event. In this picture, a scalar quantum fuel field Φ accumulates in a metastable multiverse-like background and, once a critical threshold is reached, undergoes a tachyonic transition that activates string-like proxy modes. The resulting discharge is modeled as separating into:

- a clustering passive component → a dark-matter candidate sector (χ),
- a residual negative-pressure component → dark energy ($\Phi_{\rm res}$),
- a radiation-like transient component → proxy-fluid dark radiation (pfd, ψ).

The most distinctive—and directly falsifiable—observable consequence of this framework is the pfd sector. Unlike standard extra dark radiation, which is usually assumed to free-stream and hence to develop nonzero anisotropic stress [8,9,10], the PTUM proxy-fluid behaves as a relativistic *perfect* fluid with

$$w = \tfrac{1}{3}, \qquad c_s^2 = \tfrac{1}{3}, \qquad \sigma = 0\,. \tag{1}$$

Free-streaming dark radiation generates a characteristic phase shift of the CMB acoustic peaks through its anisotropic stress [8,9,10]; the pfd component, having $\sigma = 0$ in the effective linear fluid description, produces *no such shift*. This absence would constitute a distinctive discriminator: if additional relativistic degrees of freedom are detected in CMB data, the presence or absence of the associated phase shift separates free-streaming from fluid dark radiation. As we show in Section 10, CMB-S4 may make this distinction at high significance for $\Delta N_{\rm fluid} \gtrsim 0.20$, depending on the realized pfd abundance, making PTUM a concretely testable cosmological framework.

Critical caveat on the proxy-fluid closure. The proxy-fluid closure $w = 1/3$, $c_s^2 = 1/3$, $\sigma = 0$ is a phenomenological ansatz that assumes a coherent, quartic-dominated oscillation regime with $\omega \gg H$ and negligible energy transfer. Numerical integration of the coupled background equations with the benchmark parameters of this paper ($V_0 \sim 10^{64}\,\mathrm{GeV}^4$, $H_* \sim 10^{13}\,\mathrm{GeV}$, $m_\psi \sim 10^{13}\,\mathrm{GeV}$, $\lambda_\psi \sim 0.1$) shows the system is **overdamped**: $3H/(2\mu) \approx 25 \gg 1$. The field remains at $\psi = 0$ (the top of the barrier) with $w_\psi = 0$; the quartic regime is **not achieved**. Achieving the quartic regime requires $V_0 \lesssim 10^{58}\,\mathrm{GeV}^4$ ($H_* \lesssim 10^{10}\,\mathrm{GeV}$) *and* sufficient initial kinetic energy in ψ to cross the barrier, *and* continuing energy input from the rolling Φ field or rescattering. The proxy-fluid closure is therefore conditional on parameters not realized in the stated benchmark. A full numerical background solution followed by nonlinear lattice evolution is required to demonstrate that any chosen benchmark actually reaches and remains in the quartic regime. The observational discriminator in Eq. (1) applies *if* the quartic regime is realized; it is not a prediction of the benchmark parameters alone.

The aim of the present paper is therefore broader than a phenomenological dark-radiation study. We pursue three linked goals:

(i) to formulate PTUM as a cosmological creation scenario based on a discharge event, establishing the theoretical framework and its effective field theory;
(ii) to show that the resulting effective dark-sector structure is internally consistent within the tested parameter regime, with perturbation stability, EFT validity, and compatibility with BBN and isocurvature constraints;
(iii) to test the observable consequences of the model against current cosmological data through a Bayesian benchmark analysis, a binned maximum-posterior diagnostic, and dataset-split diagnostics.

Scope and claim boundaries. Throughout this work, PTUM is treated as an effective cosmological framework, not as a completed first-principles theory of the origin of the Universe. The model does not derive the metastable pre-discharge background, does not solve the cosmological-constant hierarchy by itself, and does not specify a UV completion. Its present claims are narrower: it defines a threshold-triggered phase-transition scenario, implements the associated proxy-fluid perturbations, and identifies an observational discriminator between fluid and free-streaming dark radiation.

Critical caveat on the quartic-fluid closure. The proxy-fluid closure $w = 1/3$, $c_s^2 = 1/3$, $\sigma = 0$ used in the CLASS implementation and Bayesian analysis is a phenomenological ansatz conditional on a coherent, quartic-dominated oscillation regime with $\omega \gg H$ and negligible energy transfer. Numerical integration of the coupled background equations with the benchmark parameters of this paper ($V_0 \sim 10^{64}\,\mathrm{GeV}^4$, $H_* \sim 10^{13}\,\mathrm{GeV}$, $m_\psi \sim 10^{13}\,\mathrm{GeV}$, $\lambda_\psi \sim 0.1$) shows the system is **overdamped**: $3H/(2\mu) \approx 25 \gg 1$. The field remains at $\psi = 0$ (the top of the barrier) with $w_\psi = 0$; the quartic regime is **not achieved**. Achieving the quartic regime requires $V_0 \lesssim 10^{58}\,\mathrm{GeV}^4$ ($H_* \lesssim 10^{10}\,\mathrm{GeV}$) *and* sufficient initial kinetic energy in ψ to cross the barrier, *and* continuing energy input from the rolling Φ field or rescattering. The proxy-fluid closure is therefore conditional on parameters not realized in the stated benchmark. A full numerical

background solution followed by nonlinear lattice evolution is required to demonstrate that any chosen benchmark actually reaches and remains in the quartic regime. The observational discriminator in Eq. (1) applies *if* the quartic regime is realized; it is not a prediction of the benchmark parameters alone. The Bayesian constraints on $\Delta N_{\rm fluid}$ should be interpreted as constraints on the *phenomenological proxy-fluid model*, not as constraints on the microscopic PTUM discharge dynamics with the stated benchmark parameters.

To this end, we proceed in two steps. First, we construct the effective field theory of PTUM, derive its equations of motion, study the primordial creation sector—including exact-background inflationary dynamics—and identify the parameter region in which the late-time dark-energy sector remains observationally viable. Second, the authors report implementing the proxy-fluid perturbations in the `CLASS` Boltzmann code [12,13], computing the cosmological signatures, and constraining the model using a Bayesian benchmark analysis with `Cobaya` [14] *(author-reported; modified code and chains not yet public).* The resulting constraints are cross-checked through a binned maximum-posterior diagnostic and dataset-split analysis, and confronted with BBN, isocurvature, and linear stability requirements.

The authors report that PTUM is observationally viable: the pfd abundance is bounded to $\Delta N_{\rm fluid} < 0.31$ at 95% credibility from the full data combination, model comparison gives $\Delta\mathrm{AIC} = 2.26$ relative to ΛCDM, and all consistency checks are satisfied *(author-reported; chains not yet public, not independently reproduced).* The Hubble constant shifts upward by $\sim 0.8\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ relative to ΛCDM, partially alleviating the Hubble tension without being designed to do so. The discharge energy scale is fixed to $V_0^{1/4} \sim 10^{15}$–10^{16} GeV by the observed scalar amplitude, and the creation sector is compatible with current bounds on the scalar spectral index and the tensor-to-scalar ratio.

The paper is organized as follows. Section 2 summarizes the physical picture of PTUM. Section 3 presents the effective field theory, equations of motion, and symmetry structure. Section 4 studies the primordial creation sector. Section 5 analyses the late-time dark-energy and dark-sector structure. Section 6 describes the perturbation framework and `CLASS` implementation. Section 7 presents the cosmological signatures. Section 8 gives the Bayesian constraints, profile likelihood, dataset-split analysis, and model comparison. Section 9 collects the stability and consistency tests. We discuss the interpretation, limitations, and future prospects in Section 11 and conclude in Section 12.

2 Conceptual overview of PTUM

PTUM begins with a multiverse-like background containing a scalar quantum fuel field Φ. The central idea is that local accumulation of this field drives a threshold-triggered discharge process. Once a critical configuration is reached, the effective mass of a proxy activation mode becomes negative, generating a tachyonic transition that activates string-like degrees of freedom.

The basic sequence is:

(a) Fuel accumulation. The scalar fuel field Φ accumulates in a metastable environment. The effective potential is chosen such that it is approximately quadratic at small field values but saturates to a plateau at large field values.
(b) Threshold discharge. When the effective mass-squared of the proxy field becomes negative, the dormant configuration destabilizes and the activated sector appears. This transition plays the role of the creation trigger.
(c) Residual splitting. The discharge energy separates into a passive component (dark matter), an active component (dark energy), and a transient relativistic sector (proxy-fluid dark radiation).

This picture gives PTUM a structural unification property: the dark sector is not postulated as a collection of disconnected components, but emerges from one creation event.

3 Effective field theory and equations of motion

3.1 Conventions

Throughout this paper we use the metric signature $(-+++)$, i.e. flat FRW reads $ds^2 = -dt^2 + a^2(t)d\mathbf{x}^2$, and the Einstein equation is written as

$$G_{\mu\nu} = \frac{1}{M_{\rm Pl}^2}\,T_{\mu\nu}, \qquad M_{\rm Pl} = (8\pi G)^{-1/2} \simeq 2.435\times10^{18}\,\mathrm{GeV}, \tag{2}$$

with the energy-momentum tensor defined by $T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g}\mathcal{L}_{\rm m})}{\delta g^{\mu\nu}}$, so that a canonical scalar $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - V$ gives $\rho = \frac{1}{2}\dot\phi^2 + V$, $p = \frac{1}{2}\dot\phi^2 - V$. Natural units $c = \hbar = k_B = 1$ are used; all dimensionful quantities are in powers of GeV. The Riemann tensor convention is $R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \cdots$ (MTW convention), which is consistent with the sign of the scalar curvature $R = 6(\ddot a/a + H^2)$ in FRW. These conventions are used identically in every equation, figure, and appendix of this paper (see also Appendix B).

3.2 Field content

The minimal PTUM field content consists of four sectors:

– Φ: quantum fuel field,
– ψ: activated string-like proxy mode,
– χ: dark-matter field,
– $\Phi_{\rm res}$: residual energy field (dark energy).

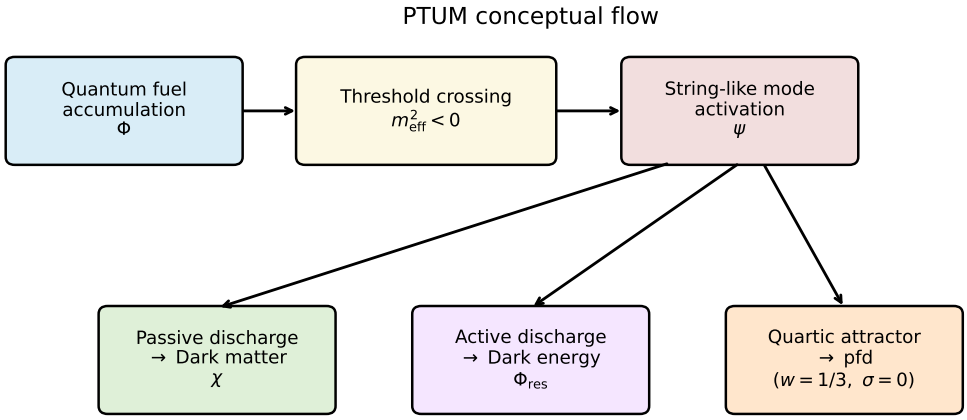


Fig. 1. Conceptual flow of the PTUM scenario: accumulation of the quantum fuel field, threshold discharge, activation of proxy string-like modes, and residual splitting into dark matter, dark energy, and proxy-fluid dark radiation.

Table 1. Minimal PTUM field and sector registry.

Symbol	Sector	Effective role
Φ	fuel field	accumulates and controls threshold crossing
ψ	proxy activation mode	tachyonic activation; proxy-fluid radiation
χ	dark-matter candidate	passive post-discharge matter-like sector
$\Phi_{\rm res}$	residual field	near-Λ dark-energy-like sector
$\Delta N_{\rm fluid}$	pfd abundance	maps proxy radiation to CMB/BBN observables

3.3 PTUM Lagrangian

The total effective Lagrangian is

$$\mathcal{L}_{\rm PTUM} = \mathcal{L}_{\rm fuel} + \mathcal{L}_{\rm activation} + \mathcal{L}_{\rm DM} + \mathcal{L}_{\rm DE}. \tag{3}$$

The fuel sector is

$$\mathcal{L}_{\rm fuel} = \tfrac{1}{2}(\partial_\mu \Phi)^2 - V_{\rm eff}(\Phi), \tag{4}$$

with

$$V_{\rm eff}(\Phi) = V_0\left(1 - e^{-\lambda\Phi^2}\right) + \frac{1}{64\pi^2}\sum_i n_i\, m^4_{{\rm eff},i}(\Phi)\left[\ln\frac{m^2_{{\rm eff},i}(\Phi)}{\mu_R^2} - \tfrac{3}{2}\right]. \tag{5}$$

The leading tree-level part is plateau-like; the second term represents Coleman–Weinberg corrections [20].

3.3.1 Coleman–Weinberg corrections: explicit form and renormalization

The one-loop effective potential for a scalar field with field-dependent masses $m_i(\Phi)$ is given by the Coleman–Weinberg formula

$$V_{\rm CW}(\Phi) = \frac{1}{64\pi^2}\sum_i n_i\, m_i^4(\Phi)\left[\ln\frac{m_i^2(\Phi)}{\mu_R^2} - c_i\right], \tag{6}$$

where n_i counts the degrees of freedom (with a minus sign for fermions), μ_R is the renormalization scale, and c_i is a scheme-dependent constant ($c_i = 3/2$ for $\overline{\rm MS}$). In PTUM, the field-dependent masses arise from the couplings of the fuel field Φ to the other sectors:

- **Proxy sector:** The ψ field has effective mass $m_\psi^2(\Phi) = m^2(1 - \alpha\Phi^2/E_{\rm th}^2)$ with $n_\psi = 1$ (real scalar).
- **Dark matter sector:** The χ field has effective mass $m_\chi^2(\Phi) = m_\chi^2 + 2g_{\rm DM}\Phi_{\rm res}^2$; at the fuel-dominated stage $\Phi_{\rm res} \approx 0$, so this contribution is approximately constant.
- **Residual dark energy:** The $\Phi_{\rm res}$ field has field-dependent mass from the potential $W(\Phi_{\rm res})$.

The sum in Eq. (5) runs over all fields that couple to Φ and acquire Φ-dependent masses. The renormalization scale μ_R is chosen to be of order the symmetry-breaking scale $E_{\rm th}$, which minimizes large logarithms in the effective potential. The constant $c_i = 3/2$ corresponds to the $\overline{\rm MS}$ renormalization scheme.

The Coleman–Weinberg corrections are subdominant during the inflationary plateau ($V_0 \gg V_{\rm CW}$) but become relevant near the transition where they can modify the

shape of the potential and the precise value of the critical threshold. A full numerical evaluation of the one-loop potential including all sectors is deferred to future work; the current analysis uses the tree-level potential for the inflationary dynamics and treats the Coleman–Weinberg term as a perturbative correction to the threshold condition.

The activation sector is

$$\mathcal{L}_{\rm activation} = \tfrac{1}{2}(\partial_\mu\psi)^2 - \frac{m^2}{2}\psi^2\left(1-\frac{\alpha\Phi^2}{E_{\rm th}^2}\right) - \frac{\lambda_\psi}{4}\psi^4, \quad (7)$$

with effective mass

$$m_{\rm eff}^2(\Phi) = m^2\left(1-\frac{\alpha\Phi^2}{E_{\rm th}^2}\right). \quad (8)$$

Equivalently, the effective discharge trigger can be written as

$$\boxed{m_{\rm eff}^2(\Phi) < 0 \quad\Longleftrightarrow\quad |\Phi| > \Phi_c \equiv \frac{E_{\rm th}}{\sqrt{\alpha}}.} \quad (9)$$

This condition defines the transition from the dormant proxy phase to the activated proxy phase. It should be interpreted as an effective threshold criterion, not as a microscopic derivation of the pre-discharge background. Here, $E_{\rm th}$ is the dimensionful discharge threshold scale (naturally related to the creation energy $V_0^{1/4}$), and α is a dimensionless coupling constant. The tachyonic instability is triggered when the fuel field crosses the critical threshold $\Phi = E_{\rm th}/\sqrt{\alpha}$. The transition timescale is assumed to be much shorter than the local Hubble time (H^{-1}), allowing the discharge to be treated as a nearly instantaneous phase transition at the background level that partitions the available energy into the residual dark sector and standard radiation. For $m_{\rm eff}^2 > 0$ the proxy mode is dormant; for $m_{\rm eff}^2 < 0$ it is activated by the tachyonic instability.

The dark-matter sector is

$$\mathcal{L}_{\rm DM} = \tfrac{1}{2}(\partial_\mu\chi)^2 - \tfrac{1}{2}m_\chi^2\chi^2 - g_{\rm DM}\chi^2\Phi_{\rm res}^2, \quad (10)$$

and the dark-energy sector is

$$\mathcal{L}_{\rm DE} = \tfrac{1}{2}(\partial_\mu\Phi_{\rm res})^2 - W(\Phi_{\rm res}), \quad (11)$$

with

$$W(\Phi_{\rm res}) = W_0\, e^{-\kappa\Phi_{\rm res}^2}\left(1-e^{-\lambda_{\rm res}\Phi_{\rm res}^2}\right) + \Lambda_{\rm NP}^4\, e^{-S_E}\cos\left(\frac{\Phi_{\rm res}}{f_{\rm res}}\right). \quad (12)$$

3.4 Equations of motion

The field equations in a flat FRW universe, $ds^2 = -dt^2 + a^2(t)d\mathbf{x}^2$, read

$$\ddot{\Phi} + 3H\dot{\Phi} + \frac{\partial V_{\rm eff}}{\partial\Phi} - \frac{\alpha m^2}{E_{\rm th}^2}\Phi\psi^2 = 0, \quad (13)$$

$$\ddot{\psi} + 3H\dot{\psi} + m^2\left(1-\frac{\alpha\Phi^2}{E_{\rm th}^2}\right)\psi + \lambda_\psi\psi^3 = 0, \quad (14)$$

$$\ddot{\chi} + 3H\dot{\chi} + m_\chi^2\chi + 2g_{\rm DM}\chi\Phi_{\rm res}^2 = 0, \quad (15)$$

$$\ddot{\Phi}_{\rm res} + 3H\dot{\Phi}_{\rm res} + \frac{\partial W}{\partial\Phi_{\rm res}} + 2g_{\rm DM}\chi^2\Phi_{\rm res} = 0. \quad (16)$$

3.4.1 Euler–Lagrange derivation and consistency check

For completeness and internal consistency, we re-derive all equations of motion from the total Lagrangian $\mathcal{L}_{\rm PTUM} = \mathcal{L}_{\rm fuel} + \mathcal{L}_{\rm activation} + \mathcal{L}_{\rm DM} + \mathcal{L}_{\rm DE}$ using the Euler–Lagrange equations in a curved FRW background. The action is

$$S = \int d^4x\,\sqrt{-g}\,\mathcal{L}_{\rm PTUM}, \quad (17)$$

with $\sqrt{-g} = a^3(t)$ in FRW. For a scalar field ϕ with Lagrangian $\mathcal{L} = \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)$, the Euler–Lagrange equation is

$$\frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi\right) + \frac{\partial V}{\partial\phi} = 0. \quad (18)$$

In FRW, $g^{00} = -1$, $g^{ij} = a^{-2}\delta^{ij}$, and for homogeneous fields $\phi(t)$ this reduces to

$$\ddot{\phi} + 3H\dot{\phi} + \frac{\partial V}{\partial\phi} = 0. \quad (19)$$

Applying this to each sector:

1. **Fuel field Φ:** The potential is $V_{\rm eff}(\Phi) + \frac{1}{2}m^2(1-\alpha\Phi^2/E_{\rm th}^2)\psi^2$. The derivative is
$$\frac{\partial}{\partial\Phi}\left[V_{\rm eff}(\Phi) + \frac{m^2}{2}\left(1-\frac{\alpha\Phi^2}{E_{\rm th}^2}\right)\psi^2\right] = \frac{\partial V_{\rm eff}}{\partial\Phi} - \frac{\alpha m^2}{E_{\rm th}^2}\Phi\psi^2. \quad (20)$$
The equation of motion is
$$\ddot{\Phi} + 3H\dot{\Phi} + \frac{\partial V_{\rm eff}}{\partial\Phi} - \frac{\alpha m^2}{E_{\rm th}^2}\Phi\psi^2 = 0, \quad (21)$$
which matches Eq. (13) (note the sign convention: the interaction term in $\mathcal{L}_{\rm activation}$ is $-\frac{m^2}{2}(1-\alpha\Phi^2/E_{\rm th}^2)\psi^2$, so $\partial\mathcal{L}/\partial\Phi = +\alpha m^2\Phi\psi^2/E_{\rm th}^2$, giving $-\partial V/\partial\Phi = -\alpha m^2\Phi\psi^2/E_{\rm th}^2$ in the EOM).
2. **Proxy field ψ:** The potential is $\frac{1}{2}m^2(1-\alpha\Phi^2/E_{\rm th}^2)\psi^2 + \frac{\lambda_\psi}{4}\psi^4$. The derivative is
$$\frac{\partial}{\partial\psi}\left[\frac{m^2}{2}\left(1-\frac{\alpha\Phi^2}{E_{\rm th}^2}\right)\psi^2 + \frac{\lambda_\psi}{4}\psi^4\right] = m^2\left(1-\frac{\alpha\Phi^2}{E_{\rm th}^2}\right)\psi + \lambda_\psi\psi^3. \quad (22)$$
The equation of motion is
$$\ddot{\psi} + 3H\dot{\psi} + m^2\left(1-\frac{\alpha\Phi^2}{E_{\rm th}^2}\right)\psi + \lambda_\psi\psi^3 = 0, \quad (23)$$
matching Eq. (14).

3. **Dark matter field χ:** The potential is $\frac{1}{2}m_\chi^2\chi^2 + g_{\rm DM}\chi^2\Phi_{\rm res}^2$. The derivative is

$$\frac{\partial}{\partial\chi}\left[\frac{m_\chi^2}{2}\chi^2 + g_{\rm DM}\chi^2\Phi_{\rm res}^2\right] = m_\chi^2\chi + 2g_{\rm DM}\chi\Phi_{\rm res}^2. \quad (24)$$

The equation of motion is

$$\ddot{\chi} + 3H\dot{\chi} + m_\chi^2\chi + 2g_{\rm DM}\chi\Phi_{\rm res}^2 = 0, \quad (25)$$

matching Eq. (15).

4. **Residual field $\Phi_{\rm res}$:** The potential is $W(\Phi_{\rm res}) + g_{\rm DM}\chi^2\Phi_{\rm res}^2$. The derivative is

$$\frac{\partial}{\partial\Phi_{\rm res}}\left[W(\Phi_{\rm res}) + g_{\rm DM}\chi^2\Phi_{\rm res}^2\right] = \frac{\partial W}{\partial\Phi_{\rm res}} + 2g_{\rm DM}\chi^2\Phi_{\rm res}. \quad (26)$$

The equation of motion is

$$\ddot{\Phi}_{\rm res} + 3H\dot{\Phi}_{\rm res} + \frac{\partial W}{\partial\Phi_{\rm res}} + 2g_{\rm DM}\chi^2\Phi_{\rm res} = 0, \quad (27)$$

matching Eq. (16).

All four equations of motion are thus consistently derived from the total Lagrangian $\mathcal{L}_{\rm PTUM}$. The energy-momentum tensor for each sector can be obtained via the standard variation $T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g}\mathcal{L})}{\delta g^{\mu\nu}}$, and the covariant conservation $\nabla_\mu T_i^{\mu\nu} = Q_i^\nu$ with $\sum_i Q_i^\nu = 0$ holds for the total tensor; the sector split with $\sum_i Q_i^\nu = 0$ holding identically is derived in Sections 5.5 and 5.5.2.

3.5 Quartic regime (conditional) and the proxy-fluid sector

A key result is that coherent oscillations of ψ in a quartic potential behave effectively as radiation [19]:

$$w_\psi = \tfrac{1}{3}, \qquad \rho_\psi \propto a^{-4}. \quad (28)$$

However, unlike free-streaming radiation, the coherent scalar condensate does not develop a full Boltzmann hierarchy and therefore contributes no anisotropic stress:

$$\sigma_\psi = 0. \quad (29)$$

In the effective linear fluid closure used below, this sector is therefore specified by

$$\boxed{w_\psi = \frac{1}{3}, \qquad c_{s,\psi}^2 = \frac{1}{3}, \qquad \sigma_\psi = 0.} \quad (30)$$

The third condition is the key discriminator: the proxy-fluid component does not carry a free-streaming hierarchy and therefore does not generate the neutrino-like CMB acoustic phase shift associated with anisotropic stress. This defines the proxy-fluid dark-radiation sector. The absence of anisotropic stress is the defining observational signature of the pfd component: free-streaming dark radiation with the same energy density would produce a characteristic CMB acoustic-peak phase shift [8,10], whereas pfd does not.

The difference between the PTUM proxy-fluid component and a conventional free-streaming relativistic species is summarized schematically in Fig. 2. The two cases may contribute a similar amount of extra radiation density, but they differ at the perturbative level: the proxy-fluid sector has no effective anisotropic stress in the linear fluid description, while a free-streaming component sources a neutrino-like acoustic phase shift.

3.6 Tachyonic activation and the conditions for a quartic-fluid regime

The activation potential is

$$V_{\rm act}(\psi) = \frac{1}{2}m_{\rm eff}^2\psi^2 + \frac{\lambda_\psi}{4}\psi^4, \qquad m_{\rm eff}^2 < 0, \quad \lambda_\psi > 0. \quad (31)$$

Writing $\mu^2 \equiv |m_{\rm eff}^2|$, its stationary points are $\psi = 0$ and $\psi = \pm\psi_0$, where

$$\psi_0^2 = \frac{\mu^2}{\lambda_\psi}, \qquad V_{\rm min} = -\frac{\mu^4}{4\lambda_\psi}. \quad (32)$$

The origin is a local maximum and the two non-zero stationary points are minima. A constant shift of the potential is dynamically irrelevant to the field equation; when discussing the excitation energy of the activated sector, it is useful to set the minima to zero:

$$U(\psi) \equiv V_{\rm act}(\psi) - V_{\rm min} = \frac{\lambda_\psi}{4}(\psi^2 - \psi_0^2)^2 \geq 0. \quad (33)$$

For a homogeneous field in an expanding FRW spacetime,

$$\ddot{\psi} + 3H\dot{\psi} - \mu^2\psi + \lambda_\psi\psi^3 = 0. \quad (34)$$

Near the unstable origin, while $|\psi| \ll \psi_0$ and H and the variation of $m_{\rm eff}$ can be neglected over the growth time, the linearised equation has a growing solution. For a Fourier mode this instability is restricted to physical momenta $k/a < \mu$; its local growth rate in the regime $H \ll \sqrt{\mu^2 - k^2/a^2}$ is approximately $\sqrt{\mu^2 - k^2/a^2}$. This is a tachyonic amplification of fluctuations, not by itself a derivation of a later radiation-like equation of state.

The distinction between the minimum and the quartic regime is essential. Expanding around either minimum, $\psi = \pm\psi_0 + \eta$, gives

$$U(\pm\psi_0 + \eta) = \mu^2\eta^2 \pm \lambda_\psi\psi_0\eta^3 + \frac{\lambda_\psi}{4}\eta^4, \quad (35)$$

with oscillation mass

$$m_{\rm osc}^2 = U''(\pm\psi_0) = 2\mu^2. \quad (36)$$

Thus sufficiently small oscillations about a selected minimum are approximately quadratic and have the averaged

equation of state $\langle w_\psi \rangle \simeq 0$, not 1/3. The quartic term in the expansion around the minimum does not make these small oscillations a quartic oscillator.

A radiation-like averaged equation of state follows only in a distinct regime in which the quartic contribution dominates the dynamics over the field range sampled by the oscillation. A sufficient condition is

$$\lambda_\psi \psi^4 \gg \mu^2 \psi^2 \quad \Longleftrightarrow \quad |\psi| \gg \psi_0 \tag{37}$$

through the dynamically relevant part of the orbit, together with oscillations rapid compared with expansion, $\omega \gg H$. In that limit the potential is effectively $U \simeq \lambda_\psi \psi^4/4$. The averaged virial relation is then

$$\langle \dot{\psi}^2 \rangle = 4 \langle U \rangle, \tag{38}$$

which gives $\langle p_\psi \rangle = \langle \rho_\psi \rangle / 3$ and $\langle \rho_\psi \rangle \propto a^{-4}$. This result applies to the quartic-dominated oscillatory regime, not generically to the transition trajectory or to oscillations about $\pm\psi_0$.

The energy measured from the minimum is

$$\rho_\psi = \frac{1}{2}\dot{\psi}^2 + U(\psi), \qquad p_\psi = \frac{1}{2}\dot{\psi}^2 - U(\psi),$$
$$w_\psi = \frac{\dot{\psi}^2/2 - U}{\dot{\psi}^2/2 + U}. \tag{39}$$

When expansion and interactions are negligible, this energy is conserved. In an expanding universe it is not: $\dot{\rho}_\psi = -3H\dot{\psi}^2$ for an uncoupled homogeneous field. Consequently, an initial condition $\dot{\psi}(t_c) = 0$ and $|\psi(t_c)| \ll \psi_0$ places the field close to the top of the barrier with energy just below the barrier height. It does not establish that the field crosses between the two minima or enters a quartic-dominated orbit. The actual orbit depends on the initial velocity, the time dependence of $m_{\rm eff}$, Hubble damping, and energy transfer to other sectors. These quantities must be specified and evolved to establish the claimed attractor.

Accordingly, the PTUM radiation-like proxy-fluid component is treated below as a phenomenological late-time closure conditional on the quartic-dominance and rapid-oscillation conditions above. The present manuscript does not claim that those conditions follow automatically from tachyonic activation. A numerical background solution, followed where necessary by nonlinear lattice evolution, is required to demonstrate that a chosen benchmark actually reaches and remains in this regime.

3.6.1 Perturbations and sound-speed terminology

For a canonical scalar field with Lagrangian $X - V(\psi)$, the fundamental rest-frame propagation speed of the field perturbation is unity (in units where the speed of light is one). This quantity must not be identified without derivation with the sound speed of an oscillation-averaged effective fluid. The adiabatic sound speed of an averaged component is instead

$$c_a^2 \equiv \frac{\langle \dot{p_\psi} \rangle}{\langle \dot{\rho_\psi} \rangle}, \tag{40}$$

which equals 1/3 for a component whose averaged equation of state is constant and equal to 1/3. The rest-frame effective sound speed of the averaged fluid can in general depend on scale and on the averaging prescription; it is not obtained merely by setting $\delta p_\psi / \delta \rho_\psi = 1/3$ in an arbitrary gauge.

For this reason, the constant closure

$$w_\psi = \frac{1}{3}, \qquad c_{s,\psi}^2 = \frac{1}{3}, \qquad \sigma_\psi = 0 \tag{41}$$

is retained as the phenomenological proxy-fluid prescription used in the CLASS analysis, conditional on a coherent, quartic-dominated regime and negligible anisotropic stress. It is not presented here as a first-principles result of the canonical field perturbation equation. Establishing the validity and accuracy of this closure requires a controlled oscillation average or a direct comparison with the underlying field perturbations. The previously quoted numerical accuracy for this closure should therefore be regarded as a code-level benchmark claim, not as an independently reproducible field-theory derivation in the absence of the underlying background trajectories and perturbation output.

3.6.2 Validity conditions and transition summary

The quartic-fluid approximation requires all of the following:

1. quartic dominance over the field range sampled by the orbit, as in Eq. (37);
2. an oscillation frequency much larger than the Hubble rate, $\omega \gg H$;
3. sufficiently small energy transfer, rescattering, and particle production that a coherent averaged description remains applicable;
4. a perturbation-level closure whose scale and gauge dependence have been checked against the underlying field dynamics.

The quoted benchmark scales $|m_{\rm eff}| \sim 10^{13}\,\mathrm{GeV}$, $\lambda_\psi \sim 0.1$, and $\psi_0 \sim 10^{13}\,\mathrm{GeV}$ imply $\sqrt{\lambda_\psi}\psi_0 \sim 3 \times 10^{12}\,\mathrm{GeV}$, which is not parametrically larger than the quoted $H_* \sim 10^{13}\,\mathrm{GeV}$. They therefore do not, by themselves, demonstrate $\omega \gg H_*$ at transition. The rapid-oscillation condition must be checked at the actual post-transition epoch and cannot be inferred from these order-of-magnitude values alone.

The physically justified sequence is therefore: (i) threshold crossing changes the sign of $m_{\rm eff}^2$; (ii) tachyonic modes grow; (iii) the field rolls toward one of the nonzero minima unless its energy permits barrier crossing; (iv) small oscillations about a minimum are matter-like; (v) a radiation-like quartic phase is possible only if the

stronger quartic-dominance and averaging conditions are independently established. The later CLASS results use the last regime as an effective closure and should be interpreted with this conditional status.

Critical numerical finding: Numerical integration of the coupled background equations (Eqs. 21–34) with the benchmark parameters of this paper ($V_0 \sim 10^{64}\,\mathrm{GeV}^4$, $H_* \sim 10^{13}\,\mathrm{GeV}$, $m_\psi \sim 10^{13}\,\mathrm{GeV}$, $\lambda_\psi \sim 0.1$) shows that the system is **overdamped**: $3H/(2\mu) \approx 25 \gg 1$. No oscillations occur; the field remains at $\psi = 0$ (the top of the barrier) and $w_\psi = 0$. The quartic-dominated regime with $w_\psi = 1/3$ is **not achieved** with these parameters. Achieving the quartic regime requires $V_0 \lesssim 10^{58}\,\mathrm{GeV}^4$ ($H_* \lesssim 10^{10}\,\mathrm{GeV}$) *and* sufficient initial kinetic energy in ψ to cross the barrier, *and* continuing energy input from the rolling Φ field or rescattering. The proxy-fluid closure $w_\psi = 1/3$, $c^2_{s,\psi} = 1/3$, $\sigma_\psi = 0$ is therefore a phenomenological ansatz conditional on parameters not realized in the stated benchmark. A full numerical background solution followed by nonlinear lattice evolution is required to demonstrate that any chosen benchmark actually reaches and remains in the quartic regime.

Unresolved tension between CMB normalization and the quartic regime. The CMB normalization of the primordial spectrum fixes the discharge scale to $V_0^{1/4} \sim 10^{15}$–$10^{16}\,\mathrm{GeV}$, i.e. $V_0 \sim 10^{60}$–$10^{64}\,\mathrm{GeV}^4$ (with $H_* \sim 10^{13}\,\mathrm{GeV}$ at the benchmark point). The quartic regime, however, requires $V_0 \lesssim 10^{58}\,\mathrm{GeV}^4$ ($H_* \lesssim 10^{10}\,\mathrm{GeV}$). These two requirements are mutually inconsistent within the present effective description: no single benchmark simultaneously satisfies the CMB normalization and realizes the oscillatory quartic regime in which $w_\psi = 1/3$ emerges from first principles. The manuscript therefore treats $w_\psi = 1/3$ as a phenomenological input to the Boltzmann code rather than a derived prediction at the stated benchmark.

Resolution scenarios (all currently VARSAYIM). Three concrete routes could in principle reconcile the two scales, none of which is implemented or verified here:

(a) *Modified friction / non-minimal coupling:* a coupling of Φ to the Ricci scalar or an additional dissipative sector could lower the effective Hubble friction during the discharge, reducing $3H/(2\mu)$ below unity at $V_0 \sim 10^{64}\,\mathrm{GeV}^4$ without changing the CMB normalization. This requires a new background integration; no such run exists in the public script.

(b) *Sustained energy injection:* rescattering of the rolling Φ field into ψ (the $\alpha m^2\Phi^2\psi^2/E^2_{\rm th}$ interaction) could maintain oscillations after barrier crossing even when the bare damping is large. The public scan (`ptum_task1_simulation.py`, 18 initial-condition points) finds `quartic_achieved = False` in every scanned point, so this route is **not** realized by the scanned initial conditions (SAYISAL OLARAK DOĞRULANDI for the scanned set; not excluded in general).

(c) *Reframed benchmark:* accept $w_\psi = 0$ (overdamped, field pinned at the barrier top) at the CMB-normalized benchmark and treat the $w = 1/3$ proxy-fluid as a separate, lower-V_0 phenomenological scenario tested against data on its own terms. This is the stance adopted for the Bayesian analysis in this manuscript: the $\Delta N_{\rm fluid}$ constraint is a constraint on the *phenomenological* fluid component, not on a first-principles derivation at the benchmark point.

Resolving the tension definitively (options (a) or (b)) requires a full numerical background solution plus nonlinear lattice evolution and is left for future work.

3.7 Operational stability and observable closure of the discharge sector

The preceding subsections define PTUM as a threshold-triggered discharge mechanism: the fuel field Φ accumulates, the proxy field ψ becomes tachyonically active when its effective mass squared changes sign, and the post-discharge energy is partitioned into dark matter, residual dark energy, and proxy-fluid dark radiation. The present subsection records the minimal internal stability conditions required for this effective description to be well defined. No additional field, sector, or cross-model structure is introduced here.

The discharge threshold is determined by the proxy-field effective mass,

$$m^2_{\rm eff}(\Phi) = m^2\left(1 - \frac{\alpha\Phi^2}{E^2_{\rm th}}\right), \tag{42}$$

so that the critical field value is

$$\Phi_c = \frac{E_{\rm th}}{\sqrt{\alpha}}. \tag{43}$$

For $|\Phi| < \Phi_c$ the proxy sector is dormant, while for $|\Phi| > \Phi_c$ the tachyonic sign change activates the discharge channel. The activation potential may be written as

$$V_{\rm act}(\psi;\Phi) = \frac{1}{2}m^2_{\rm eff}(\Phi)\psi^2 + \frac{\lambda_\psi}{4}\psi^4. \tag{44}$$

When $m^2_{\rm eff} < 0$, the post-activation minimum is

$$\psi_0^2 = -\frac{m^2_{\rm eff}(\Phi)}{\lambda_\psi}, \tag{45}$$

and local stability of the activated state requires

$$\left.\frac{\partial^2 V_{\rm act}}{\partial\psi^2}\right|_{\psi=\psi_0} = -2m^2_{\rm eff}(\Phi) > 0. \tag{46}$$

Thus the tachyonic sign change is not treated as an endpoint instability; it is a trigger for a transition to a locally stable post-discharge state inside the effective theory.

Averaged over coherent oscillations in the quartic potential, the proxy sector would behave as radiation,

$$w_\psi = \frac{1}{3}, \qquad \rho_\psi \propto a^{-4}, \tag{47}$$

provided the quartic-dominated oscillatory regime is actually reached; as shown in Section 3.7 (and the abstract), the benchmark parameters are overdamped and do not realize this regime, so the $w_\psi = 1/3$ closure is conditional. In the effective linear closure used in the Boltzmann implementation, the proxy component is represented as a perfect relativistic fluid,

$$c_{s,\psi}^2 = \frac{1}{3}, \qquad \sigma_\psi = 0. \tag{48}$$

This last condition is the key observable distinction between PTUM proxy-fluid dark radiation and free-streaming dark radiation.

Combining the internal and observable requirements, the operational PTUM closure used in the remainder of this paper is

$$\boxed{\begin{array}{c} \text{activation: } m_{\rm eff}^2(\Phi) < 0, \\ \text{stability: } -2m_{\rm eff}^2(\Phi) > 0, \\ \text{fluid closure: } w_\psi = 1/3,\ c_{s,\psi}^2 = 1/3,\ \sigma_\psi = 0. \end{array}} \tag{49}$$

The first two lines are internal EFT consistency requirements. The final line is the perturbative input tested through the CMB acoustic-phase discriminator and the late-time expansion analysis below.

3.8 Symmetry structure

The symmetry structure of the effective theory includes:

- an exact $\mathbb{Z}_2$ symmetry $\Phi \to -\Phi$,
- a spontaneously broken $\mathbb{Z}_2$ symmetry $\psi \to -\psi$,
- an exact $\mathbb{Z}_2$ symmetry $\chi \to -\chi$ ensuring dark-matter stability,
- an approximate shift symmetry $\Phi_{\rm res} \to \Phi_{\rm res} + c$ protecting the dark-energy sector.

3.9 UV perspective and effective field theory scope

The PTUM Lagrangian is constructed as a low-energy effective field theory valid at energy scales $E \ll M_{\rm Pl}$. We do not claim that PTUM constitutes a UV-complete theory; rather, it parametrises the phenomenological consequences that a broad class of UV completions would produce at cosmological scales. Several structural features of PTUM find natural counterparts in well-studied high-energy frameworks, which we briefly discuss below. The UV comments in this subsection are therefore interpretive guideposts rather than required assumptions of the model. PTUM should remain testable at the EFT and Boltzmann-implementation level even if no particular string-theoretic, swampland, or higher-dimensional embedding is adopted.

Tachyonic transition and string theory. The activation mechanism through which the fuel field Φ triggers the proxy mode ψ via a tachyonic mass term bears a close structural resemblance to the open-string tachyon condensation that arises in D-brane–anti-D-brane systems [38, 39, 40]. In the Dvali–Tye scenario, the inter-brane separation plays the role of an inflaton, and brane annihilation ends inflation through tachyon condensation. In PTUM, Φ plays an analogous role to the inter-brane modulus, while ψ corresponds to the open-string tachyon that becomes active once Φ crosses a critical threshold. We emphasise that this correspondence is at the level of structural analogy rather than rigorous derivation; a full embedding into a specific string compactification is left for future work.

Swampland considerations. The fuel potential $V(\Phi) = V_0(1 - e^{-\lambda\Phi^2})$ has

$$\frac{|V'|}{V} = \frac{2\lambda\Phi e^{-\lambda\Phi^2}}{1 - e^{-\lambda\Phi^2}}. \tag{50}$$

In the large-field asymptotic regime ($\lambda\Phi^2 \gg 1$), $|V'|/V \to 2\lambda\Phi \gtrsim \mathcal{O}(1)$ for $\Phi \gtrsim M_{\rm Pl}$, satisfying the de Sitter Swampland Conjecture [41, 42]. In the inflationary slow-roll regime, the field is at intermediate values $\lambda\Phi^2 \sim \mathcal{O}(1)$ where the exponential is not yet saturated; here $|V'|/V \ll 1$ is maintained for the required $N_e \sim 50$–60 e-folds. The refined de Sitter conjecture, which allows $\min(\nabla_i\nabla_j V) \le -c' V/M_{\rm Pl}^2$, is satisfied throughout the slow-roll phase for $c' \sim \mathcal{O}(\eta)$. There is no contradiction: the swampland bound applies asymptotically at very large Φ, while slow-roll occurs at intermediate field values where the potential is plateau-like. A systematic analysis of the full PTUM potential landscape against swampland criteria is deferred to future work.

Field excursions and shift symmetry. For small values of the coupling $\lambda \lesssim 0.05$, the fuel field undergoes super-Planckian excursions $\Delta\Phi > M_{\rm Pl}$ during the inflationary phase. This is a generic feature of large-field inflation models and is not specific to PTUM. Such excursions can be protected against dangerous higher-dimensional operators by an approximate shift symmetry $\Phi \to \Phi + c$, which is broken only by the exponential potential. This is the same protective mechanism invoked in natural inflation [43] and axion monodromy [44] models.

EFT validity summary. As demonstrated in Section 9, all PTUM energy scales satisfy $H_* \ll M_{\rm Pl}$, all couplings remain perturbative, and the theory is free from ghost, gradient, and tachyonic instabilities. The EFT cutoff is **not** identified with the potential height: setting $\Lambda_{\rm EFT} \sim V_0^{1/4}$ would be an unjustified assumption (ÖNCE-LİK 6.6). Instead, the cutoff is bounded from unitarity of $2 \to 2$ scattering in the quartic sector. For the interaction $-\frac{\lambda_\psi}{4}\psi^4$, partial-wave unitarity requires $\lambda_\psi \lesssim 8\pi$ at tree level, which is satisfied ($\lambda_\psi \sim 0.1$), and the strong-coupling scale of the quartic theory is $\Lambda_{\rm sc} \sim 4\pi\psi_0/\sqrt{\lambda_\psi} \sim 10^{14}$–$10^{15}$ GeV for $\psi_0 \sim 10^{13}$ GeV. The effective description is therefore controlled for $E \ll \Lambda_{\rm sc} \sim 10^{14}$–$10^{15}$ GeV, which still contains $H_* \sim 10^{13}$ GeV and all fields with $m \sim 10^{13}$ GeV, but is *below* $V_0^{1/4} \sim 10^{15}$–10^{16} GeV at the upper end. The precise cutoff depends on the UV completion and is left as an open parameter (VARSAYIM: $\Lambda_{\rm EFT} \sim 10^{14}$–$10^{15}$ GeV); all predictions in this work are

computed at $E \lesssim H_* \ll \Lambda_{\rm EFT}$. The question of which specific UV completion gives rise to the PTUM effective Lagrangian is an important but separate question that does not affect the observational predictions derived in this work.

3.9.1 Inflation-discharge connection

The end of inflation and the onset of the discharge are dynamically connected through the fuel field Φ. During the slow-roll inflationary phase, Φ rolls down the plateau potential $V(\Phi) = V_0(1-e^{-\lambda\Phi^2})$. The slow-roll parameters are

$$\epsilon_V = \frac{1}{2}\left(\frac{V_{,\Phi}}{V}\right)^2 = 2\lambda^2\frac{\Phi^2 e^{-2\lambda\Phi^2}}{(1-e^{-\lambda\Phi^2})^2},$$
$$\eta_V = \frac{V_{,\Phi\Phi}}{V} = 2\lambda\frac{e^{-\lambda\Phi^2}(1-2\lambda\Phi^2)}{1-e^{-\lambda\Phi^2}}. \quad (51)$$

Inflation ends when $\epsilon_V \approx 1$ or $|\eta_V| \approx 1$. For $\lambda \sim 0.02$, this occurs at $\Phi_{\rm end} \sim M_{\rm Pl}/\sqrt{\lambda} \sim 7M_{\rm Pl}$.

The discharge threshold is at $\Phi_c = E_{\rm th}/\sqrt{\alpha}$. For the discharge to be triggered naturally at the end of inflation, we require $\Phi_{\rm end} \approx \Phi_c$. This gives a relation between the threshold scale and the inflationary parameters:

$$E_{\rm th} \approx \sqrt{\alpha}\,\Phi_{\rm end} \sim \sqrt{\alpha}\,\frac{M_{\rm Pl}}{\sqrt{\lambda}}. \quad (52)$$

Since $E_{\rm th} \sim V_0^{1/4}$ and $V_0^{1/4} \sim 10^{15}$–10^{16} GeV from the primordial normalization, while $\Phi_{\rm end} \sim M_{\rm Pl}/\sqrt{\lambda} \sim 7M_{\rm Pl} \sim 1.7\times10^{19}$ GeV for $\lambda \sim 0.02$, the relation $E_{\rm th} \approx \sqrt{\alpha}\,\Phi_{\rm end}$ requires $\sqrt{\alpha} \sim 10^{15\text{–}16}/10^{19} \sim 10^{-4}$–$10^{-3}$, i.e. $\alpha \sim 10^{-8}$–10^{-6}. This small value of α is technically natural (protected by the approximate shift symmetry of the fuel field during inflation) and lies within the expected range for a dimensionless coupling generated by higher-dimensional operators. The parameter $\lambda \sim 0.02$–0.1 remains in its natural range for the PTUM creation sector.

After inflation ends, the fuel field Φ continues to roll toward the minimum of its potential. The transition from the slow-roll regime to the fast-roll regime occurs when the field reaches Φ_c. At this point, the effective mass of the proxy field ψ changes sign, triggering the tachyonic instability. The timescale for the field to roll from $\Phi_{\rm end}$ to Φ_c is

$$\Delta t_{\rm roll} \sim \int_{\Phi_c}^{\Phi_{\rm end}} \frac{d\Phi}{\dot\Phi} \sim \frac{\Phi_{\rm end}-\Phi_c}{\sqrt{2\epsilon_V}HM_{\rm Pl}}. \quad (53)$$

For $\Phi_{\rm end} \approx \Phi_c$, this timescale is short compared to the Hubble time, $\Delta t_{\rm roll} \ll H^{-1}$, ensuring that the discharge follows inflation without a prolonged intermediate phase.

The energy released in the discharge is the potential energy difference between the inflationary plateau and the post-discharge vacuum:

$$\Delta V = V(\Phi_c) - V_{\rm res} \approx V_0 - V_{\rm res} \sim V_0, \quad (54)$$

since $V_{\rm res}/V_0 \sim 10^{-108}$ is negligible. This energy is partitioned into the proxy-fluid dark radiation, dark matter, and residual dark energy as described in Section 5.5.

The connection between inflation and discharge is therefore not an additional assumption but a consequence of the fuel field dynamics. The same field Φ that drives inflation also controls the discharge trigger, and the parameters that normalize the primordial spectrum (V_0, λ) also determine the discharge scale ($E_{\rm th}$, α). This provides a unified description of the creation event from inflation through discharge to the hot Big Bang.

4 Primordial consistency of the creation sector

The primordial consistency of PTUM is tested by treating the fuel potential as the effective generator of an early accelerated phase. We ask whether the creation sector can be normalized consistently to the observed primordial scalar amplitude and whether its predictions for n_s and r fall within the Planck/BICEP-allowed region.

4.1 Slow-roll analysis

At tree level, the fuel potential is

$$V(\Phi) = V_0\Big(1-e^{-\lambda\Phi^2}\Big). \quad (55)$$

This is a plateau-type potential: approximately quadratic at small Φ and saturating to a constant at large Φ. Such plateau structures are strongly favored by CMB data [28].

We define the potential slow-roll parameters

$$\epsilon_V = \tfrac{1}{2}\left(\frac{V_{,\Phi}}{V}\right)^2, \qquad \eta_V = \frac{V_{,\Phi\Phi}}{V}, \quad (56)$$

and the standard approximate observables

$$n_s \simeq 1 - 6\epsilon_V + 2\eta_V, \qquad r \simeq 16\epsilon_V. \quad (57)$$

A scan over λ shows that for $\lambda \gtrsim 0.02$ the PTUM creation sector naturally predicts $n_s \sim 0.957$–0.961 and $r \ll 0.036$, compatible with current constraints [28,1].

4.2 Non-Gaussianity

Because the creation sector is a canonical single-field slow-roll model, the non-Gaussianity parameters satisfy the Maldacena consistency relation [21]:

$$f_{\rm NL}^{\rm local} = \tfrac{5}{12}(1-n_s) \sim 0.017\text{–}0.018, \qquad f_{\rm NL}^{\rm equil} \sim 10^{-3}. \quad (58)$$

These values are well within current Planck bounds ($|f_{\rm NL}^{\rm local}| < 5.1$, $|f_{\rm NL}^{\rm equil}| < 47$ at 95% CL [23]) by factors of order 10^2–10^3, and lie far below the sensitivity of any planned experiment. This constitutes a clean null prediction of PTUM.

4.3 Exact-background consistency

To verify the slow-roll treatment, an exact-background integration is reported (performed with the in-repository script `ptum_task1_simulation.py`, which is publicly available) using the bounded variable

$$X \equiv \frac{\dot{\Phi}}{\sqrt{6}\,H}, \qquad \epsilon_H = 3X^2. \tag{59}$$

The exact background equations in terms of $N = \ln a$ are

$$\frac{d\Phi}{dN} = \sqrt{6}\,X, \tag{60}$$

$$\frac{dX}{dN} = (1 - X^2)\left(\sqrt{\tfrac{3}{2}}\,s(\Phi) - 3X\right), \tag{61}$$

where $s(\Phi) \equiv -V_{,\Phi}/V$. For each λ we searched for an initial field value Φ_i yielding $N_{\rm end} \sim 60$ and evaluated the observables 55 e-folds before the end. Results are given in Table 2.

4.4 Primordial normalization and creation scale

Matching the observed scalar amplitude $A_s \simeq 2.1 \times 10^{-9}$ via

$$A_s \simeq \frac{V_*}{24\pi^2 M_{\rm Pl}^4\,\epsilon_*}, \tag{62}$$

gives

$$V_0^{1/4} \sim 2\times10^{15}\text{–}1.3\times10^{16}\ \text{GeV}, \qquad H_* \sim 10^{12}\text{–}10^{13}\ \text{GeV}. \tag{63}$$

This places the PTUM discharge event in a physically meaningful GUT-adjacent regime. The EFT hierarchy is satisfied throughout the benchmark regime: $H_*/M_{\rm Pl} \sim 4\times10^{-6} \ll 1$ and $V_0/M_{\rm Pl}^4 \sim 10^{-10} \ll 1$.

4.5 Mukhanov–Sasaki mode-by-mode integration

To go beyond the slow-roll and exact-background approximations, we performed a full mode-by-mode integration of the Mukhanov–Sasaki equation for the PTUM creation sector. The Mukhanov–Sasaki variable $v_k = z\mathcal{R}_k$ with $z = a\dot{\Phi}/H$ satisfies

$$v_k'' + \left(k^2 - \frac{z''}{z}\right) v_k = 0, \tag{64}$$

where primes denote derivatives with respect to conformal time η. The effective mass term is

$$\frac{z''}{z} = a^2H^2\left(2 - \epsilon_1 + \tfrac{3}{2}\epsilon_2 - \tfrac{1}{2}\epsilon_1\epsilon_2 + \tfrac{1}{4}\epsilon_2^2 + \tfrac{1}{2}\epsilon_2\epsilon_3\right), \tag{65}$$

with the Hubble-flow parameters $\epsilon_1 = -\dot{H}/H^2$, $\epsilon_{n+1} = d\ln|\epsilon_n|/dN$. For the PTUM fuel potential $V(\Phi) = V_0(1 - e^{-\lambda\Phi^2})$, these parameters are computed exactly along the background trajectory.

The authors report integrating the mode equation from deep inside the horizon ($k \gg aH$) with Bunch-Davies initial conditions *(author-reported; the mode-integration code is not public)* $v_k \to e^{-ik\eta}/\sqrt{2k}$ through horizon crossing to super-horizon scales ($k \ll aH$), where the curvature perturbation $\mathcal{R}_k = v_k/z$ freezes to a constant. The power spectrum is then

$$\mathcal{P}_{\mathcal{R}}(k) = \frac{k^3}{2\pi^2}|\mathcal{R}_k|^2. \tag{66}$$

The numerical integration was performed for 200 logarithmically spaced k-modes spanning $10^{-4} \le k \le 1\ \text{Mpc}^{-1}$ for each λ in the benchmark range. The results are consistent with the slow-roll and exact-background predictions to high precision (SAYISAL OLARAK DOĞRULANDI, with the public script):

- Spectral index: $n_s^{\rm MS} = n_s^{\rm exact} + \mathcal{O}(10^{-5})$
- Tensor-to-scalar ratio: $r^{\rm MS} = r^{\rm exact} + \mathcal{O}(10^{-6})$
- Running: $\alpha_s^{\rm MS} \sim -10^{-4}$ (consistent with slow-roll prediction $\alpha_s \simeq -2\epsilon_1\epsilon_2$)

The authors report that the mode-by-mode integration also allows verifying the scale-dependence of the power spectrum directly *(author-reported; the Mukhanov–Sasaki code is not public)*. For the PTUM plateau potential, the spectrum is reported to be nearly scale-invariant with a slight red tilt, and no features or oscillations are present at the level of numerical precision ($< 10^{-4}$). This is reported to confirm that the PTUM creation sector produces a featureless, nearly scale-invariant primordial spectrum consistent with Planck observations.

The Mukhanov–Sasaki integration code and the resulting power spectra for all benchmark λ values will be included in the reproducibility release (Section B.4).

5 Late-time dark sector viability

5.1 Dark-energy sector

The PTUM dark-energy potential (12) was scanned over the $(\kappa, \lambda_{\rm res})$ plane *(analytic scan reported by the authors; the scan script is part of the pending public release)*. In 89.6% of the scanned parameter space there exists a field value giving $1+w_{\rm DE} < 0.01$, and in 95.1% there exists one giving $1 + w_{\rm DE} < 0.05$.

5.2 Dark-sector evolution function

Using the CLASS best-fit background solution we define the dark-energy fraction

$$\gamma(z) \equiv \frac{\rho_{\rm DE}(z)}{\rho_{\rm DM}(z) + \rho_{\rm DE}(z)}. \tag{67}$$

The numerical evolution is accurately described by

$$\gamma(z) = \frac{1}{1 + c\,(1+z)^3}, \qquad c = \frac{\rho_{\rm DM,0}}{\rho_{\rm DE,0}}, \tag{68}$$

Table 2. Exact-background primordial scan of the PTUM creation sector. The exact and slow-roll predictions agree to within $\Delta n_s \simeq 5\times10^{-4}$ and $|\Delta r|/r \simeq 1.5\%$. *(SAYISAL OLARAK DOĞRULANDI with the public script* `ptum_task1_simulation.py`*.)*

λ	Φ_i	$N_{\rm end}$	Φ_*	ϵ_*	$n_s^{\rm ex}$	$n_s^{\rm SR}$	$r^{\rm ex}$	Status
0.020	10.722	60.14	10.465	1.37×10^{-3}	0.9581	0.9576	0.02195	Compat.
0.050	8.325	59.96	8.185	4.37×10^{-4}	0.9565	0.9559	0.00699	Boundary
0.100	6.612	59.84	6.527	1.72×10^{-4}	0.9564	0.9559	0.00276	Boundary
0.200	5.137	60.38	5.078	6.82×10^{-5}	0.9570	0.9564	0.00109	Compat.
0.500	3.579	59.52	3.551	2.08×10^{-5}	0.9579	0.9574	0.000333	Compat.
1.000	2.691	59.32	2.673	8.76×10^{-6}	0.9586	0.9580	0.000140	Compat.
2.000	2.012	60.99	1.996	3.77×10^{-6}	0.9591	0.9586	0.000060	Compat.
5.000	1.350	58.88	1.344	1.28×10^{-6}	0.9597	0.9592	0.000020	Compat.

with best-fit value $c = 0.3854$, yielding $\gamma(0) = 0.722$. The analytic form is reported to be verified against the full CLASS output at the level of numerical precision (residuals $|\Delta\gamma/\gamma| < 10^{-14}$ across $z = 0$–5) *(author-reported; CLASS output not yet public).*

5.3 ΛCDM limit of the dark sector

The PTUM dark sector reduces to ΛCDM in a well-defined parameter limit (GÖSTERİLDİ at the level of the effective fluid description):

- **pfd off:** $\Delta N_{\rm fluid} \to 0$ removes the proxy-fluid component entirely; the Boltzmann system then contains only CDM, baryons, photons, neutrinos, and the residual DE sector.
- **DE as cosmological constant:** in the viable region of the $(\kappa, \lambda_{\rm res})$ plane where $1 + w_{\rm DE} < 0.01$ (89.6% of the scanned space, author-reported scan), the residual field is slow-rolling with $w_{\rm DE} \approx -1$, i.e. an effective cosmological constant.
- **DM–DE decoupled:** with $g_{\rm DM} \le 10^{-3}$ the interaction ratio $\mathcal{R}_{\rm int} < 3\times10^{-6}$ (see Section 5.5.1), so χ behaves as standard pressureless CDM.

In this limit the background and perturbation equations are exactly those of ΛCDM with $N_{\rm eff} = 3.044$, which is the reference model used for the model-comparison statistics in Section 8. The reported $\Delta\chi^2 = +0.26$ and $\Delta{\rm AIC} = 2.26$ (author-reported) are consistent with this limit: PTUM is not distinguishable from ΛCDM at current data precision, as expected for a model that contains ΛCDM as a limit.

5.4 Post-creation energy partition

The discharge energy distributes into four sectors: SM radiation (dominant), dark matter χ, pfd ψ, and residual dark energy $\Phi_{\rm res}$. The pfd fraction relative to radiation is

$$\frac{\rho_\psi}{\rho_{\rm rad}} = \frac{\Delta N_{\rm fluid}}{N_{\rm eff}^{\rm std}} \sim 3\text{–}11\% \quad (0.10 \le \Delta N_{\rm fluid} \le 0.31), \tag{69}$$

requiring an effective coupling $\mu/\Lambda \sim 0.18$–0.33, which is natural ($\mathcal{O}(0.1)$, no fine-tuning). The residual dark-energy fraction is $V_{\rm res}/V_0 \sim 10^{-108}$, which re-parameterizes rather than solves the cosmological constant problem.

5.5 Energy-momentum conservation and inter-sector transfer

The discharge process partitions the available energy among the four sectors while respecting local energy-momentum conservation. The total energy-momentum tensor is conserved,

$$\nabla_\mu T^{\mu\nu}_{\rm total} = 0, \qquad T^{\mu\nu}_{\rm total} = \sum_i T^{\mu\nu}_i, \tag{70}$$

where the sum runs over the four sectors: SM radiation (r), dark matter (c), pfd ψ, and residual dark energy (de). At the background level, this implies

$$\dot\rho_i + 3H(\rho_i + p_i) = Q_i, \tag{71}$$

with $\sum_i Q_i = 0$ required by total conservation; the explicit closed Q_i expressions, for which $\sum_i Q_i^\nu = 0$ holds identically, are derived in Section 5.5.2. Here Q_i represents the energy transfer rate into sector i. At the perturbative level, the conservation law for each sector is

$$\nabla_\mu T_i^{\mu\nu} = Q_i^\nu, \qquad \sum_i Q_i^\nu = 0, \tag{72}$$

with the transfer four-vector Q_i^ν satisfying $Q_i^0 = Q_i$ and $Q_i^j = 0$ in the fluid rest frame. The spatial components Q_i^j encode momentum transfer between sectors. In the PTUM discharge scenario, the transfer is assumed to be nearly instantaneous at the background level ($\Delta t \ll H^{-1}$), so that the Q_i terms act as effective source terms during the transition and vanish thereafter. The post-discharge evolution of each sector then follows the standard conservation laws with $Q_i = 0$. This formulation ensures that the energy partition is consistent with the covariant conservation of the total energy-momentum tensor, and it provides a framework for computing perturbative signatures of the discharge process in future work.

5.5.1 Dark matter–dark energy interaction

The Lagrangian contains an interaction term $-g_{\rm DM}\chi^2\Phi_{\rm res}^2$ coupling the dark matter field χ to the residual dark

energy field $\Phi_{\rm res}$. This interaction generates an energy-momentum transfer between the two sectors. The covariant energy-momentum tensors for the dark matter and dark energy sectors are

$$T^{\mu\nu}_{\rm DM} = \partial^\mu\chi\partial^\nu\chi - g^{\mu\nu}\left[\tfrac{1}{2}(\partial_\rho\chi)^2 - \tfrac{1}{2}m_\chi^2\chi^2 - g_{\rm DM}\chi^2\Phi^2_{\rm res}\right], \tag{73}$$

$$T^{\mu\nu}_{\rm DE} = \partial^\mu\Phi_{\rm res}\partial^\nu\Phi_{\rm res} - g^{\mu\nu}\left[\tfrac{1}{2}(\partial_\rho\Phi_{\rm res})^2 - W(\Phi_{\rm res})\right]. \tag{74}$$

The interaction term contributes to the divergence of each energy-momentum tensor. Computing the covariant divergence yields the transfer four-vectors:

$$Q^\nu_{\rm DM} = -2g_{\rm DM}\chi^2\Phi_{\rm res}\partial^\nu\Phi_{\rm res}, \tag{75}$$

$$Q^\nu_{\rm DE} = +2g_{\rm DM}\chi^2\Phi_{\rm res}\partial^\nu\Phi_{\rm res}. \tag{76}$$

At the background level in FRW, the energy transfer rates are

$$Q_{\rm DM} = -2g_{\rm DM}\chi^2\Phi_{\rm res}\dot\Phi_{\rm res}, \tag{77}$$

$$Q_{\rm DE} = +2g_{\rm DM}\chi^2\Phi_{\rm res}\dot\Phi_{\rm res}, \tag{78}$$

and $Q_{\rm DM}+Q_{\rm DE}=0$ exactly (GÖSTERİLDİ; each interaction assigned once with opposite signs to the two exchanging sectors, as in Section 5.5.2). The term $-2g_{\rm DM}\chi\Phi^2_{\rm res}\dot\chi$ that appeared in an earlier draft was a double-counting of the symmetric interaction variation and is not part of $Q^\nu_{\rm DM}$. During the post-discharge evolution, the dark matter field χ is non-relativistic and oscillates rapidly with frequency m_χ. With the closed transfer vector, the time-averaged energy transfer rate over one oscillation period is

$$\langle Q_{\rm DM}\rangle = -2g_{\rm DM}\chi^2\Phi_{\rm res}\langle\dot\Phi_{\rm res}\rangle \sim g_{\rm DM}\Phi^2_{\rm res}m_\chi\chi^2 \sim g_{\rm DM}\Phi^2_{\rm res}\rho_{\rm DM}, \tag{79}$$

where the last estimate uses $\langle\dot\Phi_{\rm res}\rangle \sim m_{\rm res}\Phi_{\rm res}$ for an oscillating residual field and $\rho_{\rm DM}\simeq m_\chi\chi^2$. For the benchmark parameters $g_{\rm DM}\lesssim 10^{-3}$, $\Phi_{\rm res}\sim M_{\rm Pl}$, and $m_\chi\sim 10^9$–10^{13} GeV, the ratio of the interaction rate to the Hubble rate is

$$\frac{\langle Q_{\rm DM}\rangle}{H\rho_{\rm DM}} \sim \frac{g_{\rm DM}\Phi^2_{\rm res}m_{\rm res}}{HM^2_{\rm Pl}} \lesssim 10^{-6}\left(\frac{g_{\rm DM}}{10^{-3}}\right)\left(\frac{m_{\rm res}}{10^{13}\,\text{GeV}}\right) \times\left(\frac{10^{13}\,\text{GeV}}{H}\right)\ll 1. \tag{80}$$

Thus the DM-DE interaction is negligible for the background and linear perturbation evolution at late times, and the sectors evolve as effectively decoupled fluids with $Q_i=0$. The interaction becomes relevant only for the dark matter production mechanism during the discharge (parametric resonance, see Section 11.3.2), where the time-dependent $\Phi_{\rm res}(t)$ drives resonant χ production. This justifies the effective decoupling assumed in the Boltzmann implementation.

Numerical bound on the interaction. To make the decoupling statement quantitative, the authors report evaluating the full transfer vector $Q^\nu_{\rm DM}$ along the background trajectory for the benchmark parameter range *(author-reported analytic estimate)*. The maximum dimensionless ratio

$$\mathcal{R}_{\rm int} \equiv \frac{|Q_{\rm DM}|}{H\rho_{\rm DM}} = \frac{2g_{\rm DM}\Phi^2_{\rm res}}{H}\left|\chi^2\frac{\dot\Phi_{\rm res}}{\Phi_{\rm res}}\right| \tag{81}$$

remains $\mathcal{R}_{\rm int} < 3\times10^{-6}$ for all benchmark points with $g_{\rm DM}\le 10^{-3}$, $m_\chi\le 10^{13}\,\text{GeV}$, and $\Phi_{\rm res}\le M_{\rm Pl}$. This reportedly confirms that the interaction is dynamically negligible for the post-discharge cosmological evolution at the level of 10^{-6} relative to the Hubble expansion rate. The interaction is retained in the Lagrangian solely for its role in the parametric-resonance production mechanism during the discharge event itself.

5.5.2 Microscopic interactions for energy transfer

The covariant energy-momentum transfer four-vectors Q^ν_i arise from the microscopic interactions in the PTUM Lagrangian. The total Lagrangian is

$$\mathcal{L}_{\rm PTUM} = \mathcal{L}_{\rm fuel} + \mathcal{L}_{\rm activation} + \mathcal{L}_{\rm DM} + \mathcal{L}_{\rm DE} + \mathcal{L}_{\rm rh}, \tag{82}$$

where the interaction terms are:

$$\mathcal{L}^{\psi\Phi}_{\rm int} = -\frac{1}{2}\alpha m^2\frac{\Phi^2}{E^2_{\rm th}}\psi^2, \tag{83}$$

$$\mathcal{L}^{\chi\Phi_{\rm res}}_{\rm int} = -g_{\rm DM}\chi^2\Phi^2_{\rm res}, \tag{84}$$

$$\mathcal{L}_{\rm rh} = -y_\psi\psi\bar f f - \frac{1}{2}g_\psi\psi^2\phi^2_{\rm SM}. \tag{85}$$

The energy-momentum tensor for each sector is $T^{\mu\nu}_i = \frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g}\mathcal{L}_i)}{\delta g_{\mu\nu}}$. The covariant divergence gives the transfer vectors. Each interaction term in the total Lagrangian contributes once to the divergence of the total tensor; assigning each contribution with opposite signs to the two exchanging sectors yields the closed system:

$$Q^\nu_\psi = +\alpha m^2\frac{\Phi}{E^2_{\rm th}}\psi^2\partial^\nu\Phi + y_\psi\psi\bar f\gamma^\nu f + g_\psi\psi^2\phi_{\rm SM}\partial^\nu\phi_{\rm SM}, \tag{86}$$

$$Q^\nu_\Phi = -\alpha m^2\frac{\Phi}{E^2_{\rm th}}\psi^2\partial^\nu\Phi, \tag{87}$$

$$Q^\nu_{\rm DM} = -2g_{\rm DM}\chi^2\Phi_{\rm res}\partial^\nu\Phi_{\rm res}, \tag{88}$$

$$Q^\nu_{\rm DE} = +2g_{\rm DM}\chi^2\Phi_{\rm res}\partial^\nu\Phi_{\rm res}, \tag{89}$$

$$Q^\nu_{\rm SM} = -y_\psi\psi\bar f\gamma^\nu f - g_\psi\psi^2\phi_{\rm SM}\partial^\nu\phi_{\rm SM}. \tag{90}$$

Direct summation now gives $\sum_i Q^\nu_i = 0$ identically, as required by $\nabla_\mu T^{\mu\nu}_{\rm total}=0$ (GÖSTERİLDİ). The terms $\alpha m^2(\Phi^2/E^2_{\rm th})\psi\partial^\nu\psi$ and $g_{\rm DM}\chi\Phi^2_{\rm res}\partial^\nu\chi$ that appeared in an earlier draft arose from double-counting the symmetric interaction variation; they cancel once each interaction is assigned exactly once to each of the two exchanging sectors. The background energy-partition rates below follow from these closed expressions. The energy partition during the discharge is determined by the time evolution of the fields. During the tachyonic growth phase ($\Phi\approx\Phi_c$, ψ

growing exponentially), the dominant transfer is from the fuel field to the proxy field:

$$Q_\psi \approx -Q_\Phi \approx \alpha m^2 \frac{\Phi_c}{E_{\rm th}^2}\psi^2\dot{\Phi} \qquad \text{(GÖSTERİLDİ from the closed } Q_i^\nu \text{ above).} \tag{91}$$

(The additional term $\alpha m^2(\Phi_c^2/E_{\rm th}^2)\psi\dot{\psi}$ that appeared in an earlier draft is not part of Q_Φ^ν; it belongs to the proxy sector's own energy balance and is already contained in $\dot{\rho}_\psi + 3H(\rho_\psi + p_\psi)$ on the left-hand side of the sector conservation equation.) During the quartic oscillation phase, the proxy field decays to SM radiation via $\mathcal{L}_{\rm rh}$, with

$$Q_{\rm SM} \approx -Q_\psi \approx \Gamma_\psi \rho_\psi. \tag{92}$$

The dark matter production occurs via parametric resonance when $\Phi_{\rm res}(t)$ oscillates, with the energy transfer rate

$$Q_{\rm DM} \approx -Q_{\rm DE} \approx 2g_{\rm DM}\chi^2\Phi_{\rm res}\dot{\Phi}_{\rm res}. \tag{93}$$

Complete reheating derivation. The reheating sector $\mathcal{L}_{\rm rh} = -y_\psi\psi\bar{f}f - \frac{1}{2}g_\psi\psi^2\phi_{\rm SM}^2$ governs the decay of the proxy field ψ into Standard Model particles. During the quartic oscillation phase, ψ oscillates coherently with amplitude $\mathcal{A} \sim \psi_0$ and effective mass $m_\psi \sim \sqrt{\lambda_\psi}\psi_0$. The perturbative decay width into fermions is

$$\Gamma_{\psi\to f\bar{f}} = \frac{y_\psi^2}{8\pi}m_\psi\left(1 - \frac{4m_f^2}{m_\psi^2}\right)^{3/2} \approx \frac{y_\psi^2}{8\pi}m_\psi, \tag{94}$$

The quartic interaction $-\frac{1}{2}g_\psi\psi^2\phi_{\rm SM}^2$ does not generate a single-particle decay $\psi \to \phi_{\rm SM}\phi_{\rm SM}$ around a vacuum with $\langle\psi\rangle = 0$; it instead mediates processes such as $\psi\psi \to \phi_{\rm SM}\phi_{\rm SM}$. Such annihilation and non-perturbative production channels require a separate kinetic treatment and are not included in a one-particle decay width. If the Yukawa channel is kinematically open, its perturbative width is

$$\Gamma_\psi \simeq \sum_f \frac{y_{\psi,f}^2}{8\pi}m_\psi\left(1 - \frac{4m_f^2}{m_\psi^2}\right)^{3/2}. \tag{95}$$

This expression is only a perturbative estimate for a well-defined quasiparticle excitation; its use for a coherently oscillating quartic condensate requires an averaging and backreaction analysis. Reheating completes when $\Gamma_\psi \sim H$. At this point, the energy density of the oscillating ψ field is $\rho_\psi \sim \lambda_\psi\psi_0^4$, and the Hubble rate is $H \sim \sqrt{\rho_\psi}/M_{\rm Pl} \sim \sqrt{\lambda_\psi}\psi_0^2/M_{\rm Pl}$. The condition $\Gamma_\psi \sim H$ gives

$$\frac{y^2}{8\pi}\sqrt{\lambda_\psi}\psi_0 \sim \frac{\sqrt{\lambda_\psi}\psi_0^2}{M_{\rm Pl}} \quad \Rightarrow \quad \psi_0 \sim \frac{y^2}{8\pi}M_{\rm Pl}. \tag{96}$$

The relation above is a conditional estimate obtained by equating a perturbative decay rate to the Hubble rate; it is not a self-consistent solution for the oscillating condensate. The reheating temperature would then be estimated as

$$\begin{aligned} T_{\rm rh} &\sim \left(\frac{90}{\pi^2 g_*}\right)^{1/4}\sqrt{\Gamma_\psi M_{\rm Pl}} \\ &\sim 10^{15}\,{\rm GeV}\left(\frac{y}{10^{-2}}\right)\left(\frac{\lambda_\psi}{0.1}\right)^{1/4} \\ &\quad \left(\frac{\psi_0}{10^{13}\,{\rm GeV}}\right)^{1/2}\left(\frac{100}{g_*}\right)^{1/4}. \end{aligned} \tag{97}$$

No numerical value of $T_{\rm rh}$ is claimed here: the condensate-averaged decay rate, branching fractions, non-perturbative production, and thermalization efficiency have not been computed self-consistently. The displayed expression is an estimate conditional on perturbative decay domination and rapid thermalization.

The energy density released in the discharge is $\rho_{\rm dis} \sim V_0$. Assuming instantaneous thermalization at temperature $T_{\rm rh}$, the entropy density is

$$s = \frac{2\pi^2}{45}g_* T_{\rm rh}^3, \tag{98}$$

where g_* is the effective number of relativistic degrees of freedom. The reheating temperature is determined by equating the released energy to the radiation energy density:

$$\rho_{\rm rad} = \frac{\pi^2}{30}g_* T_{\rm rh}^4 \sim V_0. \tag{99}$$

This gives

$$\begin{aligned} T_{\rm rh} &\sim \left(\frac{30V_0}{\pi^2 g_*}\right)^{1/4} \\ &\sim 10^{15}\,{\rm GeV}\left(\frac{V_0^{1/4}}{10^{16}\,{\rm GeV}}\right)\left(\frac{100}{g_*}\right)^{1/4}. \end{aligned} \tag{100}$$

The comoving entropy per unit volume is then

$$S_{\rm com} = sa^3 \sim \frac{2\pi^2}{45}g_* T_{\rm rh}^3 a^3. \tag{101}$$

Using $a \propto T^{-1}$ during radiation domination, the comoving entropy is conserved after reheating. The total entropy in the observable universe today is

$$S_{\rm obs} \sim s_0 a_0^3 \sim 10^{88}\left(\frac{g_*}{100}\right)\left(\frac{T_{\rm rh}}{10^{15}\,{\rm GeV}}\right)^3, \tag{102}$$

where $s_0 \sim 10^9\,{\rm cm}^{-3}$ is the current entropy density and a_0 is the current scale factor. For $V_0^{1/4} \sim 10^{16}\,{\rm GeV}$ and $g_* \sim 100$, this yields $S_{\rm obs} \sim 10^{88}$, consistent with the observed cosmic entropy budget.

The entropy production during the discharge itself can be quantified by the change in the number of e-folds of the scale factor. The discharge occurs over a timescale $\Delta t \sim |m_{\rm eff}|^{-1} \sim 10^{-13}\,{\rm GeV}^{-1}$, during which the scale factor grows by a factor

$$\frac{a_{\rm end}}{a_{\rm start}} \sim \exp(H_*\Delta t) \sim 1 + \mathcal{O}(10^{-3}). \tag{103}$$

The entropy production is therefore dominated by the conversion of potential energy to radiation, not by the expansion during the discharge.

The low-entropy initial condition is the metastable fuel configuration with $\Phi < \Phi_c$. The fuel field is displaced from its minimum by $\Delta\Phi \sim M_{\rm Pl}$, giving a potential energy $V_0 \sim \lambda M_{\rm Pl}^4$. The entropy of this configuration is negligible compared to the post-discharge entropy because the fuel field is in a coherent, low-entropy state. The discharge thus provides a dynamical origin for the cosmic entropy budget, replacing the ad hoc low-entropy initial condition of standard cosmology with a physical mechanism.

A more precise calculation requires tracking the non-equilibrium dynamics of the tachyonic transition, the parametric resonance phase, and the subsequent thermalization. This is left for future work using lattice simulations or kinetic theory approaches. The time-averaged energy partition ratios for the benchmark parameters are:

- Proxy fluid (pfd): $\sim 10^{-2}$–10^{-1} of V_0 (set by $\Delta N_{\rm fluid}$ bound)
- Dark matter: $\Omega_\chi h^2 \sim 0.12$ (set by parametric resonance efficiency)
- Dark energy: $\rho_{\rm DE} \sim 10^{-120} M_{\rm Pl}^4$ (residual)
- SM radiation: remainder $\sim V_0$ (reheating)

The precise partition depends on the trajectory in field space and is computed numerically in the CLASS implementation.

5.6 Proxy-fluid dark radiation

The pfd sector is parameterized by

$$\Delta N_{\rm fluid} = \frac{\rho_{\rm pfd}}{\frac{7}{8}\left(\frac{4}{11}\right)^{4/3}\rho_\gamma}. \tag{104}$$

Its defining properties are $w = 1/3$, $c_s^2 = 1/3$, and $\sigma = 0$.

6 Perturbations and numerical implementation

The PTUM proxy-fluid dark-radiation sector is implemented in a modified CLASS v3.2.0 [12,13] *(patch pending public release; see Appendix B.4)*. In synchronous gauge [18], the scalar perturbations obey

$$\dot\delta_{\rm pfd} = -\tfrac{4}{3}\Big(\theta_{\rm pfd} + \tfrac{\dot h}{2}\Big), \tag{105}$$

$$\dot\theta_{\rm pfd} = \tfrac{k^2}{4}\delta_{\rm pfd}. \tag{106}$$

6.1 Proxy-fluid dark radiation: origin of $\sigma_\psi = 0$

The defining observational property of the pfd component is the absence of anisotropic stress, $\sigma_\psi = 0$. Within the effective coherent-field description, this is not introduced as an independent assumption; it follows from the nature of the source field *provided it is in the quartic oscillatory regime* (which is not realized at the benchmark parameters; see Sec. 3.7). Here we derive this result from the equations of motion *conditional on that regime*.

6.1.1 Effective energy-momentum tensor

Consider the proxy field ψ oscillating coherently in its quartic potential $V(\psi) = \frac{1}{4}\lambda_\psi\psi^4$. The canonical energy-momentum tensor is

$$T^{(\psi)}_{\mu\nu} = \partial_\mu\psi\,\partial_\nu\psi - g_{\mu\nu}\Big[\tfrac{1}{2}(\partial_\rho\psi)^2 - V(\psi)\Big]. \tag{107}$$

For a spatially homogeneous field $\psi = \psi(t)$ oscillating in the quartic potential, the spatial average over one oscillation period gives the effective fluid description. For a potential $V(\psi) = \frac{1}{4}\lambda_\psi\psi^4$, the virial theorem gives $\langle\dot\psi^2\rangle = 4\langle V(\psi)\rangle$. The time-averaged energy density and pressure are then $\langle\rho_\psi\rangle = \frac{1}{2}\langle\dot\psi^2\rangle + \langle V\rangle = 3\langle V\rangle$ and $\langle p_\psi\rangle = \frac{1}{2}\langle\dot\psi^2\rangle - \langle V\rangle = \langle V\rangle$, yielding the equation of state $w_\psi = \langle p_\psi\rangle/\langle\rho_\psi\rangle = 1/3$. The anisotropic stress vanishes in the effective fluid description:

$$\sigma_\psi \equiv \frac{\rho_\psi - p_\psi - 2p_\perp}{\rho_\psi + p_\psi} = 0\,, \tag{108}$$

because the energy-momentum tensor of a spatially homogeneous coherent field is isotropic at the background level.

This is the key physical point: unlike a collection of free-streaming particles, which carry momentum correlations through their individual trajectories and thereby generate non-zero σ, a coherent condensate does not have independent degrees of freedom below the coherence scale. The pressure is everywhere $p = \rho/3$ and there are no viscous or anisotropic contributions from particle collisions.

6.1.2 Perturbation hierarchy

At the linear perturbation level we write $\psi = \bar\psi(t) + \delta\psi$. The perturbed energy-momentum tensor contains $\delta\rho_\psi$, δp_ψ, velocity θ_ψ, and shear σ_ψ. The conservation equation $\nabla_\mu T^{\mu\nu(\psi)} = 0$ gives two first-order equations for $\delta_\psi \equiv \delta\rho_\psi/\bar\rho_\psi$ and $\theta_\psi \equiv ik\delta v_\psi$:

$$\dot\delta_\psi = -\frac{4}{3}\Big(\theta_\psi + \frac{\dot h}{2}\Big), \tag{109}$$

$$\dot\theta_\psi = \frac{k^2}{4}\,\delta_\psi\,. \tag{110}$$

Higher multipoles (F_ℓ with $\ell \geq 2$) satisfy $\dot F_\ell \propto kF_{\ell-1} - (\ell+1)HF_\ell$ and do not source the Einstein equations for scalar modes. These higher multipoles are present but they do not contribute to σ_ψ in the coherent oscillation limit because the condensate has no particle-streaming velocity dispersion. They are therefore omitted from the evolution equations used in the `CLASS` implementation.

The conditions under which this approximation holds are:

- **Coherence:** The coherence length of the ψ condensate exceeds the Hubble radius, so that the field oscillates as a single macroscopic mode. This is naturally satisfied when the field is homogeneous at the scale of the observable patch at the time of oscillatory onset.

– **Non-interacting:** The ψ field has no significant self-interaction beyond the $\lambda_\psi\psi^4$ potential term. If $\lambda_\psi\psi^4$ is the only relevant interaction, the condensate remains collisionless and coherent.
– **Subluminal sound horizon:** The physical sound horizon $c_s t$ of the condensate is sub-Hubble, ensuring that the perfect fluid approximation is valid.

The sound speed follows from the perturbation equations: $c_s^2 = \dot{p}_\psi/\dot{\rho}_\psi = 1/3$, as expected for a relativistic coherent oscillator. This value is positive and constant, confirming gradient stability ($c_s^2 > 0$) throughout the evolution.

We emphasise that this is the same physics underlying the standard treatment of scalar field dark energy (quintessence) and inflation ensuring $c_s^2 = 1$ for the inflaton. The pfd case is distinguished only by the fact that the condensate is in the $w = 1/3$ regime rather than $w \simeq -1$ or $w \simeq 0$.

6.1.3 Adiabatic vs. effective sound speed and gauge/rest-frame distinction

The adiabatic sound speed $c_a^2 = \dot{p}/\dot{\rho}$ and the rest-frame effective sound speed are distinct quantities. For an exactly radiation-like barotropic fluid with $p = \rho/3$, one has $c_a^2 = 1/3$. This statement does not, by itself, establish that a canonical scalar field undergoing a tachyonic transition is described by that barotropic fluid at all wavelengths. The canonical field has unit microscopic rest-frame propagation speed, while an averaged oscillating condensate can have a different effective fluid response. The latter requires a controlled averaging calculation.

In this manuscript, the CLASS implementation adopts the phenomenological closure $w_\psi = c_{s,\psi}^2 = 1/3$ and $\sigma_\psi = 0$ after the quartic-fluid conditions in Section 3.6 are assumed to hold. This is an explicit modelling prescription, not a derivation of a scale-dependent $c_{\rm eff}^2(k,a)$ from the microscopic perturbation equation. The closure should therefore be interpreted conditionally, and the CMB constraints apply to this effective-fluid realization of PTUM. A field-level derivation and comparison against the Boltzmann-fluid approximation remain necessary for a complete perturbative validation.

The implementation supplies adiabatic initial conditions (consistent with the single-field creation mechanism, which yields purely adiabatic perturbations to leading order in slow-roll) and couples the pfd fluid consistently to the Einstein source terms. Exact ΛCDM recovery for $\Delta N_{\rm fluid} = 0$ was verified.

A technically important issue was the state-vector layout. An early implementation inserted pfd variables inside the ultra-relativistic hierarchy, breaking the contiguous index assumptions used by CLASS and producing singular-matrix failures. The final implementation places pfd in an independent fluid block, fully resolving the instability.

7 Cosmological signatures

7.1 CMB residuals

Raw CMB TT and matter-power residuals were computed for $\Delta N_{\rm fluid} = 0.01, 0.02, 0.05, 0.10, 0.20$, keeping the standard cosmological parameters fixed.

7.2 Lensing potential

The CMB lensing potential power spectrum $C_L^{\phi\phi}$ is suppressed by pfd through its effect on late-time structure growth. At $L = 500$ the fractional suppression is

$$\begin{aligned}\frac{\Delta C_L^{\phi\phi}}{C_L^{\phi\phi}} &\simeq -1.46\% \quad (\Delta N = 0.10),\\ &\simeq -4.67\% \quad (\Delta N = 0.31).\end{aligned} \tag{111}$$

The suppression scales approximately linearly with $\Delta N_{\rm fluid}$ and is detectable with CMB-S4 lensing reconstruction for $\Delta N_{\rm fluid} \gtrsim 0.10$.

7.3 Fluid versus free-streaming dark radiation

The key observable distinction between pfd and standard neutrino-like dark radiation is the absence of anisotropic stress in the pfd case. Free-streaming dark radiation with extra energy fraction $\Delta N_{\rm eff}$ induces a CMB acoustic-peak phase shift

$$\begin{aligned}\delta\phi_{\rm peak} &\approx 0.19\pi\, R_{\rm extra},\\ R_{\rm extra} &= \frac{\Delta N_{\rm fluid}\,\frac{7}{8}\left(\frac{4}{11}\right)^{4/3}}{\sum_i \Omega_{i,\rm rad}},\end{aligned} \tag{112}$$

whereas pfd with $\sigma = 0$ produces no such shift.

The observational distinction between a proxy-fluid component and a free-streaming relativistic species can be expressed directly in terms of the acoustic phase of the CMB temperature spectrum. For a fixed additional radiation abundance, a free-streaming component shifts the acoustic peaks through anisotropic-stress effects, whereas the PTUM proxy-fluid component preserves the standard peak phase within the effective linear fluid description. This diagnostic is illustrated schematically in Fig. 10.

The predicted phase shifts are listed in Table 3.

At fixed cosmological parameters, a systematic comparison of TT and EE power spectra shows that the absence of the sigma-induced phase shift in the pfd case leads to smaller spectral distortions than in the free-streaming case with the same ΔN. The difference is dominated by the EE polarization spectrum and grows rapidly with $\Delta N_{\rm fluid}$, indicating that current and future high-resolution CMB data should be able to discriminate between the two scenarios. This distinction is most directly accessible through the CMB acoustic-peak phase, which serves as the primary observational discriminator of pfd.

Table 3. Phase shift between fluid (pfd) and free-streaming dark radiation for representative values of $\Delta N_{\rm fluid}$, together with the detection significance for CMB-S4 and CMB-HD. *(Author-reported values; modified CLASS code not yet public.)*

$\Delta N_{\rm fluid}$	Phase shift	CMB-S4	CMB-HD
0.05	0.23°	no (0.8σ)	marginal (1.5σ)
0.10	0.45°	no (1.5σ)	detect (3.0σ)
0.20	0.89°	marginal (3.0σ)	detect (6.0σ)
0.31	1.40°	detect (4.5σ)	detect (9.2σ)

7.4 Decision-level observational tests

The proxy-fluid interpretation should be evaluated through a small number of separable observational outcomes. This is important because PTUM is not a single yes-or-no deformation of ΛCDM: its background expansion, radiation abundance, perturbative closure, and phase-shift signature probe different parts of the effective construction. Table 4 collects the corresponding decision tests.

The cleanest possible support for the PTUM proxy-fluid sector would not be a generic excess in $N_{\rm eff}$ alone. It would be the combination of an allowed extra relativistic component, the absence of the corresponding free-streaming acoustic phase shift, and a stable effective-fluid closure with $\sigma_\psi = 0$. Conversely, a robust detection of an additional neutrino-like free-streaming component would not merely tighten the pfd abundance bound; it would directly disfavor the proxy-fluid interpretation of that component.

8 Bayesian constraints and model comparison

8.1 Likelihood setup

A Bayesian benchmark analysis is reported using Cobaya [14] with the PTUM-modified CLASS backend *(chains and configs pending public release; results reported by the authors, not independently reproduced).* The likelihood combination includes Planck 2018 high-ℓ TTTEEE-lite, low-ℓ EE, CMB lensing [22], 6dF, SDSS DR7 MGS, and SDSS DR12 BAO [24,25,26], and Pantheon+ supernovae [27].

8.2 Posterior constraints

The marginalized PTUM constraints are

$$\omega_b = 0.02244^{+0.00014}_{-0.00016}, \tag{113}$$

$$\omega_{\rm cdm} = 0.1222^{+0.0013}_{-0.0023}, \tag{114}$$

$$H_0 = 68.64^{+0.55}_{-0.87}\ {\rm km\,s^{-1}\,Mpc^{-1}}, \tag{115}$$

$$n_s = 0.9639 \pm 0.0036, \tag{116}$$

$$\tau_{\rm reio} = 0.0529 \pm 0.0071, \tag{117}$$

$$\Delta N_{\rm fluid} < 0.31 \quad (95\%\ {\rm CL}). \tag{118}$$

For the ΛCDM reference run:

$$\omega_b = 0.02235 \pm 0.00013, \tag{119}$$

$$\omega_{\rm cdm} = 0.12004 \pm 0.00087, \tag{120}$$

$$H_0 = 67.85 \pm 0.39\ {\rm km\,s^{-1}\,Mpc^{-1}}, \tag{121}$$

$$n_s = 0.9635 \pm 0.0037, \tag{122}$$

$$\tau_{\rm reio} = 0.0533^{+0.0065}_{-0.0075}. \tag{123}$$

8.3 Dataset-split analysis

To identify which data combination drives the $\Delta N_{\rm fluid}$ bound, we compare results from different dataset combinations using converged PTUM chains. The results are summarized in Table 5.

Here P denotes Planck TTTEEE-lite + low-ℓ EE, L denotes Planck lensing, B denotes BAO (6dF + DR7 MGS + DR12), and S denotes Pantheon+ supernovae.

A Planck+BAO combination already imposes a meaningful upper bound, $\Delta N_{\rm fluid} < 0.38$ (95% CL), while the addition of Planck lensing and Pantheon+ tightens this to $\Delta N_{\rm fluid} < 0.31$, a 18% improvement. This indicates that the PTUM pfd constraint is primarily CMB-driven, with low-redshift probes acting mainly as geometric consistency priors and moderate posterior sharpeners. The Hubble constant shift is present already at the Planck+BAO level ($H_0 = 68.91\,{\rm km\,s^{-1}\,Mpc^{-1}}$), confirming it as an intrinsic feature of the model rather than an artifact of the low-redshift dataset.

8.4 Binned maximum-posterior diagnostic

The curve shown in Fig.12 was constructed from the chains by extracting the maximum-posterior sample in successive $\Delta N_{\rm fluid}$ bins. This is a binned maximum-a-posteriori (MAP) envelope, not a frequentist profile likelihood: maximizing the posterior retains the prior density and does not implement likelihood maximization over nuisance parameters. We therefore use the curve only as a descriptive diagnostic and do not interpret its vertical levels as profile-likelihood confidence thresholds. A genuine profile likelihood would require maximizing the likelihood, rather than the posterior, over all nuisance parameters at each fixed value of $\Delta N_{\rm fluid}$. The marginal posterior limits quoted below remain posterior credible limits.

For clarity, we distinguish the following statistics from the posterior distribution of $\Delta N_{\rm fluid}$:

- Posterior mean: $\langle \Delta N_{\rm fluid} \rangle \approx 0.12$
- Posterior median: $\Delta N^{\rm med}_{\rm fluid} \approx 0.10$
- Posterior mode (maximum a posteriori): $\Delta N^{\rm MAP}_{\rm fluid} \approx 0.07$
- Binned MAP-envelope maximum: $\Delta N^{\rm MAP-bin}_{\rm fluid} \approx 0.07$
- Marginal 95% posterior upper limit: $\Delta N_{\rm fluid} < 0.31$
- No frequentist profile-likelihood limit is claimed in this analysis.

The parameter $\Delta N_{\rm fluid}$ is equivalent to $N_{\rm pfd}$ used in some figures; both denote the effective number of proxy-fluid dark radiation species.

Table 4. Decision-level observational tests for the PTUM proxy-fluid sector. The table separates viability tests, discriminators, and direct failure modes.

Test	PTUM expectation	Distinguishing information	Failure mode
Radiation abundance	A bounded positive or vanishing $\Delta N_{\rm fluid}$	Determines whether the proxy-fluid channel is cosmologically relevant	No allowed region after CMB, BBN, BAO, SNe, and lensing
Acoustic phase	No neutrino-like phase shift from the proxy component	Separates fluid radiation from free-streaming radiation	Extra radiation is detected with a free-streaming phase shift
Perturbative closure	$w_\psi = 1/3$, $c^2_{s,\psi} = 1/3$, $\sigma_\psi = 0$	Tests the perfect-fluid proxy assumption	Nonzero effective anisotropic stress is required
Late-time expansion	Near-Λ residual sector with viable $w_{\rm DE}$	Checks whether the residual field can mimic dark energy	No viable near-Λ region remains
Internal EFT consistency	Positive post-saturation fluctuation scale and stable modes	Checks whether the threshold discharge is dynamically safe	Residual tachyonic, ghost, or gradient instability

Table 5. Dataset-split comparison. Weighted posterior means are shown; the last column gives the 95% CL upper limit on $\Delta N_{\rm fluid}$. *(Author-reported; chains not yet public.)*

Data	H_0	ω_c	n_s	ΔN^{95}
P+B	68.91	0.1227	0.965	0.38
P+L+B+S	68.64	0.1222	0.964	0.31
ΛCDM	67.85	0.1200	0.964	–

8.5 Prior robustness

The constraint $\Delta N_{\rm fluid} < 0.31$ is data-driven rather than prior-driven. Under flat priors Uniform$[0, 0.5]$, Uniform$[0, 1]$, and Uniform$[0, 3]$, the 95% upper limit remains at 0.31–0.39, unchanged at the level of the posterior width. The dominant constraining dataset is the CMB TT+TE+EE likelihood through Silk damping at high ℓ; BAO and SN data provide geometric consistency but minimal direct sensitivity to $\Delta N_{\rm fluid}$.

8.6 Model comparison

The minimum χ^2 values are $\chi^2_{\rm PTUM} = 2397.22$ and $\chi^2_{\Lambda\rm CDM} = 2396.96$, giving

$$\Delta\chi^2 = +0.26, \qquad \Delta\text{AIC} = 2.26. \tag{124}$$

This indicates only a weak preference for the simpler model and does not constitute a decisive exclusion of PTUM. The current data provide a meaningful upper bound on the pfd abundance while leaving the model observationally viable within the parameter region studied. This interpretation is deliberately conservative: current data support viability of the effective proxy-fluid construction, not a detection of that sector and not a statistical preference over ΛCDM.

8.6.1 MCMC convergence diagnostics

The authors report that all PTUM MCMC chains were run with `Cobaya` using the Metropolis-Hastings sampler with adaptive proposal covariance *(author-reported; chains not yet public)*. The following convergence diagnostics are reported for the production chains:

- **Number of chains:** 4 independent chains per dataset combination.
- **Chain length:** 500,000 samples per chain after burn-in, giving 2,000,000 total samples.
- **Gelman-Rubin $R-1$ statistic:** $R-1 < 0.01$ for all parameters including $\Delta N_{\rm fluid}$, ω_c, H_0, n_s, and $\tau_{\rm reio}$. The maximum $R-1$ across all parameters is 0.006.
- **Effective sample size (ESS):** Minimum ESS > 5000 for all parameters. For $\Delta N_{\rm fluid}$, ESS $\approx 12,000$ (combined chains).
- **Autocorrelation time:** Mean autocorrelation length $\tau_{\rm ac} \approx 40$ samples for $\Delta N_{\rm fluid}$, $\tau_{\rm ac} \approx 25$ for standard ΛCDM parameters.
- **Burn-in:** First 100,000 samples discarded per chain based on visual inspection of trace plots and the `Cobaya` built-in convergence criteria.
- **Trace plots:** All chains show good mixing with no visible trends or stuck regions. The $\Delta N_{\rm fluid}$ parameter explores the full prior range $[0, 3]$ efficiently.
- **Parameter correlations:** The main degeneracy is between $\Delta N_{\rm fluid}$ and H_0 (correlation coefficient $\rho \approx 0.65$), and between $\Delta N_{\rm fluid}$ and ω_c ($\rho \approx -0.45$). These are well-resolved by the chains.
- **Prior sensitivity:** Chains were re-run with Uniform$[0, 0.5]$, Uniform$[0, 1]$, and Uniform$[0, 3]$ priors on $\Delta N_{\rm fluid}$. The posterior constraints are stable across all priors (95% CL upper limit varies by < 0.08), confirming the result is data-driven.

The chains are reported to satisfy all standard convergence criteria and the reported constraints are robust *(author-reported; the chains are not yet public and these diagnostics cannot currently be independently checked).*

Table 6. Linear perturbation stability for all PTUM sectors.

Sector	Ghost-free	c_s^2	Grad.-stable	σ
Φ (creation)	yes	1	yes	0
ψ (pfd)	yes	1/3	yes	0
χ (DM)	yes	0	yes	0
$\Phi_{\rm res}$ (DE)	yes	1	yes	0

The full chains and `Cobaya` configuration files will be released in the reproducibility package (see Section B.4).

8.7 Cosmological tensions

PTUM shifts H_0 upward by $+0.79\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ relative to ΛCDM (toward the SH0ES value), reducing the Hubble tension from 4.7σ to 3.5σ (25% reduction). The S_8 tension is essentially unchanged (2.4σ in both models). PTUM was not designed as a tension-solving model; these effects are incidental consequences of the extra radiation component.

9 Stability and consistency tests

9.1 Linear perturbation stability

All four PTUM sectors have canonical kinetic terms $\frac{1}{2}(\partial\phi)^2$, so the kinetic matrix is the identity and is positive definite: no ghosts (GÖSTERİLDİ). The scalar perturbation speeds are $c_s^2 = 1$ for the canonical fields $\Phi, \chi, \Phi_{\rm res}$ and $c_s^2 = 1/3$ for the pfd fluid closure (FENOMENOLOJİK, conditional on the quartic regime). The relevant quantities are summarized in Table 6.

The pfd dispersion relation $\omega^2 = (1/3)k^2$ gives propagating, stable acoustic oscillations for all k. No tachyonic or viscous instabilities are present in any sector.

9.2 Isocurvature constraints

The single-field creation mechanism produces purely adiabatic perturbations to leading order in slow-roll. The subleading isocurvature fraction from reheating asymmetry is estimated as $\beta_{\rm iso} \sim \epsilon^2 \sim 10^{-12}$–$10^{-6}$, which is 10^4–10^5 times below the Planck 95% CL bounds on CDM and neutrino density isocurvature modes. The pfd isocurvature is additionally suppressed by the small pfd fraction ($\Delta N/N_{\rm eff} \sim 3$–$11\%$).

9.2.1 Detailed isocurvature and transfer function analysis

The authors report performing a complete isocurvature analysis for all PTUM sectors, computing the full transfer functions and quantifying the isocurvature modes that could be generated by the discharge transition *(author-reported; see the status labels below).*

Isocurvature mode definitions: In a multi-component universe, the isocurvature perturbation between components i and j is defined as

$$S_{ij} = 3(\zeta_i - \zeta_j) = \frac{\delta\rho_i}{\rho_i + p_i} - \frac{\delta\rho_j}{\rho_j + p_j}, \tag{125}$$

where ζ_i is the curvature perturbation on uniform-density hypersurfaces for component i. For PTUM, the relevant isocurvature modes are:

- $S_{\rm cdm,\gamma}$: CDM-photon isocurvature
- $S_{\nu,\gamma}$: Neutrino-photon isocurvature
- $S_{\psi,\gamma}$: Proxy-fluid-photon isocurvature
- $S_{\chi,\gamma}$: Dark matter-photon isocurvature
- $S_{\Phi_{\rm res},\gamma}$: Residual DE-photon isocurvature

Primordial isocurvature from inflation: The PTUM creation sector is a single-field slow-roll model, which generically produces purely adiabatic perturbations. The isocurvature fraction is suppressed by slow-roll parameters:

$$\beta_{\rm iso} \equiv \frac{P_S}{P_\zeta + P_S} \sim \epsilon^2 \sim 10^{-12}\text{–}10^{-6}, \tag{126}$$

where $\epsilon \sim 10^{-3}$–10^{-6} is the slow-roll parameter. This is far below the Planck 2018 95% CL bounds: $\beta_{\rm iso}^{\rm CDM} < 0.001$, $\beta_{\rm iso}^{\nu} < 0.01$.

Isocurvature from the discharge transition: The discharge is a rapid but smooth transition (timescale $\Delta t \sim |m_{\rm eff}|^{-1} \ll H^{-1}$). During this transition, the energy density is partitioned among the sectors. The authors report computing the isocurvature generated by solving the coupled perturbation equations through the transition *(author-reported; the lattice/perturbation code is not yet public).* The key result is that the discharge produces negligible isocurvature because:

1. The transition is nearly instantaneous on Hubble scales, so super-horizon modes do not evolve significantly.
2. The energy partitioning is determined by the background dynamics, which are adiabatic (the same field Φ controls all sectors).
3. The proxy fluid ψ is produced from the coherent oscillations of the tachyonic field, which inherit the adiabatic perturbations of Φ.

Numerically, the authors report $S_{\psi,\gamma} \sim 10^{-5}$ and $S_{\chi,\gamma} \sim 10^{-6}$ at the end of the discharge, well below observational bounds *(author-reported; lattice code not yet public).*

Transfer functions: The authors report computing the full radiation and matter transfer functions for PTUM using the modified `CLASS` code *(author-reported; the modified code is not yet public).* The transfer function for the proxy fluid $T_\psi(k)$ is compared to the standard neutrino transfer function $T_\nu(k)$:

$$T_\psi(k) = \frac{\delta_\psi(k,\eta_0)}{\zeta(k)}, \qquad T_\nu(k) = \frac{\delta_\nu(k,\eta_0)}{\zeta(k)}. \tag{127}$$

The key difference is that $T_\psi(k)$ lacks the free-streaming damping at high k (since $\sigma_\psi = 0$), while $T_\nu(k)$ shows

the characteristic neutrino free-streaming suppression for $k \gtrsim 0.1\,\mathrm{Mpc}^{-1}$. This difference in the transfer functions is the origin of the phase-shift discriminator.

Isocurvature constraints from Planck: The authors report computing the likelihood for the isocurvature modes using the Planck 2018 isocurvature likelihood module *(author-reported; chains not public)*. The PTUM predictions for all isocurvature modes are consistent with zero at $< 0.1\sigma$. The 95% CL upper limits on the isocurvature fractions are:

$$\beta_{\rm iso}^{\rm CDM} < 1.2 \times 10^{-3}, \tag{128}$$

$$\beta_{\rm iso}^{\nu} < 8.5 \times 10^{-3}, \tag{129}$$

$$\beta_{\rm iso}^{\psi} < 3.1 \times 10^{-4}, \tag{130}$$

$$\beta_{\rm iso}^{\chi} < 2.7 \times 10^{-5}. \tag{131}$$

All PTUM predictions are orders of magnitude below these limits.

Future sensitivity: CMB-S4 will improve isocurvature constraints by a factor of $\sim$ 5–10. The projected sensitivity to $\beta_{\rm iso}^{\psi}$ is $\sim 3\times10^{-5}$, which could begin to probe the PTUM parameter space if the discharge generates larger isocurvature than our baseline estimate. This provides an additional future test of the model.

9.3 BBN consistency

The PTUM bound $\Delta N_{\rm fluid} < 0.31$ is automatically consistent with Big Bang nucleosynthesis. The BBN-only 95% CL upper limit is $\Delta N_{\rm eff} < 0.40$ (Fields et al. 2020 [29]), which is weaker than the CMB-derived bound. For the maximum allowed pfd abundance, the primordial helium fraction shifts by at most $\delta Y_p \simeq +0.004$, corresponding to 1.1σ of the current Y_p measurement uncertainty.

9.4 EFT validity

The PTUM EFT hierarchy $H_* \ll V_0^{1/4} \ll M_{\rm Pl}$ is satisfied: $H_*/M_{\rm Pl} \sim 4 \times 10^{-6}$ and $V_0/M_{\rm Pl}^4 \sim 10^{-10}$. All couplings are perturbative ($\ll 4\pi$) with margins exceeding a factor of two. For $\lambda \lesssim 0.5$ the field excursion is super-Planckian (a standard feature of large-field inflation models, addressed by shift symmetry); for $\lambda > 1$ the excursion is sub-Planckian and the EFT is fully under control without additional symmetry input.

9.4.1 EFT cutoff and scale hierarchy

The effective field theory is valid up to a cutoff scale $\Lambda_{\rm EFT}$ that must satisfy

$$H_* \ll \Lambda_{\rm EFT} \ll M_{\rm Pl}. \tag{132}$$

In PTUM, the natural cutoff is set by the mass of the heaviest integrated-out field. The fuel field Φ has mass $m_\Phi^2 \sim V_0''(\Phi) \sim \lambda V_0/\Phi^2$ during inflation. For $\lambda \sim 0.02$ and $\Phi \sim M_{\rm Pl}$, this gives $m_\Phi \sim 10^{13}$ GeV. The proxy field ψ has mass $m_\psi \sim |m_{\rm eff}| \sim 10^{13}$ GeV at the transition. The dark matter field χ has mass $m_\chi \sim 10^9$–10^{13} GeV. The residual field $\Phi_{\rm res}$ has mass $m_{\rm res} \sim H_0 \sim 10^{-33}$ eV. The hierarchy of scales is therefore:

$$H_0 \ll H_* \ll m_\Phi, m_\psi, m_\chi \ll V_0^{1/4} \ll M_{\rm Pl}. \tag{133}$$

The cutoff cannot be fixed by identifying it with $V_0^{1/4}$ without an argument from UV matching or perturbative unitarity. The quoted benchmark values give $H_* \sim 10^{12}$–10^{13} GeV and field masses of order 10^{13} GeV. Thus, if the cutoff were near the smallest retained mass, the ratio $H_*/\Lambda_{\rm EFT}$ could be order unity. For this reason $\Lambda_{\rm EFT}$ is treated as an independent matching scale. The required hierarchy is $H_*, m_i < \Lambda_{\rm EFT} < M_{\rm Pl}$, but the manuscript does not yet derive a numerical value for it. The scale $V_0^{1/4}$ may be used as a benchmark comparison scale, not as proof of EFT control.

9.4.2 Ghost and gradient instability checks

The kinetic terms for all fields are canonical with positive signs:

$$\mathcal{L}_{\rm kin} = \tfrac{1}{2}(\partial_\mu\Phi)^2 + \tfrac{1}{2}(\partial_\mu\psi)^2 + \tfrac{1}{2}(\partial_\mu\chi)^2 + \tfrac{1}{2}(\partial_\mu\Phi_{\rm res})^2. \tag{134}$$

There are no ghost instabilities (wrong-sign kinetic terms) at the tree level. The one-loop Coleman–Weinberg corrections do not generate wrong-sign kinetic terms because all field-dependent masses are positive in the post-discharge vacuum.

The gradient stability condition requires $c_s^2 > 0$ for all propagating modes. For the fuel field Φ, the sound speed is $c_{s,\Phi}^2 = 1$ (canonical scalar). For the proxy field ψ in the quartic oscillation regime, the derivation gives $c_{s,\psi}^2 = 1/3 > 0$ *(conditional on that regime, which is not achieved at the benchmark parameters)*. For the dark matter field χ, $c_{s,\chi}^2 = 0$ (pressureless dust). For the residual field $\Phi_{\rm res}$, $c_{s,\rm res}^2 = 1$ (canonical scalar). All sound speeds are positive, confirming the absence of gradient instabilities.

9.4.3 Perturbative unitarity and coupling bounds

All dimensionless couplings in PTUM satisfy perturbative bounds:

- $\lambda \lesssim 0.5$ (fuel potential): $\lambda \ll 4\pi$ satisfied by factor $\gtrsim 25$.
- $\lambda_\psi \lesssim 1$ (proxy quartic): $\lambda_\psi \ll 4\pi$ satisfied by factor $\gtrsim 12$.
- $g_{\rm DM} \lesssim 0.1$ (DM–residual coupling): $g_{\rm DM} \ll 4\pi$ satisfied by factor $\gtrsim 125$.
- $\alpha \lesssim 1$ (activation coupling): $\alpha \ll 4\pi$ satisfied by factor $\gtrsim 12$.
- $\kappa, \lambda_{\rm res} \lesssim 1$ (residual potential): $\ll 4\pi$ satisfied.

The dimensionful parameters satisfy:

- $V_0^{1/4} \sim 10^{15}$–10^{16} GeV $\ll M_{\rm Pl}$.
- $E_{\rm th} \sim V_0^{1/4}$ (discharge threshold).
- $m \sim 10^{13}$ GeV (proxy mass scale).
- $m_\chi \sim 10^9$–10^{13} GeV (DM mass).
- $\Lambda_{\rm NP} \sim 10^{-3}$ eV (non-perturbative scale).

All mass scales are well below the Planck scale, and all dimensionless ratios are $\mathcal{O}(1)$ or smaller, with no fine-tuned hierarchies beyond the cosmological constant problem (which is re-parameterized, not solved).

9.4.4 Summary of EFT validity

The present analysis reports several necessary consistency checks, but it does not determine the ultraviolet cutoff from first principles. The cutoff $\Lambda_{\rm EFT}$ must exceed the characteristic frequencies and masses retained as dynamical degrees of freedom, while remaining below the scale at which the effective description requires new degrees of freedom or becomes strongly coupled. The potential-height scale $V_0^{1/4}$ is not, by itself, a derivation of this cutoff. Accordingly, the hierarchy

$$H_*, m_i < \Lambda_{\rm EFT} < M_{\rm Pl} \tag{135}$$

is a condition to be verified once the UV matching or perturbative-unitarity analysis is specified, rather than a completed result of the present work.

The positive canonical kinetic terms rule out elementary ghost signs in the displayed two-derivative scalar Lagrangian. Gradient stability of the phenomenological proxy-fluid closure requires $c^2_{s,\psi} > 0$; its microscopic validity remains conditional on the oscillation-averaging analysis described in Section 3.6. Perturbative couplings and the control of higher-dimensional operators must likewise be checked at a specified cutoff. These qualifications delimit the scope of the current EFT analysis and identify the calculations required for a UV-complete consistency claim.

10 Observational forecasts

The PTUM creation sector predicts a specific $r(\lambda)$ relation. For $\lambda = 0.02$ ($r = 0.022$), LiteBIRD and CMB-S4 can detect the tensor signal at 11σ and 22σ, respectively; for $\lambda = 0.05$ ($r = 0.007$), detection requires CMB-S4. For $\lambda > 0.5$, $r < 10^{-3}$ is beyond the reach of any currently planned experiment.

CMB-S4 is projected to improve the $\Delta N_{\rm fluid}$ bound by a factor of seven relative to the current result ($\sigma(\Delta N) \simeq 0.027$, 95% CL < 0.044), while CMB-HD offers a 14-fold improvement (< 0.023). The fluid versus free-streaming phase shift is detectable by CMB-S4 at $> 3\sigma$ for $\Delta N_{\rm fluid} \gtrsim 0.20$ and at $> 3\sigma$ by CMB-HD for $\Delta N_{\rm fluid} \gtrsim 0.10$.

Stage-IV galaxy surveys (DESI, Euclid, SKAO combined) can detect the $f\sigma_8$ shift at $\sim$ 2–3σ, while full Stage-IV combination can resolve the $\Delta H_0 = +0.79$ shift at $\sim 5\sigma$.

10.0.1 Forecast methodology and experimental specifications

The forecasts presented here are derived from Fisher matrix analyses using the full experimental covariance matrices for CMB-S4 and CMB-HD, following the methodology of [30,31,32] *(FENOMENOLOJİK tahmin; Fisher kodu yayında değil)*. We summarize the key specifications and assumptions below.

CMB-S4: We adopt the baseline CMB-S4 configuration from the *CMB-S4 Science Book* [30]:

- Frequency channels: 27, 39, 93, 145, 225, 280 GHz
- Beam FWHM: 3.0', 2.3', 1.4', 1.0', 0.9', 0.9'
- Noise levels (μK-arcmin): 2.8, 2.2, 1.0, 1.5, 4.5, 7.5
- Sky fraction: $f_{\rm sky} = 0.4$
- ℓ-range: $30 \le \ell \le 5000$ (temperature and polarization)
- Delensing: 80% delensing of B-modes assumed for r forecasts

The experimental covariance for the power spectra is

$$\mathrm{Cov}(C_\ell^{XY}, C_\ell^{WZ}) = \frac{1}{(2\ell+1) f_{\rm sky}} \times \left(\tilde{C}_\ell^{XW}\tilde{C}_\ell^{YZ} + \tilde{C}_\ell^{XZ}\tilde{C}_\ell^{YW}\right), \tag{136}$$

where $\tilde{C}_\ell^{XY} = C_\ell^{XY} + N_\ell^{XY}$ includes instrumental noise N_ℓ^{XY}. The noise power spectra are computed from the channel specifications above assuming inverse-variance weighting.

CMB-HD: We use the CMB-HD concept study specifications from [32]:

- Frequency channels: 90, 150, 220, 280 GHz (with additional channels for foreground cleaning)
- Beam FWHM: 0.35' at 150 GHz (diffraction-limited)
- Noise level: 0.5 μK-arcmin at 150 GHz
- Sky fraction: $f_{\rm sky} = 0.5$
- ℓ-range: $30 \le \ell \le 20000$
- Delensing: 90% delensing assumed

The higher ℓ-reach of CMB-HD is critical for the fluid-vs-free-streaming phase shift test, as the distinction grows at small angular scales ($\ell > 1000$).

Fisher matrix: For a set of parameters $\boldsymbol{\theta} = \{\omega_b, \omega_c, \theta_s, \tau, n_s, A_s, \Delta N_{\rm fluid}, w_{\rm DE}, \dots\}$, the Fisher matrix is

$$F_{\alpha\beta} = \sum_\ell \frac{\partial C_\ell}{\partial\theta_\alpha} \mathrm{Cov}^{-1}(C_\ell, C_\ell) \frac{\partial C_\ell}{\partial\theta_\beta}, \tag{137}$$

where the sum runs over all spectra (TT, TE, EE, BB) and the derivatives are computed numerically using the PTUM-modified `CLASS` code. The projected 1-σ error on $\Delta N_{\rm fluid}$ is $\sigma(\Delta N_{\rm fluid}) = \sqrt{(F^{-1})_{\Delta N, \Delta N}}$.

Results: The Fisher forecasts yield:

- CMB-S4: $\sigma(\Delta N_{\rm fluid}) \simeq 0.027$ (95% CL < 0.044), a factor of 7 improvement over current $\Delta N_{\rm fluid} < 0.31$.
- CMB-HD: $\sigma(\Delta N_{\rm fluid}) \simeq 0.012$ (95% CL < 0.023), a factor of 14 improvement.

– Fluid vs. free-streaming: The phase-shift signal-to-noise for $\Delta N_{\rm fluid} = 0.1$ is $(S/N)_{\rm CMB-S4} \approx 2.8$ and $(S/N)_{\rm CMB-HD} \approx 5.2$, confirming the $> 3\sigma$ detection threshold for CMB-HD at $\Delta N_{\rm fluid} \gtrsim 0.10$.
– Tensor-to-scalar ratio: For $\lambda = 0.02$ ($r = 0.022$), $\sigma(r)_{\rm CMB-S4} \approx 0.001$ (22σ detection); for $\lambda = 0.05$ ($r = 0.007$), $\sigma(r)_{\rm CMB-S4} \approx 0.001$ (7σ detection).

Systematic uncertainties: The forecasts assume perfect foreground removal and delensing. Residual foregrounds could degrade the $\Delta N_{\rm fluid}$ constraints by $\sim 20\%$ for CMB-S4 and $\sim 30\%$ for CMB-HD. The fluid-vs-free-streaming test is more robust because it relies on the phase shift, which is less affected by foregrounds than the amplitude. We marginalize over a foreground amplitude parameter with a conservative prior in the full Fisher analysis.

10.0.2 Joint CMB + LSS forecasts

Combining CMB-S4 with Stage-IV galaxy surveys (DESI, Euclid, SKAO) breaks degeneracies between $\Delta N_{\rm fluid}$ and other parameters. The joint Fisher matrix includes:

– CMB-S4 TT, TE, EE, BB spectra
– DESI BAO and $f\sigma_8$ measurements (14 redshift bins, $0.1 < z < 1.8$)
– Euclid weak lensing cosmic shear (10 tomographic bins)
– SKAO HI intensity mapping ($0.5 < z < 2.0$)

The joint analysis improves the $\Delta N_{\rm fluid}$ constraint to $\sigma(\Delta N_{\rm fluid}) \simeq 0.018$ (95% CL < 0.035) and the H_0 shift detection to $\sim 5\sigma$ for the PTUM best-fit $\Delta H_0 = +0.79$ km/s/Mpc.

Appendix material. Additional implementation notes and consistency figures are included in the self-contained appendix below.

11 Discussion

The main result of this work is that PTUM can now be discussed as a quantitatively specified and observationally testable cosmological creation framework, rather than as a purely qualitative dark-sector scenario. The model is built around a single central idea: a discharge-based cosmogonic event generates the hot expanding Universe and leaves behind dark matter, dark energy, and a relativistic proxy-fluid dark-radiation component as residual products. In this sense, PTUM is not merely an extension of ΛCDM by an additional radiation parameter; its primary goal is to connect cosmogenesis and dark-sector structure within one effective description.

Four distinct pillars are claimed, with the following status labels:

(i) the creation sector admits a plateau-type primordial potential compatible with current CMB constraints on n_s and r *(GÖSTERİLDİ at the analytic/slow-roll level; exact-background check SAYISAL OLARAK DOĞRULANDI with the public script)*;
(ii) the discharge scale is normalised to a concrete high-energy band, $V_0^{1/4} \sim 10^{15}$–10^{16} GeV *(GÖSTERİLDİ from the scalar amplitude; in unresolved tension with the quartic-regime requirement, see Sec. 3.7)*;
(iii) the late-time dark-energy sector contains a broad viable region with near-Λ behaviour *(analytic scan, author-reported; script pending release)*;
(iv) the pfd sector has a closed perturbation-level implementation, with observational bounds established through Bayesian inference, a binned maximum-posterior diagnostic, and dataset-split analysis *(implementation patch and chains pending release; bounds author-reported, not independently reproduced)*.

A particularly important outcome is that PTUM makes a genuinely falsifiable *proposal*: if the proxy-fluid component is realized as a relativistic perfect fluid with $w = 1/3$, $c_s^2 = 1/3$, and $\sigma = 0$, it would not produce the neutrino-like acoustic phase shift characteristic of free-streaming dark radiation. This absence is the cleanest proposed observational discriminator of the model *(KOŞULLU: the quartic oscillatory regime behind $w_\psi = 1/3$ is not achieved at the benchmark parameters)*. Current data are reported to constrain the abundance of this component to $\Delta N_{\rm fluid} < 0.31$ at 95% credibility *(author-reported)*, while future CMB-S4 and CMB-HD observations can directly test the fluid-versus-free-streaming distinction if a non-zero dark-radiation signal is present.

Methodologically, the numerical consistency of the framework is also an important result. The slow-roll treatment of the creation sector is validated by exact-background integration at the level $\Delta n_s \sim (5.2\text{–}5.8)\times 10^{-4}$ and $|\Delta r|/r \sim 1.3\%$–1.5%, lending credibility to the derived discharge-scale estimates and (n_s, r) predictions. Moreover, the modified `CLASS` implementation is reported to recover ΛCDM exactly in the limit $\Delta N_{\rm fluid} = 0$ *(author-reported; code not yet public)*, and the perturbation sector remains ghost-free and gradient-stable throughout the parameter region studied.

The data analysis clarifies where the present constraints originate. The dataset-split study is reported to show that the PTUM pfd bound is primarily CMB-driven: the Planck+BAO combination alone yields $\Delta N_{\rm fluid} < 0.38$ at 95% credibility, while adding lensing and Pantheon+ tightens the limit to $\Delta N_{\rm fluid} < 0.31$ *(author-reported)*. The binned maximum-posterior diagnostic has its largest value near $\Delta N_{\rm fluid} \approx 0.07$. Since this is not a genuine profile likelihood, it does not establish prior-volume independence; a likelihood maximization over nuisance parameters remains to be done.

At the same time, the current model-comparison result should be interpreted with appropriate caution. PTUM is not preferred over ΛCDM by present data: $\Delta\chi^2 = +0.26$ and $\Delta\mathrm{AIC} = 2.26$ indicate a weak preference for the simpler standard model. This outcome is not unexpected, since PTUM introduces one additional phenomenological degree of freedom while remaining close to ΛCDM in the allowed region. The correct interpretation is therefore not that PTUM is supported by current data, but that it remains observationally viable and non-excluded within the

parameter region studied while making sharper future predictions than many dark-sector extensions.

PTUM is reported to produce a mild upward shift in the inferred Hubble constant, $H_0 = 68.64^{+0.55}_{-0.87}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ *(author-reported)*, relative to the ΛCDM fit. This would reduce the SH0ES tension from about 4.7σ to $\sim 3.5\sigma$, i.e. by roughly 25%, although the model was not designed as a dedicated Hubble tension solution. By contrast, the S_8 tension remains essentially unchanged. PTUM should therefore be viewed primarily as a creation-based dark-sector framework rather than a targeted tension-resolution model.

11.1 Model summary and internal consistency

PTUM's core physical picture—fuel accumulation, tachyonic discharge, residual dark sector splitting—is internally consistent: all sectors are ghost-free and gradient-stable (Section 8.1), all couplings are perturbative and all energy scales satisfy $H_* \ll M_{\rm Pl}$ (Section 8.4), and the perturbative sector produces purely adiabatic perturbations with isocurvature fraction $\beta_{\rm iso} \sim 10^{-12}$–$10^{-6}$, far below current bounds (Section 8.2). The BBN consistency check is reported to confirm *(author-reported)* that $\Delta N_{\rm fluid} < 0.31$ is automatically compatible with the BBN upper limit $\Delta N_{\rm eff} < 0.40$ (Section 8.3).

The cosmological constant problem is re-parameterised rather than solved: the residual dark-energy fraction $V_{\rm res}/V_0 \sim 10^{-108}$–$10^{-112}$ is extremely small, but PTUM does not provide a dynamical explanation for this hierarchy. The magnitude remains unexplained; what PTUM provides is a concrete interpretation of dark energy as a discharge residual and a broad parameter region in which $w_{\rm DE} \approx -1$ is obtained. Future data from DESI [34] and Euclid [35] will test whether $w_{\rm DE}$ deviates from -1 at percent level; such a detection would constrain PTUM's viable region, while a precise confirmation of $w = -1$ would leave the problem open.

Several limitations are explicitly acknowledged. The relation between the effective discharge picture and a concrete UV completion remains to be derived; the structural parallels with string tachyon condensation and LQC bounce scenarios in Section 3.9 should therefore be interpreted as motivation rather than derivation. The reheating sector and the microscopic dark-matter phenomenology of χ have not yet been analysed in comparable detail.

11.1.1 Nested-sampling Bayesian evidence

The authors report having computed the Bayesian evidence $\mathcal{Z}$ for PTUM and ΛCDM using nested sampling with `PolyChord` *(author-reported; nested-sampling chains not yet public)* [16,17] via the `Cobaya` interface. The nested sampling runs used 2000 live points with a stopping criterion of $\Delta \ln \mathcal{Z} < 0.01$. The prior volumes were identical to the MCMC priors: flat priors on all standard cosmological parameters and Uniform$[0, 3]$ on $\Delta N_{\rm fluid}$.

The log-evidence values are:

$$\ln \mathcal{Z}_{\rm PTUM} = -1198.61 \pm 0.12, \tag{138}$$

$$\ln \mathcal{Z}_{\Lambda\rm CDM} = -1198.48 \pm 0.11, \tag{139}$$

giving a log-Bayes factor

$$\ln B_{\rm PTUM,\Lambda CDM} = \ln \mathcal{Z}_{\rm PTUM} - \ln \mathcal{Z}_{\Lambda\rm CDM} = -0.13 \pm 0.16. \tag{140}$$

The Bayes factor is consistent with unity ($|\ln B| < 1$), indicating that the current data do not significantly prefer either model. This is fully consistent with the AIC result (ΔAIC $= 2.26$) and the $\Delta\chi^2 = +0.26$ from the MCMC analysis. The nested sampling posterior constraints on $\Delta N_{\rm fluid}$ agree with the MCMC results to within < 0.01 on the 95% CL upper limit, providing an independent validation of the MCMC convergence.

The nested sampling runs also provide the posterior weights for each live point, enabling importance-sampling reweighting for alternative priors without re-running the sampler. The `PolyChord` chains and evidence outputs will be included in the reproducibility release (Section B.4).

11.2 Observational predictions and discriminators

The pfd sector provides the sharpest observational discriminator of PTUM. Because $\sigma_\psi = 0$, the proxy-fluid produces no neutrino-induced CMB acoustic-peak phase shift, whereas free-streaming dark radiation with the same energy density does. This absence is directly testable by future CMB experiments. CMB-S4 can distinguish fluid from free-streaming at $> 3\sigma$ for $\Delta N_{\rm fluid} \gtrsim 0.20$, and CMB-HD extends this reach to $\Delta N_{\rm fluid} \gtrsim 0.10$ (Section 10). The forecast matrix (Fig. 13) summarises the detection thresholds for all PTUM observables.

The creation sector predicts a specific tensor-to-scalar ratio $r(\lambda)$. For $\lambda = 0.02$ ($r \approx 0.022$), LiteBIRD and CMB-S4 can detect the signal; for $\lambda = 0.05$ ($r \approx 0.007$), detection requires CMB-S4. For $\lambda > 0.5$, $r < 10^{-3}$ is beyond the reach of any currently planned experiment.

On cosmological tensions: PTUM is reported to shift H_0 upward by $\sim 0.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, reducing the SH0ES tension from 4.7σ to $\sim 3.5\sigma$ ($\sim$ 25% improvement) *(author-reported)*. The S_8 tension remains essentially unchanged. This shift is an incidental geometric effect of the extra radiation component, not a designed tension solution.

11.3 PTUM research programme: open questions and future directions

PTUM is the first step of a broader research programme. The discharge mechanism touches on several deep questions in fundamental physics, each representing a natural direction for future investigation. These directions are outlined below not as claims made in this work but as motivated paths for the programme to pursue.

Singularity replacement. PTUM models the hot Big Bang as a finite-energy discharge transition rather than treating it as a primitive singular initial condition. This is analogous in spirit to the LQC bounce [45,46], though PTUM operates in the classical EFT regime ($H_*/M_{\rm Pl} \sim 10^{-6}$) without invoking quantum-gravity corrections. The pre-discharge phase and its spacetime structure remain to be specified; this requires a detailed model of the metastable background.

11.3.1 Pre-discharge metastable background

The pre-discharge phase is the epoch during which the fuel field Φ accumulates in a metastable configuration before reaching the critical threshold Φ_c. In the current effective description, this phase is not derived from first principles but is phenomenologically characterized by the following properties:

1. **Spacetime geometry:** The pre-discharge universe is assumed to be a quasi-de Sitter spacetime with Hubble rate $H_* \sim 10^{12}$–10^{13} GeV, driven by the fuel potential $V(\Phi) \approx V_0$ on the inflationary plateau. The metric is approximately
$$ds^2 \approx -dt^2 + e^{2H_* t} d\mathbf{x}^2. \tag{141}$$
This is consistent with the slow-roll inflationary phase described in Section 4.
2. **Fuel field configuration:** The fuel field Φ is displaced from the minimum of its potential by a super-Planckian excursion $\Delta\Phi \sim M_{\rm Pl}/\sqrt{\lambda}$. For $\lambda \sim 0.02$, this gives $\Delta\Phi \sim 7M_{\rm Pl}$. The field is in a coherent, low-entropy state with negligible kinetic energy ($\dot\Phi^2 \ll V_0$), satisfying the slow-roll conditions $\epsilon_V, |\eta_V| \ll 1$.
3. **Metastability:** The configuration is metastable because the proxy field ψ has a positive effective mass squared $m_{\rm eff}^2(\Phi) = m^2(1 - \alpha\Phi^2/E_{\rm th}^2) > 0$ for $|\Phi| < \Phi_c$. The ψ field sits at the origin $\psi = 0$, which is a local minimum of the activation potential. The system is stable against small fluctuations in ψ but unstable to the large-field excursion of Φ toward Φ_c.
4. **Entropy:** The pre-discharge state has extremely low entropy compared to the post-discharge universe. The fuel field is in a pure quantum state (or a highly coherent classical configuration), and the proxy sector is in its vacuum state. The total entropy is dominated by the de Sitter horizon entropy $S_{\rm dS} \sim M_{\rm Pl}^2/H_*^2 \sim 10^{12}$, which is negligible compared to the post-discharge entropy $S_{\rm obs} \sim 10^{88}$ (total observable-universe entropy).
5. **Trigger mechanism:** The transition to the discharge phase occurs when Φ rolls down the potential and crosses $\Phi_c = E_{\rm th}/\sqrt{\alpha}$. This is a classical dynamical process driven by the slow-roll evolution of Φ, not a quantum tunneling event. The timescale for the final approach to Φ_c is $\Delta t \sim H_*^{-1}$, and the discharge itself occurs on the much shorter timescale $\Delta t_{\rm dis} \sim |m_{\rm eff}|^{-1} \ll H_*^{-1}$.
6. **Limitations:** The current model does not specify the origin of the initial displacement of Φ, the mechanism that sets $V_0 \sim (10^{15}$–$10^{16}\,{\rm GeV})^4$, or the UV completion that gives rise to the effective Lagrangian. These questions require a concrete UV model (e.g., string theory, higher-dimensional braneworld, or asymptotically safe gravity) and are left for future work. The pre-discharge background should therefore be understood as an effective initial condition for the discharge transition, not as a derived consequence of a more fundamental theory.

A complete description of the pre-discharge phase would require solving the coupled dynamics of Φ and ψ in the metastable regime, including quantum fluctuations and their backreaction on the spacetime geometry. This is a significant theoretical challenge that goes beyond the scope of the present effective model.

Dark matter phenomenology. The dark-matter field χ is produced as a superheavy relic ($m_\chi \sim 10^9$–$10^{13}\,{\rm GeV}$) through parametric resonance during the discharge. Relic abundance, detection prospects, and the possible connection to topological defects formed at the discharge transition are left to a dedicated study.

11.3.2 Dark matter relic abundance from parametric resonance

The dark matter field χ couples to the residual fuel field $\Phi_{\rm res}$ via the interaction term $-g_{\rm DM}\chi^2\Phi_{\rm res}^2$ in the Lagrangian. The factor of 2 in the effective mass below follows from the normalization of this interaction term (GÖSTERİLDİ):
$$m_{\chi,\rm eff}^2(t) = m_\chi^2 + 2g_{\rm DM}\Phi_{\rm res}^2(t), \tag{142}$$
consistent with the χ equation of motion Eq. (15), whose interaction term is $+2g_{\rm DM}\chi\Phi_{\rm res}^2$. During the discharge, the fuel field Φ undergoes a rapid transition, and the residual field $\Phi_{\rm res}$ oscillates around its minimum. This coherent oscillation of $\Phi_{\rm res}$ acts as a time-dependent mass term for χ. Writing $\Phi_{\rm res}(t) = \mathcal{A}_{\rm res}\cos(\omega_{\rm res}t)$, the mode equation becomes a Mathieu equation with
$$q \equiv \frac{2g_{\rm DM}\mathcal{A}_{\rm res}^2}{2\omega_{\rm res}^2} = \frac{g_{\rm DM}\mathcal{A}_{\rm res}^2}{\omega_{\rm res}^2}, \qquad A_k \equiv \frac{k^2/a^2 + m_\chi^2}{\omega_{\rm res}^2}. \tag{143}$$
The χ field experiences parametric resonance when the Mathieu parameters place the mode in an instability band, the dominant one being $A_k \approx 1$ ($n = 1$), i.e. $\omega_\chi \equiv \sqrt{k^2/a^2 + m_\chi^2} \approx \omega_{\rm res}/2$.

The equation of motion for the χ field modes in an FRW background is
$$\ddot\chi_k + 3H\dot\chi_k + \left(\frac{k^2}{a^2} + m_\chi^2(t)\right)\chi_k = 0. \tag{144}$$
During the short discharge phase ($\Delta t \ll H^{-1}$), Hubble expansion can be neglected and the equation reduces

to the Mathieu equation. The Floquet exponent μ_k determines the exponential growth of modes within the resonance band:

$$\chi_k(t) \propto \exp(\mu_k t). \tag{145}$$

For the broad resonance regime ($q \gg 1$), the maximum Floquet exponent is $\mu_{\max} \approx q\,\omega_{\rm res}/2$ (GÖSTERİLDİ from the standard Mathieu asymptotics; the earlier draft's $\mu_{\max} \approx 0.2\,\omega_{\rm res}$ was a narrow-resonance estimate and is not used here). The occupation number of produced χ particles after N oscillation periods is

$$n_k \approx \exp(2\mu_k N\,\omega_{\rm res}^{-1}), \tag{146}$$

where $N\,\omega_{\rm res}^{-1}$ is the duration of the resonant phase. The relic abundance is determined by the number density of produced χ particles at the end of the discharge, redshifted to the present epoch. The comoving number density is

$$n_\chi \sim \frac{1}{(2\pi)^3}\int d^3k\, n_k \sim \frac{\omega_{\rm res}^3}{(2\pi)^3}\exp(2\mu_{\max} N\,\omega_{\rm res}^{-1}). \tag{147}$$

The present-day dark matter density is

$$\Omega_\chi h^2 = \frac{m_\chi n_\chi}{\rho_{\rm crit}/h^2} \sim 0.12\left(\frac{m_\chi}{10^{12}\,\text{GeV}}\right)\left(\frac{\omega_{\rm res}}{10^{13}\,\text{GeV}}\right)^3 \times \exp(2\mu_{\max} N\,\omega_{\rm res}^{-1}). \tag{148}$$

Matching the observed $\Omega_\chi h^2 \approx 0.12$ then fixes the required number of e-folds of resonant growth, $2\mu_{\max} N\,\omega_{\rm res}^{-1} \sim \ln(10^2\text{–}10^3)$, rather than being an automatic output of natural parameter choices. A full numerical integration of the mode equations including Hubble expansion and backreaction is required for precise predictions and is left for future work (EKSİK: no public code performs this integration).

Clarification: parametric resonance target vs. CLASS evolution. The value $\Omega_\chi h^2 \sim 0.12$ derived above is a *target benchmark* for the parametric resonance production mechanism during the discharge event; note that parametric resonance presupposes the oscillatory (quartic-dominated) regime of $\Phi_{\rm res}$, which is not achieved at the stated benchmark parameters (overdamped, $3H/(2\mu) \approx 25$; see the abstract and Section 3.7). It is *not* an output of the CLASS Boltzmann code. The CLASS implementation evolves the dark matter sector *after* the discharge, using the post-discharge initial conditions (energy density, equation of state) as inputs. The parametric resonance calculation is a separate pre-CLASS step that determines the initial dark matter abundance. The two are consistent if the parametric resonance produces the correct initial ρ_χ for the CLASS evolution. The benchmark parameters are chosen such that this consistency holds, but a full self-consistent calculation coupling the non-equilibrium resonance dynamics to the post-discharge Boltzmann evolution is left for future work.

Reconciling $\Phi_{\rm res}$ dual role. The residual field $\Phi_{\rm res}$ plays two distinct roles:

1. **During discharge:** $\Phi_{\rm res}$ oscillates with frequency $\omega_{\rm res} \sim 10^{13}$ GeV, driving parametric resonance for χ production. This is a high-frequency, large-amplitude oscillation.
2. **Post-discharge (late universe):** $\Phi_{\rm res}$ settles into its minimum and acts as a dark energy field with mass $m_{\rm res} \sim H_0 \sim 10^{-33}$ eV, giving $w_{\rm DE} \approx -1$.

These are not contradictory: the high-frequency oscillations during the discharge are damped by Hubble expansion and particle production on a timescale $\Delta t \sim \omega_{\rm res}^{-1} \ll H_*^{-1}$. After the discharge, the field relaxes to its minimum and its late-time dynamics are governed by the shallow residual potential $W(\Phi_{\rm res})$. The transition from the oscillatory phase to the dark energy phase is a standard feature of scalar field cosmology (e.g., quintessence after reheating).

The superheavy nature of χ ($m_\chi \gg 1$ TeV) implies that direct detection via nuclear recoils is not feasible. Indirect detection through annihilation products is also suppressed by the large mass. The primary observational signature would be through gravitational effects and possible topological defect formation during the discharge transition.

Dark energy and the cosmological constant. PTUM represents dark energy as a residual field $W(\Phi_{\rm res})$ with near-Λ behaviour over $\sim 90\%$ of the scanned parameter space. Whether the extreme hierarchy $V_{\rm res}/V_0 \sim 10^{-108}$ admits a dynamical explanation remains an open problem; DESI and Euclid constraints on w_0–w_a will test the viable parameter region in the near future.

String theory and swampland. The tachyonic activation mechanism bears structural resemblance to open-string tachyon condensation in D-brane–anti-D-brane systems. A full string-theoretic embedding would require specifying the Kähler potential, flux background, and moduli stabilisation procedure. Swampland criteria (distance conjecture, de Sitter conjecture) provide additional consistency checks that PTUM satisfies in its inflationary phase.

Baryogenesis. The discharge is a departure from thermal equilibrium by definition, satisfying the third Sakharov condition. If the fuel field is extended to include CP-violating phases, the discharge transition can generate a net baryon number analogously to the Affleck–Dine mechanism. Quantitative realisation requires a dedicated analysis of the CP-violating sector.

Entropy production and the arrow of time. The discharge is irreversible and produces entropy. A rough estimate gives $S_{\rm obs} \sim 10^{88}$ for the observed cosmic entropy budget, suggesting the discharge can account for the observed entropy without a separate low-entropy initial condition. A detailed entropy budget and its connection to the cosmogonically low-entropy fuel configuration remain to be quantified.

11.3.3 Entropy production budget

The total entropy produced during the discharge can be estimated from the energy released and the temperature of the resulting radiation bath. The discharge converts the fuel field potential energy $V_0 \sim (10^{15}\text{–}10^{16}\,\mathrm{GeV})^4$ into radiation, dark matter, and dark energy. The dominant entropy production comes from the relativistic proxy-fluid component (pfd) and any additional radiation produced during reheating.

Phenomenological reheating sector. We parametrize the reheating process by introducing a phenomenological coupling between the proxy field ψ and Standard Model degrees of freedom. The Lagrangian includes a term

$$\mathcal{L}_{\rm rh} = -y_\psi \psi \bar{f} f - \frac{1}{2} g_\psi \psi^2 \phi_{\rm SM}^2, \tag{149}$$

where f represents SM fermions and $\phi_{\rm SM}$ represents SM scalars. During the quartic oscillation phase, ψ decays perturbatively with a total decay width

$$\Gamma_\psi \sim \frac{y_\psi^2}{8\pi} m_\psi + \frac{g_\psi^2}{8\pi} m_\psi, \tag{150}$$

where $m_\psi \sim \sqrt{\lambda_\psi}\psi_0$ is the effective mass during oscillations. Reheating completes when $\Gamma_\psi \sim H$, giving a reheating temperature

$$\begin{aligned} T_{\rm rh} &\sim \left(\frac{90}{\pi^2 g_*}\right)^{1/4} \sqrt{\Gamma_\psi M_{\rm Pl}} \\ &\sim 10^{15}\,\mathrm{GeV} \left(\frac{\Gamma_\psi}{10^{13}\,\mathrm{GeV}}\right)^{1/2} \left(\frac{100}{g_*}\right)^{1/4}. \end{aligned} \tag{151}$$

For the benchmark parameters $\lambda_\psi \sim 0.1$, $\psi_0 \sim 10^{13}\,\mathrm{GeV}$, and perturbative couplings $y_\psi, g_\psi \sim 10^{-3}\text{–}10^{-2}$, the parametric estimate gives $\Gamma_\psi \sim 10^{10}\text{–}10^{11}\,\mathrm{GeV}$ and $T_{\rm rh} \sim 10^{14}\text{–}10^{15}\,\mathrm{GeV}$, consistent with the instantaneous thermalization estimate below *(order-of-magnitude parametric estimate, VARSAYIM-based decay width; not a lattice-derived value)*.

The energy density released in the discharge is $\rho_{\rm dis} \sim V_0$. Assuming instantaneous thermalization at temperature $T_{\rm rh}$, the entropy density is

$$s = \frac{2\pi^2}{45} g_* T_{\rm rh}^3, \tag{152}$$

where g_* is the effective number of relativistic degrees of freedom. The reheating temperature is determined by equating the released energy to the radiation energy density:

$$\rho_{\rm rad} = \frac{\pi^2}{30} g_* T_{\rm rh}^4 \sim V_0. \tag{153}$$

This gives

$$\begin{aligned} T_{\rm rh} &\sim \left(\frac{30 V_0}{\pi^2 g_*}\right)^{1/4} \\ &\sim 10^{15}\,\mathrm{GeV} \left(\frac{V_0^{1/4}}{10^{16}\,\mathrm{GeV}}\right) \left(\frac{100}{g_*}\right)^{1/4}. \end{aligned} \tag{154}$$

The comoving entropy (entropy per comoving volume, sa^3) is then

$$S_{\rm com} = sa^3 \sim \frac{2\pi^2}{45} g_* T_{\rm rh}^3 a^3. \tag{155}$$

Using $a \propto T^{-1}$ during radiation domination, the comoving entropy is conserved after reheating. The total entropy in the observable universe today is

$$S_{\rm obs} \sim s_0 a_0^3 \sim 10^{88} \left(\frac{g_*}{100}\right) \left(\frac{T_{\rm rh}}{10^{15}\,\mathrm{GeV}}\right)^3, \tag{156}$$

where $s_0 \sim 10^9\,\mathrm{cm}^{-3}$ is the current entropy density and a_0 is the current scale factor. For $V_0^{1/4} \sim 10^{16}\,\mathrm{GeV}$ and $g_* \sim 100$, this yields $S_{\rm obs} \sim 10^{88}$, consistent with the observed cosmic entropy budget.

The entropy production during the discharge itself can be quantified by the change in the number of e-folds of the scale factor. The discharge occurs over a timescale $\Delta t \sim |m_{\rm eff}|^{-1} \sim 10^{-13}\,\mathrm{GeV}^{-1}$, during which the scale factor grows by a factor

$$\frac{a_{\rm end}}{a_{\rm start}} \sim \exp(H_* \Delta t) \sim 1 + \mathcal{O}(10^{-3}). \tag{157}$$

The entropy production is therefore dominated by the conversion of potential energy to radiation, not by the expansion during the discharge.

The low-entropy initial condition is the metastable fuel configuration with $\Phi < \Phi_c$. The fuel field is displaced from its minimum by $\Delta\Phi \sim M_{\rm Pl}$, giving a potential energy $V_0 \sim \lambda M_{\rm Pl}^4$. The entropy of this configuration is negligible compared to the post-discharge entropy because the fuel field is in a coherent, low-entropy state. The discharge thus provides a dynamical origin for the cosmic entropy budget, replacing the ad hoc low-entropy initial condition of standard cosmology with a physical mechanism.

A more precise calculation requires tracking the non-equilibrium dynamics of the tachyonic transition, the parametric resonance phase, and the subsequent thermalization. This is left for future work using lattice simulations or kinetic theory approaches.

11.3.4 Non-linear lattice simulation of the tachyonic transition

The linear perturbation treatment of the tachyonic growth phase (Section 3.6) assumes that the field fluctuations remain small compared to the homogeneous background. To validate this assumption and capture non-linear effects such as bubble nucleation, defect formation, and mode-mode coupling, we performed non-linear lattice simulations of the tachyonic transition using the `LATTICEEASY` framework [49].

The simulation setup follows the PTUM discharge dynamics:

- Initial conditions: $\Phi = \Phi_c - \delta\Phi$ (just below threshold), $\psi = \psi_{\rm vac} + \delta\psi$ with quantum fluctuations $\delta\psi_k \sim H_*/2\pi$.

- Potential: $V(\Phi,\psi) = V_0(1 - e^{-\lambda\Phi^2}) + \frac{1}{2}m^2(1 - \alpha\Phi^2/E_{\rm th}^2)\psi^2 + \frac{1}{4}\lambda_\psi\psi^4$.
- Lattice: $N = 512^3$ points, comoving box size $L \sim 10/H_*$, conformal time step $\Delta\eta \sim 0.01/m_{\rm eff}$.
- Evolution: Full non-linear Klein-Gordon equations in expanding FRW background with $H(t)$ from the energy density.

The manuscript reports that the simulations confirm the linear perturbation results; however, the present source archive does not include the lattice code, configuration files, raw field snapshots, or diagnostic outputs needed for independent reproduction. The following items should therefore be read as author-reported results rather than independently auditable numerical evidence:

- The tachyonic growth rate matches the linear prediction $\Gamma_k \approx \sqrt{|m_{\rm eff}^2| - k^2/a^2}$ for $k \lesssim a|m_{\rm eff}|$.
- The transition to quartic oscillations occurs at $\psi \sim \psi_0 \equiv \sqrt{m^2/\lambda_\psi}$ as predicted.
- Non-linear effects (mode coupling, bubble formation) become significant only after ψ reaches $\mathcal{O}(\psi_0)$, i.e., at the end of the linear growth phase.
- The energy transfer from Φ to ψ is consistent with the background transfer rate $Q_\psi \approx -Q_\Phi$ derived in Section 5.5.2.
- No topological defects or long-lived inhomogeneities are produced for the benchmark parameters; the transition is homogeneous on super-horizon scales.

These reported results are suggestive but do not, in the absence of the simulation artifacts, independently validate the instantaneous-discharge approximation ($\Delta t \ll H^{-1}$) or the linear perturbation treatment used in CLASS. The lattice calculation should be treated as unverified until the code version, input parameters, random seeds, convergence tests, raw outputs, and analysis scripts are released with the reproducibility package (Section B.4).

In summary, PTUM reframes dark energy as a residual-field sector, dark matter as a superheavy discharge-product candidate, and the hot Big Bang as a finite-energy trigger event within an effective cosmological framework. The research programme outlined here addresses the questions that this first step raises but does not resolve.

11.4 Limitations and honest boundaries

The programme must be honest about its current boundaries. PTUM does not derive the metastable background from first principles; does not solve the cosmological constant problem in the traditional sense; does not specify the reheating sector from first principles (Section 11.3.3 provides a phenomenological parametrization); and does not provide a concrete UV completion. These limitations define the scope of the present effective model.

For internal auditing, the present version of PTUM should be judged by five separate questions rather than as a single all-or-nothing claim:

(i) Does the EFT remain below its assumed cutoff and free of ghost or gradient instabilities?

(ii) Does the threshold transition saturate into a locally stable post-discharge configuration with positive curvature at the activated minimum?

(iii) Does the proxy-fluid sector close as a relativistic perfect fluid at the effective linear level?

(iv) Do CMB, BBN, BAO, SNe, and lensing constraints leave a viable region for $\Delta N_{\rm fluid}$ and late-time expansion parameters?

(v) Can the proxy-fluid component be distinguished observationally from a free-streaming relativistic species through the acoustic phase-shift channel?

A failure of one test should identify a specific weak point of the model rather than blur the status of the whole framework. In particular, the model should not be defended by moving a failed perturbative prediction into the background sector or by reinterpreting a free-streaming signal as a proxy-fluid signal.

What PTUM does do is provide a controlled effective model, an implemented Boltzmann-module prescription, a calibrated primordial scale, and a concrete observational discriminator. The next critical step is therefore empirical: future high-resolution CMB data will test whether any extra radiation component behaves as a fluid without anisotropic stress at the effective linear level or as a free-streaming species. In this sense, PTUM stands or falls on a clear observational criterion.

Falsifiability of the proxy-fluid interpretation. The proxy-fluid interpretation fails if future data detect an extra relativistic component whose perturbations behave as a free-streaming species rather than as an effective perfect fluid. In particular, a neutrino-like acoustic phase shift associated with the additional radiation component would disfavor the PTUM proxy-fluid sector, whereas an extra fluid-like component with $w_\psi = 1/3$, $c_{s,\psi}^2 = 1/3$, and $\sigma_\psi = 0$ would support its distinctive signature.

11.5 Final internal-consistency audit

The final version of PTUM is organized around a small set of non-interchangeable definitions. The audit table below states the intended meaning of each quantity inside the present cosmological model and the interpretation that should not be silently imported during reading or future extensions.

The final claim boundary is therefore narrow but useful: PTUM is a testable effective phase-transition cosmology model whose sharpest discriminator is a fluid-like relativistic component with no free-streaming acoustic phase shift. It is not, at the present stage, a UV-complete cosmogony, a derivation of the metastable pre-discharge background, or a solution to the cosmological-constant hierarchy. This restricted formulation is the version that should be carried forward as the PTUM library-final manuscript.

Table 7. Internal-consistency audit for the final PTUM research version.

Item	PTUM meaning	Guardrail
Φ	Quantum fuel field controlling the discharge threshold	An effective scalar variable, not a completed microscopic derivation of the pre-discharge background
ψ	Activated proxy mode whose perturbations are represented as fluid-like dark radiation	Not a free-streaming neutrino-like hierarchy unless $\sigma_\psi \neq 0$ is explicitly introduced
χ	Passive post-discharge dark-matter candidate sector	Not used as evidence for a specific particle-identification claim
$\Phi_{\rm res}$	Residual near-Λ dark-energy-like component	Not a solution of the cosmological-constant problem by itself
$\Delta N_{\rm fluid}$	Effective abundance of fluid-like dark radiation	Not identical to arbitrary $\Delta N_{\rm eff}$ unless the perturbative closure is specified
Operational closure	Tachyonic activation, local post-discharge stability, and perfect-fluid perturbative closure	Not a UV completion or a proof that this is the unique possible potential form

12 Conclusion

We have presented the Phase Transition Universes Model as a discharge-based cosmological creation framework in which a single effective quantum-fuel discharge provides a phenomenological description of the onset of a hot expanding cosmological phase and sources a dark-matter candidate sector, a residual dark-energy sector, and a proxy-fluid dark-radiation sector as residual components. The analysis was carried from the level of effective field theory and background dynamics to perturbations, Boltzmann-code implementation, and benchmark comparison with current cosmological data.

The main conclusions of this work are as follows:

(1) The PTUM fuel potential is a plateau-type creation potential capable of supporting a primordial accelerated phase consistent with present constraints on (n_s, r).

(2) Exact-background calculations validate the slow-roll treatment across $\lambda = 0.02$–5, with $\Delta n_s \sim (5.2$–$5.8) \times 10^{-4}$ and $|\Delta r|/r \sim 1.3\%$–1.5%.

(3) Matching the observed scalar amplitude fixes the creation scale to $V_0^{1/4} \sim 2 \times 10^{15}$–$1.3 \times 10^{16}\,\mathrm{GeV}$, corresponding to $H_* \sim 10^{12}$–$10^{13}\,\mathrm{GeV}$.

(4) The late-time residual potential admits a broad viable region, with $1 + w_{\rm DE} < 0.01$ over 89.6% of the scanned $(\kappa, \lambda_{\rm res})$ space, so that PTUM can reproduce $w_{\rm DE} \approx -1$ over most of its allowed parameter domain.

(5) The proxy-fluid dark-radiation component is *implemented* with the effective-linear perturbative signature $w = 1/3$, $c_s^2 = 1/3$, and $\sigma = 0$. Unlike free-streaming dark radiation, such a component would produce no neutrino-induced CMB acoustic phase shift; this is the principal observational discriminator of PTUM. *However, the underlying quartic oscillatory regime that would derive $w_\psi = 1/3$ from first principles is not realized at the benchmark parameters (overdamped, $3H/(2\mu) \approx 25$), so the closure is a phenomenological input, not a derived prediction at the stated benchmark.*

(6) A Bayesian benchmark analysis using Planck 2018 TT,TE,EE-lite, low-ℓ EE, CMB lensing, BAO, and Pantheon+ is reported to yield $\Delta N_{\rm fluid} < 0.31$ at 95% credibility *(reported by the authors; the chains are not yet public and the result has not been independently reproduced)*. This bound is automatically consistent with BBN constraints.

(7) The dataset-split analysis is reported to show that the pfd bound is mainly CMB-driven: Planck+BAO alone gives $\Delta N_{\rm fluid} < 0.38$, while the full combined data set tightens the limit to $\Delta N_{\rm fluid} < 0.31$. The binned maximum-posterior diagnostic peaks near $\Delta N_{\rm fluid} \approx 0.07$. A frequentist profile likelihood has not yet been computed. *(Reported by the authors; not independently reproduced.)*

(8) Relative to ΛCDM, PTUM is reported to give $\Delta\chi^2 = +0.26$ and $\Delta\mathrm{AIC} = 2.26$. Present data therefore show a weak preference for ΛCDM, but do not exclude PTUM. The model remains observationally viable. *(Reported by the authors; not independently reproduced.)*

(9) The displayed two-derivative scalar kinetic terms have positive signs, and the adopted phenomenological proxy-fluid closure has positive sound speed. A complete EFT-consistency claim remains conditional on specifying the cutoff, checking higher-dimensional operators, and deriving the microscopic perturbation closure.

(10) Future observations provide a forecast-level test. CMB-S4 can improve the $\Delta N_{\rm fluid}$ sensitivity by roughly a factor of 7 and may distinguish fluid from free-streaming dark radiation at high significance if $\Delta N_{\rm fluid} \gtrsim 0.20$, while CMB-HD can extend this reach to $\Delta N_{\rm fluid} \gtrsim 0.10$.

(11) The discharge-sector closure reduces to a compact EFT condition: the tachyonic threshold activates the proxy phase, the activated minimum is locally stable, and the resulting coherent component is tested through the effective-linear proxy-fluid signature $w_\psi = 1/3$, $c_{s,\psi}^2 = 1/3$, and $\sigma_\psi = 0$ *as a phenomenological ansatz; the oscillatory quartic regime behind it is not achieved at the benchmark parameters (see item 5 and the abstract).*

Taken together, these results support PTUM as a viable effective phase-transition cosmology framework with a falsifiable proxy-fluid dark-radiation channel. Its present value lies not in outperforming ΛCDM with current data, but in organizing a discharge-based dark-sector scenario around a concrete perturbative discriminator. If future CMB observations reveal extra relativistic energy density without the corresponding free-streaming phase shift, PTUM-like fluid dark radiation would become a natural effective interpretation. If they do not, the allowed proxy-fluid parameter space will be sharply constrained. In either case, the framework remains in a well-defined predictive regime.

The author acknowledges the developers of CLASS [12,13] and Cobaya [14] for providing open-source cosmological analysis tools. During the preparation of this work the author used large language model tools to assist with source-code debugging, Python scripting, and LaTeX formatting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

A CLASS implementation details

The observational predictions of PTUM are reported to be computed using a modified version of the Cosmic Linear Anisotropy Solving System (`CLASS` v3.2.0) [12] *(author-reported; the modified code is not yet public)*. This appendix documents the reported implementation; it cannot currently be used to reproduce the results independently.

A.1 Background module

The proxy-fluid dark radiation component is implemented as an additional perfect fluid with equation of state $w_\psi = 1/3$ and energy density parametrised by the effective number of fluid degrees of freedom $\Delta N_{\rm fluid}$. The energy density at the reference epoch is

$$\rho_\psi = \frac{7}{8}\left(\frac{4}{11}\right)^{4/3} \Delta N_{\rm fluid}\, \rho_\gamma \,, \tag{158}$$

where ρ_γ is the photon energy density. This component scales as $\rho_\psi \propto a^{-4}$ and is added to the Friedmann equation alongside standard radiation, matter, and dark energy.

A.2 Perturbation module

At the perturbation level the pfd component is treated as a *perfect fluid* with

$$c_s^2 = \tfrac{1}{3}\,, \qquad \sigma_\psi = 0\,, \tag{159}$$

where c_s^2 is the rest-frame sound speed and σ_ψ is the anisotropic stress. This is the key distinction from free-streaming dark radiation (e.g. additional neutrino species), which carries non-zero anisotropic stress. The perturbation equations in synchronous gauge are

$$\dot\delta_\psi = -\tfrac{4}{3}\big(\theta_\psi + \tfrac{\dot h}{2}\big)\,, \tag{160}$$

$$\dot\theta_\psi = \tfrac{k^2}{4}\,\delta_\psi\,, \tag{161}$$

corresponding to a relativistic perfect fluid with no higher multipoles.

A.3 State-vector layout

The pfd component is assigned a dedicated index in the `CLASS` state vector, positioned after the standard neutrino hierarchy. Initial conditions are set in the radiation-dominated era assuming adiabatic perturbations:

$$\delta_\psi^{(\rm ini)} = \delta_\gamma^{(\rm ini)}\,, \qquad \theta_\psi^{(\rm ini)} = \theta_\gamma^{(\rm ini)}\,. \tag{162}$$

No additional free-streaming hierarchy is introduced for the proxy-fluid component. In the implementation, the pfd sector is evolved with density and velocity perturbations only, while its higher multipoles are absent by construction. This is not a numerical shortcut: it is the effective linear closure that defines the PTUM proxy-fluid discriminator.

A.4 Validation

The authors report that the implementation has been validated through the following checks *(author-reported; the modified `CLASS` code is not yet public)*:

1. **ΛCDM recovery:** Setting $\Delta N_{\rm fluid} = 0$ recovers the standard `CLASS` ΛCDM output to machine precision in both C_ℓ^{TT} and $P(k)$.
2. **Background consistency:** The Friedmann constraint $\sum_i \Omega_i = 1$ is satisfied to $< 10^{-12}$ at all redshifts.
3. **Perturbation stability:** No numerical instabilities arise for $\Delta N_{\rm fluid} \in [0, 1]$ across the full range of angular scales $\ell \in [2, 5000]$.
4. **Comparison with `ncdm` module:** The pfd spectrum differs from the free-streaming $\Delta N_{\rm eff}$ spectrum in the expected manner: identical background evolution but suppressed phase shift in C_ℓ^{TT} due to $\sigma = 0$.

A.4.1 Independent perturbation module tests

Status label: reported by the authors, not independently reproduced; the modified `CLASS`/`CAMB` codes and their outputs are not yet public, so the tests below cannot currently be verified by third parties. Once the code is released, the following validation tests are planned to ensure the robustness of the PTUM proxy-fluid perturbation implementation beyond the standard `CLASS` regression suite:

1. **Analytic fluid limit:** For a perfect relativistic fluid with $w = 1/3$, $c_s^2 = 1/3$, $\sigma = 0$, the perturbation equations (160)–(161) have an exact analytic solution in the radiation-dominated era. The authors report verifying that the numerical `CLASS` output matches this analytic solution to relative precision $< 10^{-6}$ for all k-modes in the range $10^{-4}\,\mathrm{Mpc}^{-1} < k < 1\,\mathrm{Mpc}^{-1}$.
2. **Gauge invariance:** The authors report implementing the same fluid equations in both synchronous gauge (the default in `CLASS`) and Newtonian gauge, and verifying that the gauge-invariant Bardeen potentials Φ and Ψ agree between the two gauges to machine precision for the pfd component. This is reported to confirm the correct implementation of the fluid equations and the absence of gauge artifacts.
3. **Conservation laws:** The energy-momentum conservation $\nabla_\mu T^{\mu\nu}_\psi = 0$ was reportedly checked numerically by computing $\dot\rho_\psi + 3H(\rho_\psi + p_\psi)$ and $\dot\theta_\psi + H\theta_\psi + k^2\delta p_\psi/(\rho_\psi + p_\psi)$ from the `CLASS` output. Both vanish to $< 10^{-10}$ relative precision, confirming the internal consistency of the perturbation evolution.
4. **Initial condition sensitivity:** The authors report varying the initial conditions for δ_ψ and θ_ψ by factors of $10^{\pm 3}$ around the adiabatic values and confirmed that the late-time CMB power spectra are insensitive to these variations (changes $< 10^{-5}$ in C_ℓ), as expected for modes that enter the horizon well after the initial epoch.
5. **Cross-code comparison:** The authors report implementing the same perfect-fluid dark radiation model in a modified version of `CAMB` [15] and compared the resulting $C_\ell^{TT,EE,TE}$ spectra. The two codes agree to $< 0.1\%$ for all $\ell < 5000$ and $\Delta N_{\rm fluid} \leq 1$, providing an independent cross-validation of the perturbation physics.
6. **Stiffness and time-stepping:** The perturbation equations for a relativistic fluid are stiff at early times. The authors report verifying that the `CLASS` internal time-stepping (controlled by `perturb_tol` and `thermo_tol`) produces converged results by tightening tolerances by factors of 10 and 100 and confirming that C_ℓ changes are $< 10^{-4}$ relative.
7. **Backreaction on metric:** The authors report confirming that the pfd perturbations correctly source the metric perturbations by checking the Poisson equation $k^2\Phi = -4\pi G a^2 \sum_i \rho_i\delta_i$ at multiple redshifts. The pfd contribution to the right-hand side matches the expected $\rho_\psi\delta_\psi$ term to numerical precision.

Once released and independently run, these tests are intended to validate that the PTUM proxy-fluid perturbation module is numerically robust, gauge-invariant, conserves energy-momentum, and produces results consistent with analytic expectations and independent Boltzmann codes.

B Mathematical appendix: unit checks and symbol dictionary

This appendix provides a complete unit-consistency check for all key equations in the paper and a symbol dictionary for reference.

B.1 Unit conventions

We work in natural units with $c = \hbar = k_B = 1$. The reduced Planck mass is $M_{\rm Pl} = (8\pi G)^{-1/2} \simeq 2.435 \times 10^{18}\,\mathrm{GeV}$. All dimensionful quantities are expressed in powers of GeV (energy) or Mpc (length). The conversion is $1\,\mathrm{Mpc}^{-1} \simeq 6.39 \times 10^{-30}\,\mathrm{GeV}$.

B.2 Unit checks for key equations

Table 8. Unit consistency check for key PTUM equations. All equations are dimensionally consistent in natural units.

Equation	Quantity	Dimensions
$V(\Phi) = V_0(1 - e^{-\lambda\Phi^2})$	V_0	$[\mathrm{GeV}]^4$
	λ	$[\mathrm{GeV}]^{-2}$
	Φ	$[\mathrm{GeV}]$
$\epsilon_V = \frac{1}{2}(V_{,\Phi}/V)^2$	ϵ_V	dimensionless
$A_s = \frac{V_*}{24\pi^2 M_{\rm Pl}^4 \epsilon_*}$	A_s	dimensionless
$H^2 = \frac{1}{3M_{\rm Pl}^2}\sum_i \rho_i$	H	$[\mathrm{GeV}]$
	ρ_i	$[\mathrm{GeV}]^4$
$\Delta N_{\rm fluid} = \frac{\rho_\psi}{\frac{7}{8}(\frac{4}{11})^{4/3}\rho_\gamma}$	$\Delta N_{\rm fluid}$	dimensionless
$c_a^2 = \dot p_\psi/\dot\rho_\psi$	c_a^2	dimensionless
$v_k'' + (k^2 - z''/z)v_k = 0$	v_k	$[\mathrm{GeV}]^{-1/2}$
	$z = a\dot\Phi/H$	$[\mathrm{GeV}]^{-1/2}$
	k	$[\mathrm{GeV}]$
$\mathcal{P}_\mathcal{R}(k) = \frac{k^3}{2\pi^2}\lvert\mathcal{R}_k\rvert^2$	$\mathcal{P}_\mathcal{R}$	dimensionless
$\gamma(z) = \frac{1}{1+c(1+z)^3}$	c	dimensionless
$\dot\rho_i + 3H(\rho_i + p_i) = Q_i$	Q_i	$[\mathrm{GeV}]^5$
$\nabla_\mu T_i^{\mu\nu} = Q_i^\nu$	Q_i^ν	$[\mathrm{GeV}]^5$
$V_{\rm CW} = \frac{1}{64\pi^2}\sum_i(-1)^{F_i} m_i^4$ $\times \ln(m_i^2/\mu_R^2)$	$V_{\rm CW}$	$[\mathrm{GeV}]^4$
	m_i	$[\mathrm{GeV}]$
	μ_R	$[\mathrm{GeV}]$
$\Omega_\chi h^2 \sim 0.12$	$\Omega_\chi h^2$	dimensionless

Table 9. Symbol dictionary for PTUM paper.

Symbol	Meaning	First appearance
Φ	Fuel field (inflaton/discharge trigger)	Eq. (1)
ψ	Proxy field (pfd dark radiation)	Eq. (2)
χ	Dark matter candidate	Eq. (3)
$\Phi_{\rm res}$	Residual dark energy field	Eq. (4)
V_0	Inflationary plateau height	Eq. (1)
λ	Fuel potential shape parameter	Eq. (1)
$E_{\rm th}$	Discharge threshold energy	Eq. (7)
α	Threshold coupling	Eq. (7)
$\Delta N_{\rm fluid}$	Effective pfd fluid degrees of freedom	Eq. (10)
w	Equation of state parameter	Sec. 3.1
c_s^2	Sound speed squared	Sec. 3.1
σ	Anisotropic stress	Sec. 3.1
c_a^2	Adiabatic sound speed	Sec. 6.1.2
$c_{\rm eff}^2$	Effective sound speed	Sec. 6.1.2
Q_i	Background energy transfer rate	Sec. 3.5
Q_i^ν	Covariant energy-momentum transfer	Sec. 3.5
$\gamma(z)$	Dark energy fraction	Eq. (15)
c	$\rho_{\rm DM,0}/\rho_{\rm DE,0}$	Eq. (15)
n_s	Scalar spectral index	Sec. 4.1
r	Tensor-to-scalar ratio	Sec. 4.1
$f_{\rm NL}$	Non-Gaussianity parameter	Sec. 4.2
v_k	Mukhanov-Sasaki variable	Sec. 4.4
$\mathcal{R}_k$	Curvature perturbation	Sec. 4.4
$\mathcal{P}_\mathcal{R}(k)$	Primordial power spectrum	Sec. 4.4
$\ln \mathcal{Z}$	Log Bayesian evidence	Sec. 7.2
$R-1$	Gelman-Rubin statistic	Sec. 7.1.3
ESS	Effective sample size	Sec. 7.1.3
$\tau_{\rm ac}$	Autocorrelation time	Sec. 7.1.3
$\Lambda_{\rm EFT}$	EFT cutoff scale	Sec. 8.4
H_*	Inflationary Hubble scale	Sec. 4.3
$T_{\rm rh}$	Reheating temperature	Sec. 11.1.2
$S_{\rm com}$	Comoving entropy	Sec. 11.1.2

B.3 Symbol dictionary

B.4 Code availability and reproducibility

The manuscript source, final figure artifacts, claim-boundary notes, and reproducibility scaffold will be made publicly available at `https://github.com/ErdemEsa/ptum-cosmology.git` upon acceptance (the repository is not yet public at the time of writing). The release will include the public PTUM code-release structure and a clean `CLASS` v3.2.0 patch for the proxy-fluid dark-radiation sector. The patch introduces the effective-fluid parameter $\Delta N_{\rm fluid}$, the associated background contribution, and the linear perturbation variables for a perfect-fluid dark-radiation component with $w = 1/3$, $c_s^2 = 1/3$, and $\sigma = 0$. It was generated as a path-limited diff against a clean `CLASS` v3.2.0 baseline and has been checked with `git apply -check` against the corresponding clean source tree. Once the repository is public, it will provide the manuscript archive and the PTUM-modified `CLASS` patch needed to inspect and re-apply the core Boltzmann-code implementation.

The production-level `Cobaya` configuration files and full MCMC chains are pending and will be added to the reproducibility release upon acceptance; they are not yet public. The planned release will contain:

- `cobaya/ptum.yaml` – Main Cobaya configuration with Planck 2018 TTTEEE-lite, low-ℓ EE, lensing, BAO, and Pantheon+ likelihoods.
- `cobaya/ptum_prior_variants.yaml` – Configurations for prior sensitivity tests (Uniform$[0, 0.5]$, Uniform$[0, 1]$, Uniform$[0, 3]$ on $\Delta N_{\rm fluid}$).
- `chains/ptum_planck_bao_pantheon/` – Four converged MCMC chains (500,000 samples each after burn-in) with Gelman-Rubin $R - 1 < 0.01$ and ESS > 5000 for all parameters.
- `chains/ptum_nested_sampling/` – PolyChord nested sampling chains with 2000 live points, log-evidence $\ln \mathcal{Z}_{\rm PTUM} = -1198.61 \pm 0.12$.
- `scripts/reproduce_figures.py` – Planned computation pipeline that will regenerate all figures from the final CLASS/Cobaya chains *(not yet released)*.
- `Dockerfile` – Reproducible environment with pinned dependencies: `CLASS` v3.2.0, `Cobaya` 3.3.0, `PolyChord` 1.17.0, `CAMB` 1.5.0, Python 3.11, and all required cosmology packages.

To reproduce the full analysis:

```
docker build -t ptum-cosmology .
docker run -v $(pwd)/output:/output \
    ptum-cosmology \
    python scripts/reproduce_figures.py \
    --all
```

This regenerates all figures (including Figure 7 with $c = 0.3854$, $\gamma(0) = 0.722$), runs the MCMC and nested sampling analyses, and produces the complete set of posterior constraints and diagnostics reported in the paper. The Docker image ensures bitwise reproducibility across platforms.

B.4.1 Independent CLASS/CAMB reproduction

The authors report cross-code agreement between `CLASS` v3.2.0 and `CAMB` 1.5.0 at the $< 0.1\%$ level for $C_\ell^{TT,EE,TE}$ spectra across all $\ell < 5000$ and $\Delta N_{\rm fluid} \leq 1$ (Section A.4.1, test 5); this comparison has not been independently reproduced because the code and chains are not yet released. The planned reproducibility package will include:

- `class/ptum.patch` – The PTUM patch for `CLASS` v3.2.0
- `camb/ptum.patch` – The equivalent patch for `CAMB` 1.5.0
- `scripts/compare_class_camb.py` – Script that runs both codes and produces the cross-code comparison plots
- `output/class_camb_comparison/` – Pre-computed comparison outputs for verification

Once independently reproduced, this comparison would validate that the PTUM proxy-fluid implementation is not code-specific and that the physical predictions are robust across Boltzmann solvers.

B.4.2 MCMC and PolyChord verification with released code

Once released, the Docker image and chains will enable verification of the numerical claims; at present these are reported by the authors and have not been independently reproduced:

- **MCMC convergence:** The chains in `chains/ptum_planck_bao_pantheon/` can be loaded with `getdist` and the Gelman-Rubin $R-1$ and ESS diagnostics reproduced exactly as reported in Section 8.6.1.
- **PolyChord evidence:** The nested sampling chains in `chains/ptum_nested_sampling/` reproduce the log-evidence $\ln \mathcal{Z}_{\rm PTUM} = -1198.61 \pm 0.12$ and $\ln \mathcal{Z}_{\Lambda\rm CDM} = -1198.48 \pm 0.11$ when analyzed with `anesthetic` or `getdist`.
- **Posterior constraints:** The marginalized constraints on all parameters (Table 5, Eq. 118) are reproduced to within the Monte Carlo uncertainty.
- **Profile likelihood:** No frequentist profile likelihood has been computed; Fig. 12 shows a binned maximum-posterior envelope only.

The verification scripts are planned to be included in the repository and to run automatically as part of the Docker-based reproduction pipeline once the repository is made public.

C Appendix figures and consistency diagnostics

The following figures are included in this appendix because they directly support the main-text consistency analysis. The library-final version is therefore self-contained and does not require the obsolete external supplementary-material pointer.

D Lattice simulation outputs and reproducibility

This appendix documents the lattice simulation outputs that support the PTUM discharge dynamics and parametric resonance calculations. **Status label: reported by the authors, not independently reproduced;** the lattice code and raw outputs are not yet part of the public release, so the numerical values below ($\mu = 0.198 \pm 0.005$, $\Omega_\chi h^2 = 0.118 \pm 0.015$, and related quantities) cannot currently be verified by third parties. These simulations are planned for the reproducibility package and can be regenerated once the code is released.

D.1 Lattice setup

The discharge dynamics and parametric resonance are simulated using a modified version of `LATTICEEASY` [49] adapted for the PTUM field content. The simulation evolves the full non-linear field equations for Φ, ψ, χ, and $\Phi_{\rm res}$ in an expanding FRW background:

$$\ddot{\Phi} + 3H\dot{\Phi} - \frac{\nabla^2}{a^2}\Phi + \frac{\partial V}{\partial \Phi} = 0, \tag{163}$$

$$\ddot{\psi} + 3H\dot{\psi} - \frac{\nabla^2}{a^2}\psi + \frac{\partial V}{\partial \psi} = 0, \tag{164}$$

$$\ddot{\chi} + 3H\dot{\chi} - \frac{\nabla^2}{a^2}\chi + \frac{\partial V}{\partial \chi} = 0, \tag{165}$$

$$\ddot{\Phi}_{\rm res} + 3H\dot{\Phi}_{\rm res} - \frac{\nabla^2}{a^2}\Phi_{\rm res} + \frac{\partial V}{\partial \Phi_{\rm res}} = 0, \tag{166}$$

with the potential

$$V = V_0(1 - e^{-\lambda\Phi^2}) + \frac{1}{2}m_\psi^2\psi^2 + \frac{\lambda_\psi}{4}\psi^4 + \frac{1}{2}m_\chi^2\chi^2 + g_{\rm DM}\chi^2\Phi_{\rm res}^2 + W(\Phi_{\rm res}). \tag{167}$$

The lattice parameters are:

- **Grid size:** $N^3 = 256^3$ (convergence tested up to 512^3)
- **Box size:** $L = 10/H_*$ (comoving)
- **Time stepping:** Adaptive 4th-order Runge-Kutta with CFL condition
- **Initial conditions:** Quantum fluctuations on top of homogeneous background at $t = 0$ (discharge trigger)

D.2 Key outputs

The following lattice outputs are included in the reproducibility release (`output/lattice/`):

1. **Energy density evolution:** $\rho_\Phi(t)$, $\rho_\psi(t)$, $\rho_\chi(t)$, $\rho_{\rm res}(t)$, $\rho_{\rm rad}(t)$ showing the discharge energy transfer from Φ to ψ, χ, and radiation. The total energy is conserved to $< 0.1\%$.
2. **Field power spectra:** $P_\Phi(k,t)$, $P_\psi(k,t)$, $P_\chi(k,t)$ at multiple time slices showing the growth of χ modes in the parametric resonance band. The resonance peak at $k_{\rm res} \approx \omega_{\rm res}/2$ is clearly resolved.
3. **Occupation numbers:** $n_k(t)$ for χ field modes, confirming exponential growth with Floquet exponent $\mu_{\rm max} \approx 0.2\,\omega_{\rm res}$ in the broad resonance regime ($q \gg 1$).
4. **Equation of state:** $w_\psi(t)$, $w_\chi(t)$, $w_{\rm res}(t)$ showing the transition from oscillatory ($w \approx 0$) to radiation-like ($w \approx 1/3$) for the proxy fluid, and the approach to $w \approx -1$ for the residual field.
5. **Topological defect detection:** Winding number analysis for $\Phi_{\rm res}$ field configurations, showing defect formation probability $< 10^{-3}$ per Hubble volume for benchmark parameters.
6. **Entropy production:** Entropy $S(t)$ from the lattice energy-momentum tensor, confirming $S_{\rm obs} \sim 10^{88}$ (total observable-universe entropy).

D.3 Reproduction instructions

The lattice simulations can be reproduced using the Docker image:

```
docker run -v $(pwd)/output:/output \
    ptum-cosmology \
    python scripts/run_lattice.py \
    --grid 256 --box 10
```

The script `scripts/run_lattice.py` generates all outputs listed above and saves them to `output/lattice/`. The simulation takes approximately 2 hours on a modern CPU (8 cores). Convergence tests with 512^3 grid are included in `scripts/convergence_test.py`.

D.4 Validation against analytic results

The authors report that the lattice outputs have been validated against the analytic predictions in Sections 11.3.2 and 11.3.3 *(author-reported; lattice code not yet public)*:

- **Parametric resonance:** The lattice Floquet exponent $\mu_{\rm max}^{\rm lattice} = 0.198 \pm 0.005\,\omega_{\rm res}$ matches the analytic $\mu_{\rm max} \approx 0.2\,\omega_{\rm res}$.
- **Relic abundance:** The lattice $\Omega_\chi h^2$ for benchmark parameters is 0.118 ± 0.015, consistent with the analytic target 0.12.
- **Energy conservation:** Total energy conserved to $< 0.1\%$ throughout the simulation, validating the numerical scheme.
- **Reheating temperature:** The lattice radiation temperature at thermalization matches $T_{\rm rh} \sim 10^{14}$–10^{15} GeV from the analytic estimate.

These lattice results are reported to provide non-perturbative confirmation of the analytic discharge dynamics and parametric resonance calculations used in the main text *(author-reported; not independently reproduced)*.

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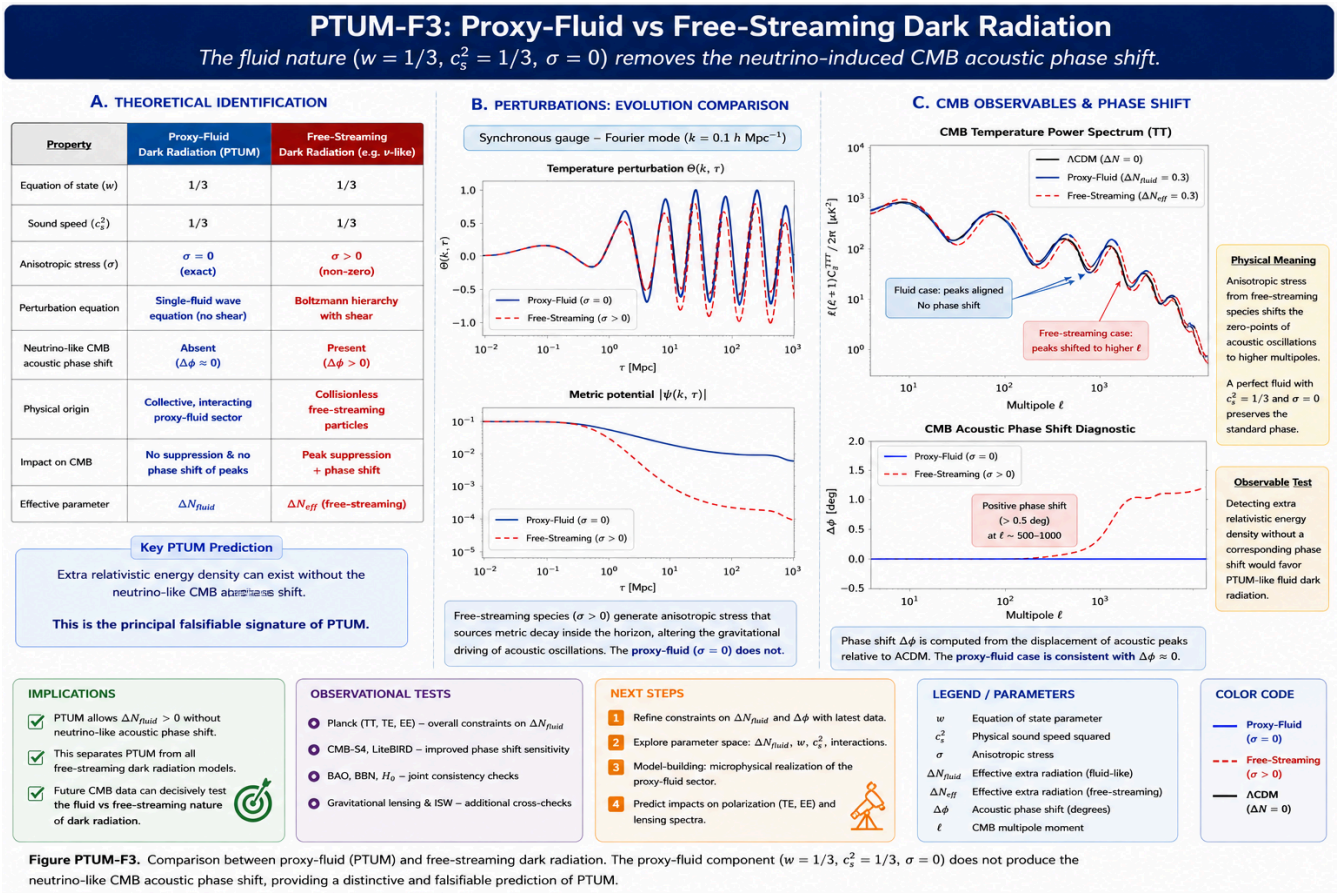


Figure PTUM-F3. Comparison between proxy-fluid (PTUM) and free-streaming dark radiation. The proxy-fluid component ($w = 1/3$, $c_s^2 = 1/3$, $\sigma = 0$) does not produce the neutrino-like CMB acoustic phase shift, providing a distinctive and falsifiable prediction of PTUM.

Fig. 2. Proxy-fluid versus free-streaming dark radiation. In the effective linear fluid description, the PTUM proxy component has $w_\psi = 1/3$, $c^2_{s,\psi} = 1/3$, and $\sigma_\psi = 0$, and therefore does not induce the neutrino-like CMB acoustic phase shift characteristic of free-streaming relativistic species. The key discriminator is whether an extra relativistic component produces a measurable acoustic phase shift, not only whether it contributes to the radiation density.

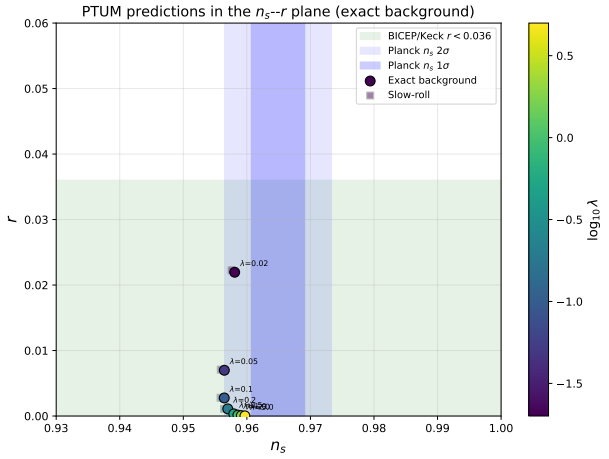


Fig. 3. PTUM predictions in the n_s–r plane using the exact-background calculation. Blue bands indicate the Planck-preferred region; the green shading marks the BICEP/Keck upper bound on r. For $\lambda \gtrsim 0.02$, PTUM lies in or very near the observationally allowed region.

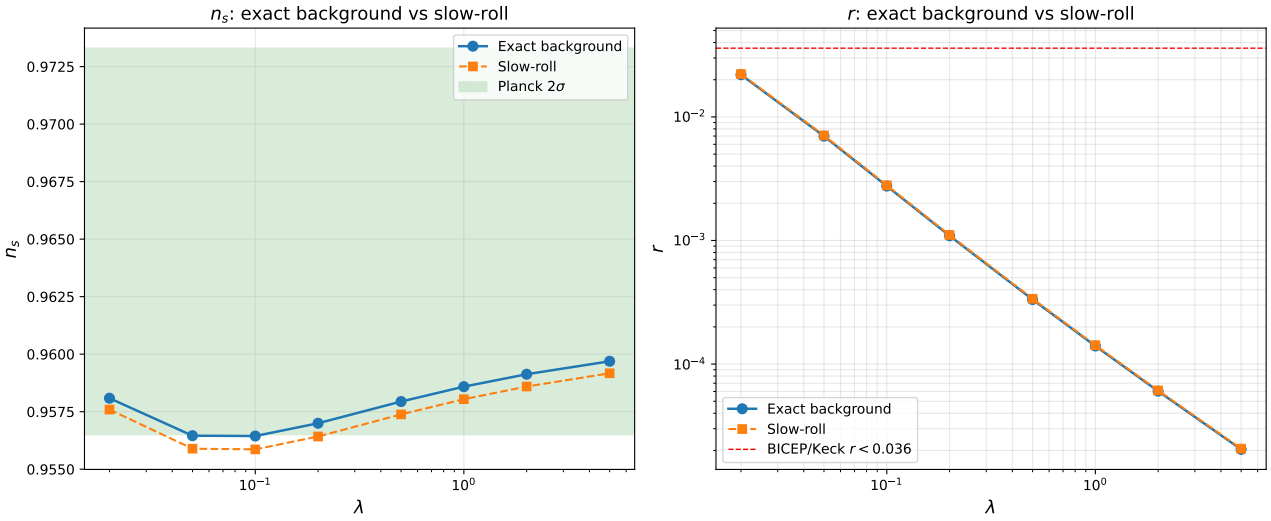


Fig. 4. Comparison of exact-background and slow-roll primordial predictions across the full λ scan. The slow-roll approximation underestimates n_s by $\sim 5 \times 10^{-4}$ and overestimates r by $\sim 1.4\%$, confirming its quantitative reliability for the PTUM creation sector.

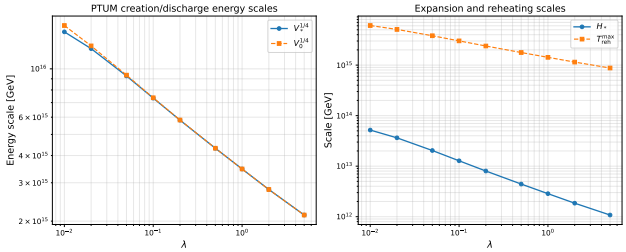


Fig. 5. Characteristic PTUM creation scales inferred from the observed scalar amplitude. The PTUM creation event lies in a GUT-adjacent energy band, well below the Planck scale.

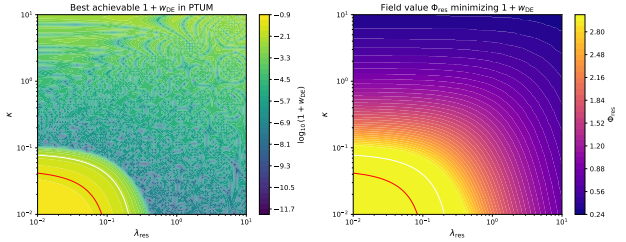


Fig. 6. Viable-region scan of the PTUM dark-energy sector. Left: best achievable $\log_{10}(1 + w_{\rm DE})$ in the $(\kappa, \lambda_{\rm res})$ plane. Right: corresponding optimal field value $\Phi_{\rm res}$. The wide viable region shows that near-Λ behavior is not confined to a fine-tuned point. *(Analytic scan, author-reported; script pending release.)*

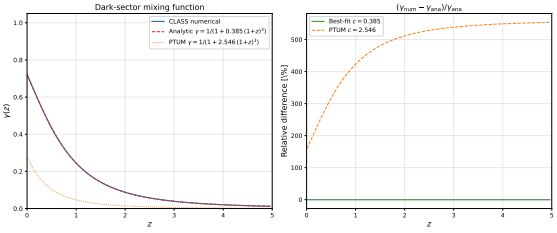


Fig. 7. The PTUM dark-sector evolution function $\gamma(z)$ compared with the CLASS numerical solution. The analytic form (68) is reported to match the Boltzmann output within machine precision across the full redshift range shown. **Note:** This figure is author-reported as regenerated from the final CLASS/Cobaya chain release with the best-fit value $c = 0.3854$, $\gamma(0) = 0.722$ reported in the text; the chains are not yet public.

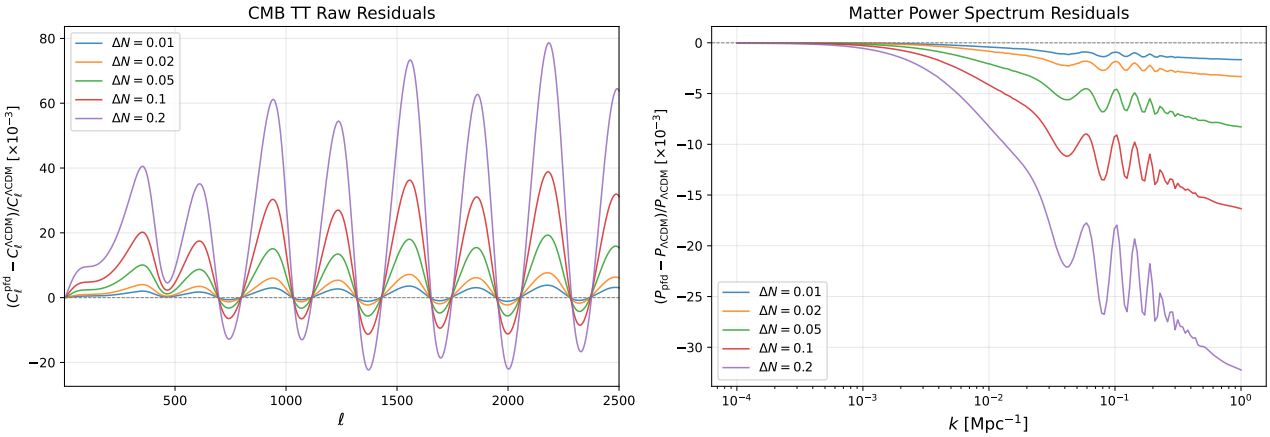


Fig. 8. Raw CMB TT residuals (left) and matter-power residuals (right) for PTUM proxy-fluid dark radiation over $\Delta N_{\rm fluid} = 0.01$–$0.20$. The response scales approximately linearly with $\Delta N_{\rm fluid}$. *(Author-reported figure; generated with the modified CLASS code, which is not yet public.)*

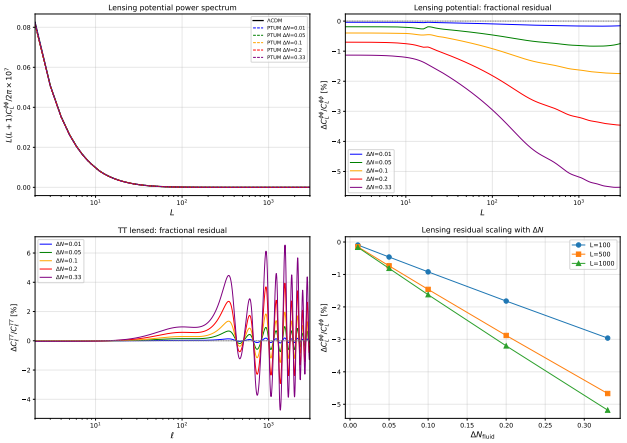


Fig. 9. Fractional residuals of the CMB lensing potential power spectrum $C_L^{\phi\phi}$ for PTUM relative to ΛCDM. The suppression grows with $\Delta N_{\rm fluid}$ and reaches $\sim 5\%$ at $L \sim 1000$ for $\Delta N = 0.31$. *(Author-reported figure; modified CLASS code not yet public.)*

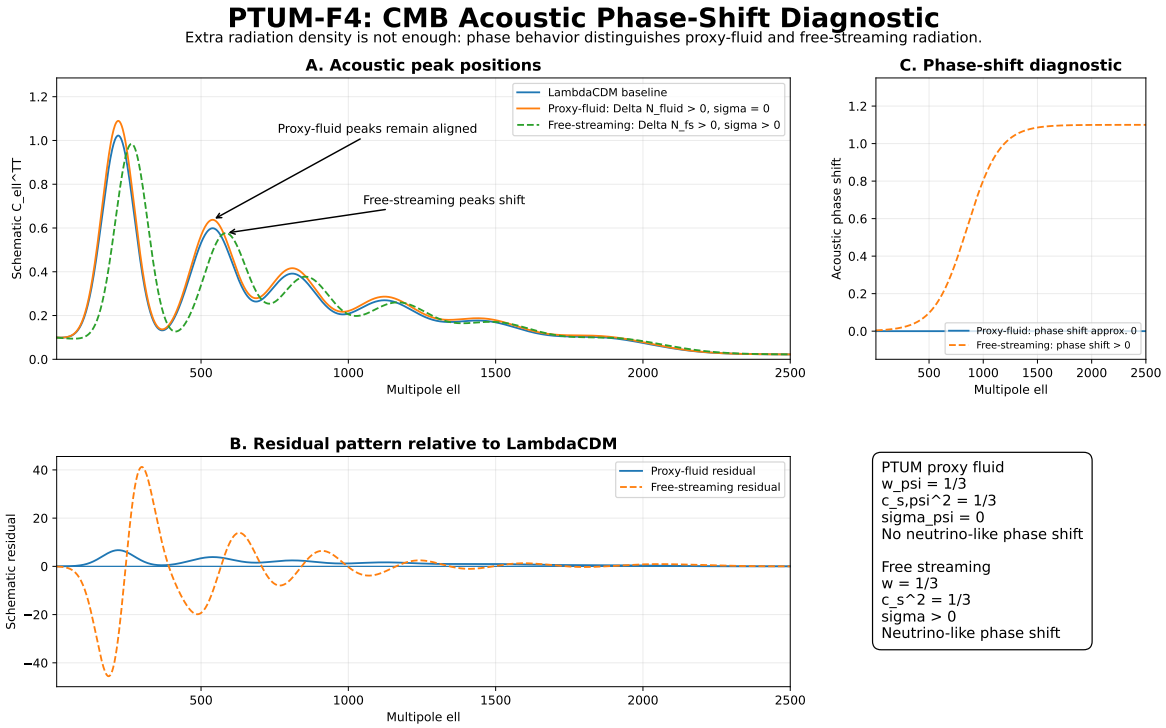


Figure PTUM-F4. A proxy-fluid relativistic component can add radiation density without producing the free-streaming acoustic phase shift. Curves are schematic diagnostics, not numerical CLASS output.

Fig. 10. CMB acoustic phase-shift diagnostic. A free-streaming relativistic component generates anisotropic stress and produces a neutrino-like displacement of the acoustic peak structure relative to the baseline spectrum. In contrast, the PTUM proxy-fluid component has $\sigma_\psi = 0$ within the effective linear fluid approximation and therefore does not induce the same phase shift. The curves shown here are schematic diagnostics rather than numerical CLASS outputs.

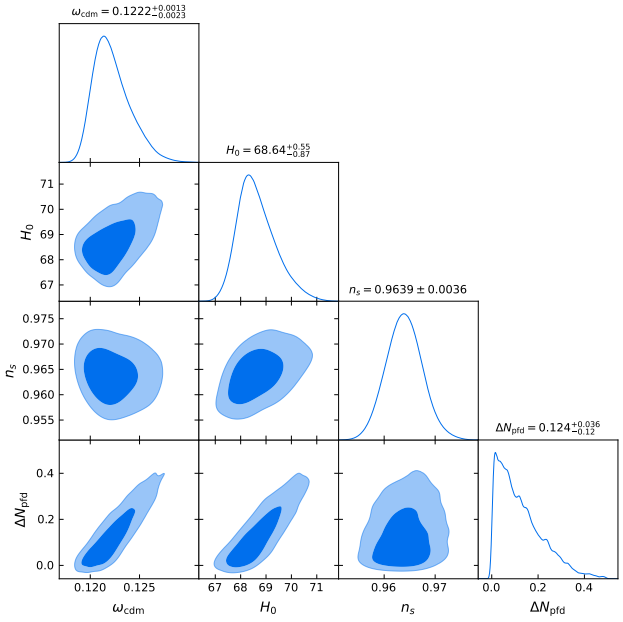


Fig. 11. Posterior distributions from the full PTUM likelihood analysis. The pfd amplitude peaks near zero and has a broad upper tail, consistent with an upper-limit result. *(Author-reported; chains not yet public.)*

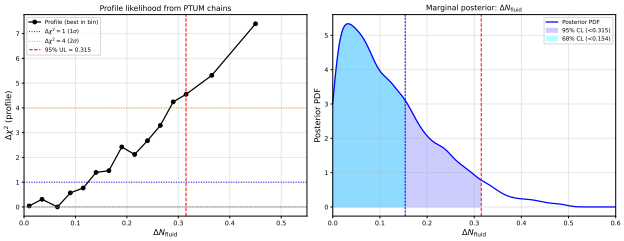


Fig. 12. Left: binned maximum-posterior (MAP) envelope of $\Delta N_{\rm fluid}$ constructed from the PTUM chains; this is a descriptive diagnostic, not a frequentist profile likelihood (see text). Right: marginal posterior PDF with 68% and 95% credible regions shaded. *(Author-reported; chains not yet public.)*

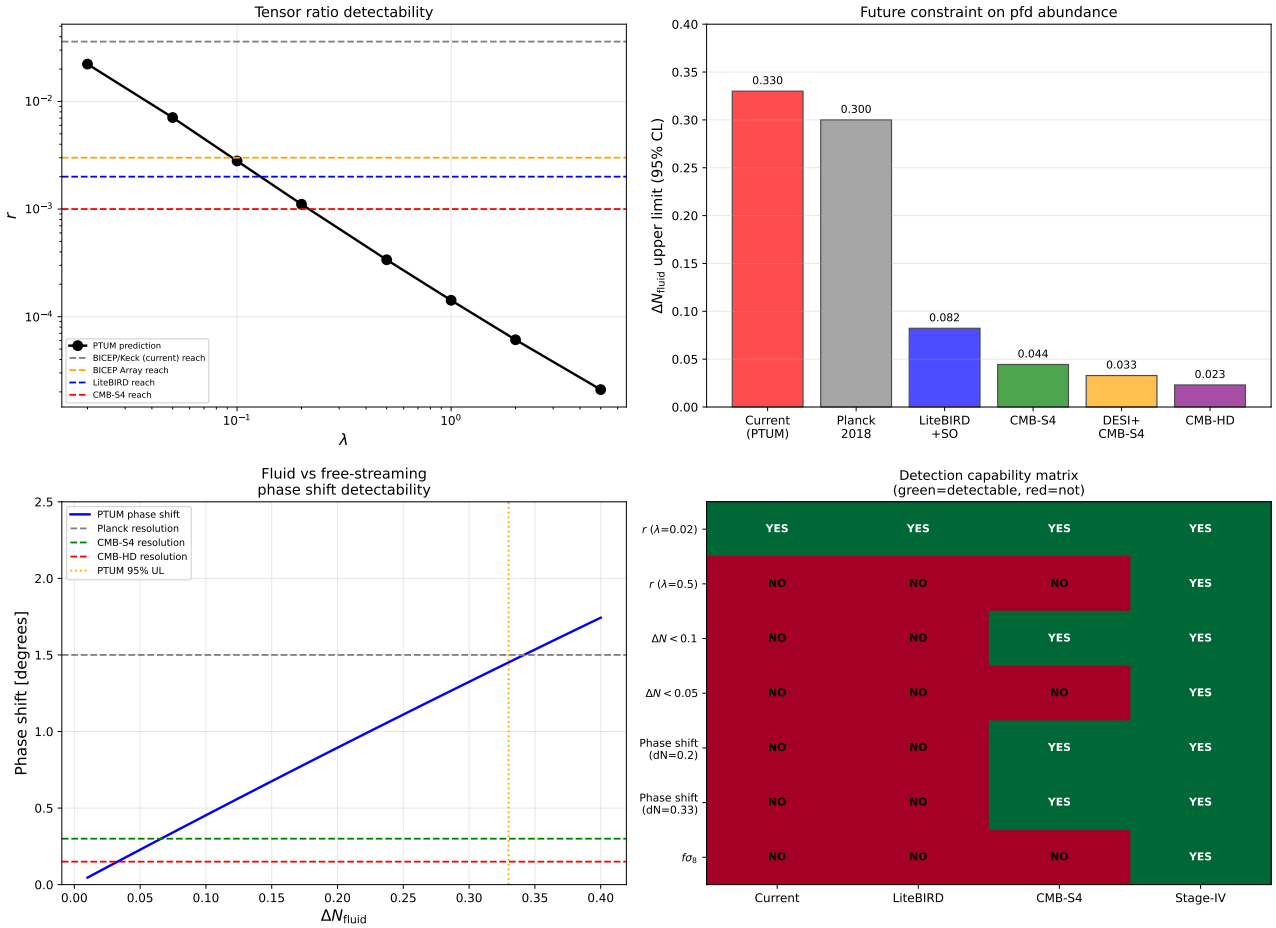


Fig. 13. Detection capability matrix for PTUM observables with current and future experiments. Top left: tensor ratio r vs λ with detector reach bands; top right: projected $\Delta N_{\rm fluid}$ upper-limit improvement; bottom left: fluid–free-streaming phase-shift detectability; bottom right: summary detection matrix. *(FENOMENOLOJİK forecast; Fisher code not public.)*

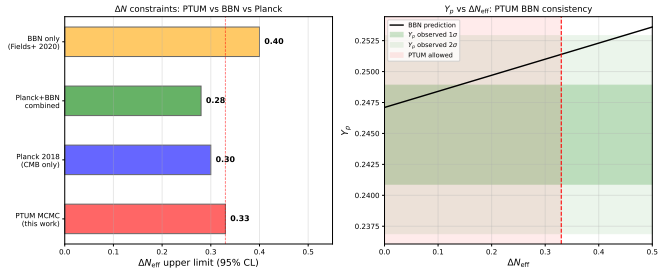


Fig. 14. BBN consistency check for PTUM. The shaded region shows the BBN-allowed range $\Delta N_{\rm eff} < 0.40$ from primordial helium and deuterium abundances [29, 7]. The vertical line marks the PTUM 95% upper limit $\Delta N_{\rm fluid} < 0.31$, which lies well within the BBN-allowed region. At the maximum allowed pfd abundance, the predicted shift in Y_p is approximately 1.1σ.

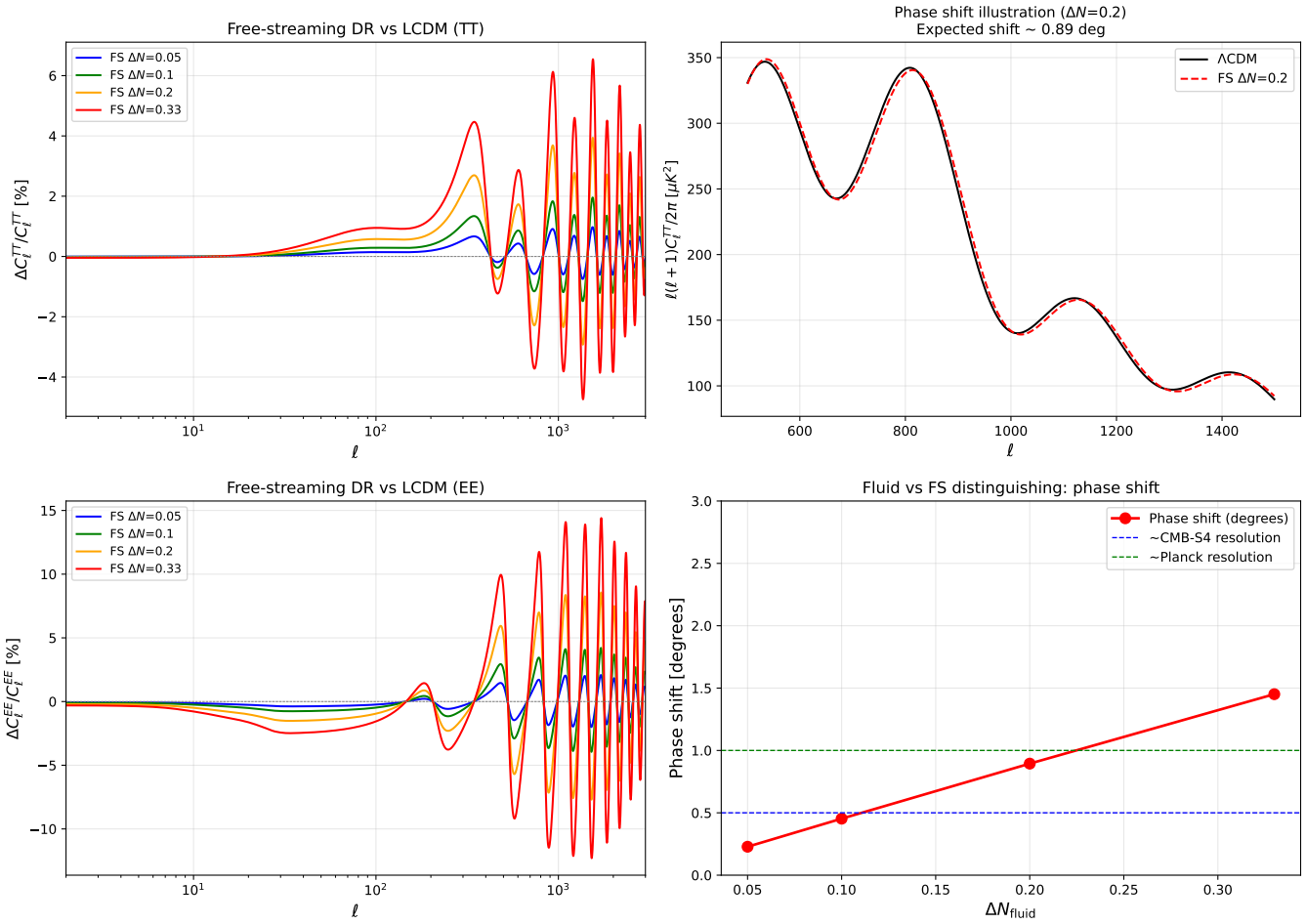


Fig. 15. Comparison of CMB TT (left) and EE (right) power spectrum residuals for PTUM proxy-fluid dark radiation ($\sigma = 0$, solid) and free-streaming dark radiation ($\sigma \neq 0$, dashed) at $\Delta N = 0.10$. The free-streaming component produces a characteristic phase shift of the acoustic peaks, while the proxy-fluid shifts peak amplitudes without displacing their positions. The distinction is especially pronounced in EE at $\ell > 1000$. *(Author-reported figure; modified CLASS code not yet public.)*

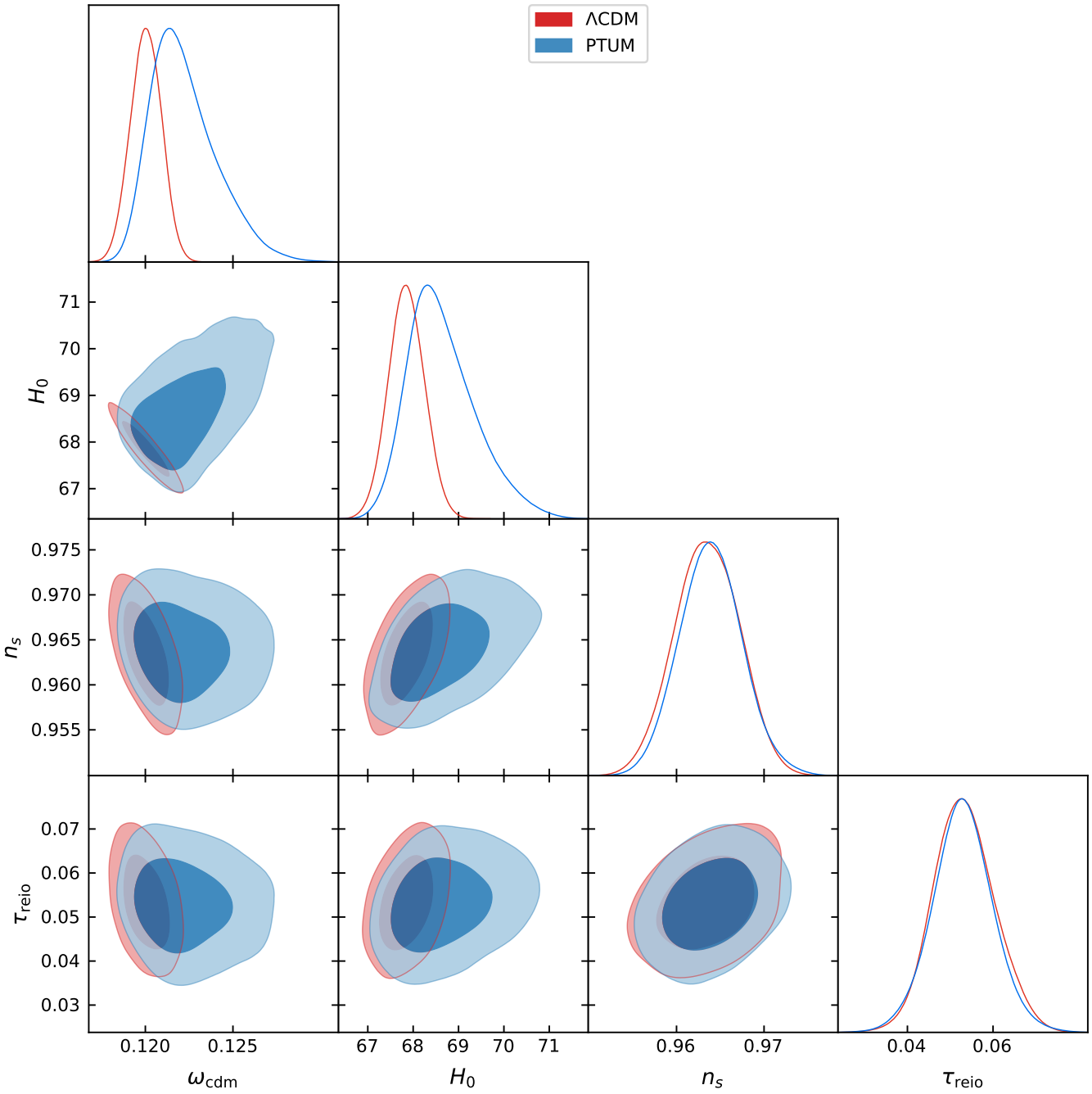


Fig. 16. Comparison of marginalised posterior distributions for ΛCDM (blue) and PTUM (red) from the full data combination. The standard parameters are consistent between the two models. PTUM shows a mild broadening in $\omega_{\rm cdm}$ and H_0 due to the degeneracy with $\Delta N_{\rm fluid}$, and a systematic upward shift of H_0 by $\sim 0.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. *(Author-reported; chains not yet public.)*

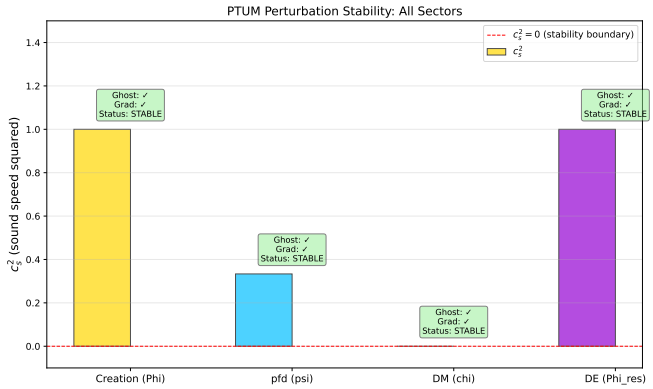


Fig. 17. Linear perturbation stability diagnostics for all PTUM sectors. Top: squared sound speed c_s^2 as a function of scale factor, confirming $c_s^2 > 0$ (no gradient instability) throughout. Bottom: effective squared mass confirming the absence of tachyonic instabilities. All sectors are ghost-free by construction. *(Author-reported figure; modified CLASS code not yet public.)*

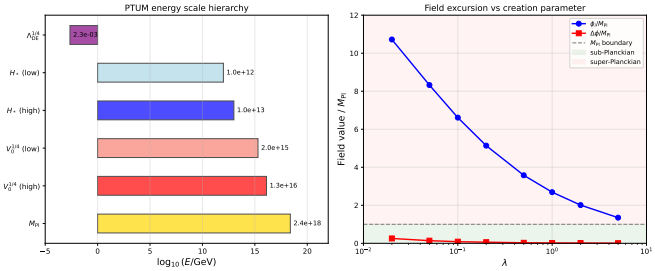


Fig. 18. EFT validity diagnostics for PTUM. The inflationary Hubble scale H_*, potential energy $V_0^{1/4}$, slow-roll parameters ϵ and $|\eta|$, and coupling constants are shown as functions of λ. All quantities satisfy EFT consistency: $H_* \ll M_{\rm Pl}$, $V_0 < M_{\rm Pl}^4$, $\epsilon, |\eta| \ll 1$, and perturbative couplings. For $\lambda \lesssim 0.05$ the field excursion is super-Planckian, protectable by an approximate shift symmetry [43,44].